

Name: Bol
Datum: 03.08.2004

Name:
Änderungsdatum:

INTEGRAL HYDRAULIK
Anlagen + Systeme

Geräteliste zu Schaltplan: **46.081 Narayangonj**

HSLM 1000/45/2

Pos.	Stck.	ID-Nr.	Zeichnung-Nr.	Benennung	Typenbezeichnung	Lieferant
1	1	15690	035-1060-150-048	Geräteschrank / hydraulik cabinet	Nennvolumen 1000 l	Integral Hydraulik
2.10;2.20	2	11982	411-1319-014-031	Innenzahnradp. / internal gear pump	PT4-31,5	Integral Hydraulik
	2	12196	350-1061-514-027	Pumpenträger / bell housing	PK350/10/19	KTR
	2	12200	425-2295-048-032	Kupplung / coupling	R42/55.48-32	KTR
3.10;3.20	2	12203	001-2443-009-063	E-Motor / electric motor	18,5KW;B5;380V;50Hz;1500min-1;IP55;ISO F/B	Siemens
4.1;4.2;4.3	3	12038	100-2417-025-250	Manometer / pressure gauge	MG 100/250B	Wika
	3	12215	004-2453-212-200	Messschlauch / hose	S100-AC-AC-0200	Hydrotechnik
5.2;5.5	2	12577	001-1332-009-020	Blockkugelhahn / shutoff valve	BKH-20S-SW9-1123	Böhmer
5,4	1	13014	001-1332-009-028	Blockkugelhahn / shutoff valve	BHG V315.320/1	Böhmer
5.7	1	12216	001-1332-009-034	Kugelhahn / shutoff valve	Best.Nr.:101004	AS
	1	11090	034-2208-000-000	Verschlussschr. / screw connector	DIN 906-G3/4/4-PH	
6.10;6.20	2	12327	301-1330-013-640	Druckbegrenzungs v./pressure relief valve	DBG-6S-210/140	Integral Hydraulik
6.2;6.5	2	12327	301-1330-013-640	Druckbegrenzungs v./pressure relief valve	DBG-6S-210/160	Integral Hydraulik
6.1;6.4	2	11728	302-1330-013-640	Druckbegrenzungs v./pressure relief valve	DBG-6S-210/210B (bauteilgeprüft)	Integral Hydraulik
	1	12218	001-1312-009-211	Reihenanschlußplatte NG 10 / connection plate NG 10	RAP-U1/2-10-3/4-02	MeWesta
7.2;7.5	2	12504	001-1342-009-067	2/2-Wege-Sitzventil / valve	AS22101a-115VAC	Wandfluh
7.6;7.10	2	12504	001-1342-009-067	2/2-Wege-Sitzventil / valve	AS22101a-115VAC	Wandfluh
7.3	1	18204	001-1342-009-106	2/2-Wege-Sitzventil-Patrone / valve	SBV11-12N-C-O-120AGH	IMAV
7.3.1	1	12236	001-1342-009-080	Ventilgehäuse / manifold	2LH-12A-B06	IMAV
7.1	1	18205	551-1340-028-000	Wegeventil / direk tional valve	W4A-10M000-AC115	Integral Hydraulik
	3	12074	509-1340-014-902	Anschlußplatte NG 10 / connection plate NG 10	G 3/4	Integral Hydraulik
	7	12543	001-1305-009-037	Betriebsanzeige / power light	Nr.3124018 mit LED gelb 110V	Murr Elektronik
9.1;9.2	2	12485	020-1531-014-014	Rückschlagventil / check valve	RHD20-PS (mit Zusatzbohrung)	Ermeto

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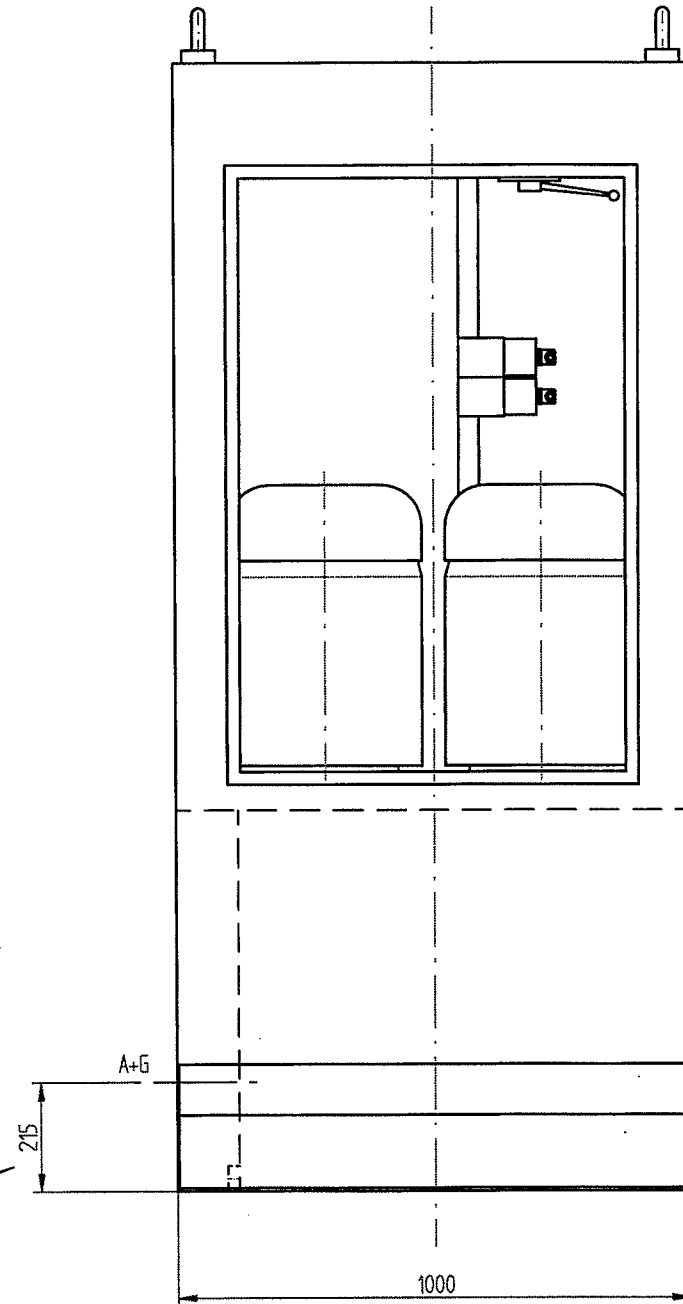
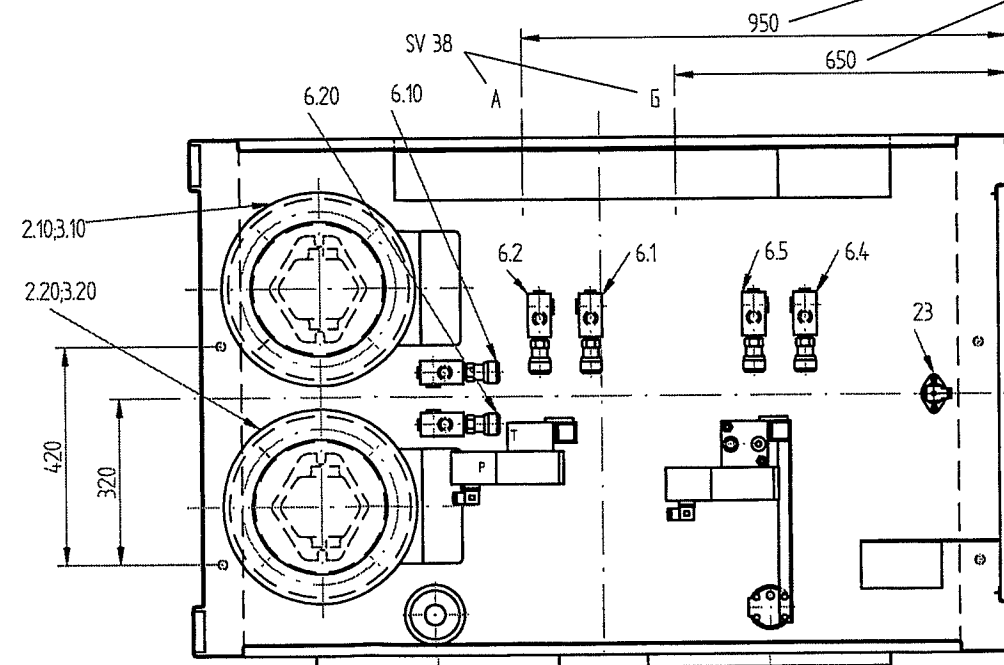
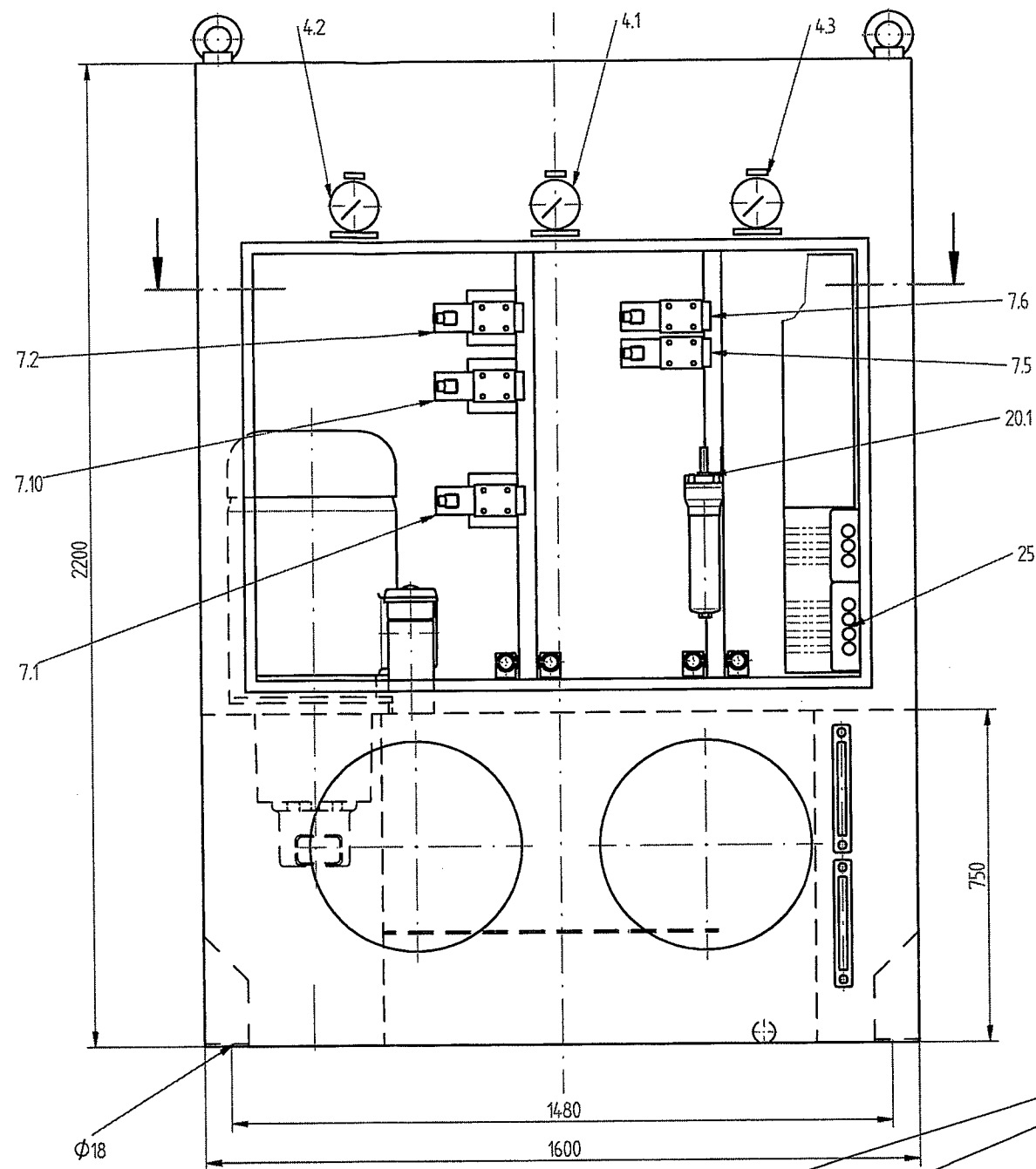
INTEGRAL HYDRAULIK
Anlagen + Systeme

Geräteliste zu Schaltplan: **46.081 Narayangonj**

HSLM 1000/45/2

Pos.	Stck.	ID-Nr.	Zeichnung-Nr.	Benennung	Typenbezeichnung	Lieferant
9.3	1	12485	020-1531-014-014	Rückschlagventil / check valve	RHD20-PS (mit Zusatzbohrung)	Ermeto
9.10;9.20	2	12485	020-1531-014-014	Rückschlagventil / check valve	RHD20-PS (mit Zusatzbohrung)	Ermeto
18.2;18.5	2	12189	022-1334-024-022	Drosselventil / flow restrictor	FT257/2-3/4	Berendsen
18.6	1	12189	022-1334-024-022	Drosselventil / flow restrictor	FT257/2-3/4	Berendsen
18.1;18.10	2	12189	022-1334-024-022	Drosselventil / flow restrictor	FT257/2-3/4	Berendsen
19	1	12056	001-1327-009-456	BelüftungsfILTER / air breather	Pi 0145 Mic	Mahle
20.1	1	16365	001-1327-009-578	Druckfilter / filter unit	Pi 30025-12	Mahle
	3	16368	001-1327-009-579	Filterelement / filter element	Pi 24025 DN Sm-x16	Mahle
23	1	18239	001-1305-009-624	Niveauüberwachung / zero overlap	SSM.1.B5.600.5.S1	Engler
	3	12197	313-2487-000-000	Ölstandsanzeige / oil level indicator		UCC
	2	12433	475-1260-014-028	Reinigungsdeckel / inspection cover	RD 475-V449-6	Raja
25	1			DreifachTaster / switch		
26	1			Klemmenkasten / terminel box		

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Zeichen	Änderungen	Ä.-M.-Nr.	Datum	Name
Datum	Name	Zeichnungs Nr.		
03.08.04	Bol	046-1462-012-081		
gepr.				
N. gepr.				

INTEGRAL HYDRAULIK
Anlagen + Systeme

Ges.-Gew.	900 kg	Betreff:	NARAYAGONJ
Gez.		Ersatz fuer	
Gepr.		Ersetzt durch	
Gez.		File-Nr.:	
Gepr.			
Oberflächenzeichen Mittelnautweg-R in um Masse ohne Toleranzangaben nach DIN 7168 Gütegrad m41el			
Maßstab	Hydraulikschrank		
1:10	Typ HSLM 1000 /45-2		
	HYDRAULIK CABINET		
G46.081			
Urheberschutz nach DIN34 vorbehalten Total BL BLNr.			

LOESCHE

CAD EDV

Installation instructions Start-up, upkeep and maintenance General hazard information

1. PIPE-LINES

1.1 GENERAL

For the installation and erection of hydraulic lines, the following are necessary: hand tools, bending rig, grinder and as well as the possibility to store erection materials.

Discussions to reach agreement between the Client and the Contractor with respect to the pipe routing are also a part of the preparation.

The installation of hydraulic piping should take place in clean surroundings in order to avoid contamination of the hydraulic system, for which the working area must be protected against dust and other contaminants.

During installation, connections to the piping and to the equipment and cylinders must remain closed. Before making connections, connectors and the corresponding connecting surfaces are to be checked for cleanliness.

The installation of the hydraulic piping network must be carried out carefully, whereby special attention is to be given to:

- Cleanliness of the work
- Straight, direct and easy-to-understand pipe routing
- Installation of the piping without tension
- During the installation, attention must be paid to
- ensure no dirt gets into the hydraulic system
- A straight-line pipe routing is to be preferred to make maintenance and repairs much easier
- Piping is to be relocated without tension

After the pipe routing has been established, the work at the erection site can begin with the installation of the pipe clamps necessary for the pipe routing. The spacing between two pipe clamps (restraints) should amount to 1 to 1,5 m. It is very important that the piping is installed without tension.

To make the installation as well as later maintenance and repair easier, the locations of connections should be offset with the necessary spacing to the next pipe.

Contacts between the pipe and construction elements should be avoided, especially with the installation of stainless pipes.

The pipe is best cut to length with the help of an appropriate machine. Should it be necessary to saw a pipe to length, it is absolutely necessary, that the pipe be deburred and leaned. This can be done using available compressed air, with which the pipe is blown out. The bending of pipes is to be preferred to the use of threaded fittings. Bending is faster, cleaner and also cheaper.

Threaded angle fittings are connections which can work loose and result in leaks.

1.2 INSTRUCTIONS FOR WELDED PIPES AND CONNECTIONS

When welding on the threaded fittings foreseen for the hydraulic system, a weld nipple is welded to the pipe. The method of welding corresponds to the usual technical possibilities today, whereby there are no problems with leak-tightness between the pipe and the weld nipple.

O-rings are used to provide a sealing between the weld nipple and the threaded fittings, so that no problems are to be expected here with proper work. The holding function between the weld nipple and threaded fitting is made with the help of a threaded collar.

The welding process to be recommended is the TIG process (tungsten inert gas), whereby the electrical current is sent through a fixed tungsten electrode surrounded by Argon gas. The filler material is introduced by hand or automatically, whereby a forming gas is introduced through the pipe in order to avoid burning through the wall into the inside. Because of the use of both gases, the TIG welding process is very clean, meaning that no slag is formed. Nevertheless, the welding seams should be pickled after welding in order to remove slag and any welding beads from the pipe.

After being pickled, the pipe must be passivated.

After the pipe installation is completed, the individual lines are flushed. The use of a separate flushing unit is to be preferred.

One can assume that flushing will require approx. 1 hour per pipe. The flushing medium must be compatible with the operating medium. The flushing of the pipe is made with the cylinder and hydraulic equipment clamped off. When the flushing is completed, the piping is connected to the hydraulic system. All connections that can be loosened must be tested for tightness, meaning that all threaded collars must be tightened by hand also using an appropriate wrench.

The system is ready for operation when this work is concluded.

1.3 ASSEMBLY INSTRUCTIONS FOR PROGRESSIVE RING FITTINGS

1.3.1 General

The basic feature of the progressive ring is that precisely designed geometry permits the limited, progressive penetration of the ring into the tube. The front cutting edge has already begun to cut into the tube before the second cutting edge contacts the tube. When the optimum prescribed depth for both cutting edges has been reached, the top edge prevents further penetration. Because of the stop edge and the shape of the two cutting edges, the forces arising are distributed along the entire taper length, thus increasing joint strength. This distribution of the forces and the specially designed internal contour at the collar end of the ring guarantee increased reliability for flexure loading. Vibrations are thereby dampened at the entry into the ring. During assembly, the stop edge causes the tightening forces to be increased up to a clearly perceptible fixed point. (assembly fixed point). After assembly is completed, the visible thrown-up collar must completely fill out the space in front of the first cutting edge.

1.3.2 Preparation of the fitting components

Oil thread and taper of the fitting as well as ring and thread of the nut. Slide nut and ring onto the end of the tube. Do not slide ring on backwards. Tighten nut by hand into tight.

Always press the tube against stop in inner cone of the fitting body. Mark a line on nut and tube in order to observe the required number of turns. Tighten the nut by approximately 1½ turns. (tube must not turn with nut). Stop edge limits tightening by increasing force required.

1.3.3 Inspection of assembly

Loosen nut, check to make sure that a visible collar fills out the space in front of the first cutting edge (it does not matter if the ring can be rotated on the tube end). All connections that can be loosened must be tested for tightness, meaning that all threaded collars must be tightened by hand also using an appropriate wrench. The system is ready for operation when this work is concluded.

1.3.4 Re-assembly

For every re-assembly the nut must be retightened without applying more force than for the first assembly. The fitting body must be firmly held during all assemblies.

1.4 FLUSHING OF THE SYSTEM

After the pipe installation is completed, the individual lines are flushed. The use of a separate flushing unit is to be preferred. One can assume that flushing will require approx. 1 hour per pipe. The flushing medium must be compatible with the operating medium. The flushing of the pipe is made with the cylinder and hydraulic equipment clamped off. When the flushing is completed, the piping is connected to the hydraulic system.

2. PROVISIONS FOR START-UP PROCEDURES

2.1 PIPELINES

The connections between the pipelines of the equipment and the machines (at the consumers) must be made by trained hydraulic experts. The pipes and fittings used must comply with the pressure rating the equipment is designed for. All the pipelines must be held by exactly fitting mounting brackets. The pipes must be seamless hydraulic pipes pursuant to DIN 2391/C, St. 35.4, NBK and bonderized. Corresponding hydraulic pipes made of high-grade steel may also be used.

Tubing must be effected as straight and short as possible. Use of angles and knees is to be avoided. The number of screwed joints must be kept as low as possible, because every screwed connection represents a leakage risk. The dimensions of all connecting pipes must comply with the specifications given in the circuit diagram. The use of pipe with too narrow an inside diameter or with a length longer than stipulated leads to pressure losses that may negatively affect the performance of the equipment.

2.2 CLEANLINESS

Without exception all the pipes used for the oil-hydraulic equipment must be clean inside and free of all rust or dirt particles.

After installation they may not be welded or soldered any more. All pipes must be installed carefully, exactly, and free of stress.

Sharp edges and burrs at the end of the pipes are to be removed.

If threads at fittings are sealed with liquid sealing agents, care shall be taken that the nose of the thread remains free, otherwise hardened particles of the sealing may find their way into the system.

Remove any vent plugs in the connections of the equipment only directly prior to connecting the relevant lines. This helps to keep the equipment clean.

2.3 MOTORS AND ACCUMULATORS

When installing the electric motor, it is indispensable to take care of the alignment between pump and motor shaft. This applies in particular to type B 3.

However, even in the case of complete, ready-assembled equipments of this type the equipment may suffer from distortion caused by transport. Therefore it should always be checked before start-up.

The values for voltage and amperage stated on the electric motors and components must correspond with the factual energy supply. After connecting the electric motor, the sense of rotation is to be checked. This is done by a short start. The sense of rotation is shown by an arrow on the pump. If the pump runs in the wrong direction for more than a few seconds, it will lead to serious damage caused by dry run.

Gas-filled accumulators must be brought to the correct prefill pressure. The position of the deaerating and shut-off valves is to be checked according to the circuit diagram. Unless otherwise stated, nitrogen is used as a gas medium. Manufacturer's provisions must unconditionally be complied with when filling, mounting, and maintaining the accumulators.

2.4 HYDRAULIC FLUID

The hydraulic fluid is to be handled very carefully. Particular attention shall be paid to the use of the proper fluid and to the fact that it is absolutely clean. Special care shall be used if the oil is taken from a barrel. It shall be checked whether the contents are free of any contamination such as rust or condensing water.

The medium shall be filled through the filler neck. If possible a pump-over unit with a fine filter is used for filling.

All parts shall be thoroughly scavenged before the equipment is run under full speed and working pressure. If at all possible, the same fluid is used for this purpose that is later used for normal operation. The consumption connections are bridged by hose connections and very sensitive devices are replaced by scavenging plates during the scavenging. If the pump belonging to the equipment has to be used for scavenging, care has to be taken that it only delivers clean fluid and working pressure is low. If a scavenging liquid is used, it must be compatible with the hydraulic fluid. If, however, the hydraulic fluid is immediately used for scavenging, it is an advantage to leave it in the equipment because an oil change may enable fresh impurities to enter the tank. After scavenging new filter elements are to be inserted.

During the first start-up the liquid level falls because the oil distributes itself all over the pipe system of the equipment. Therefore hydraulic fluid must be filled up as quickly as possible. Later on the liquid level must also be checked at regular intervals. The fluid must on no account remain below the lower marking.

3. START-UP PROCEDURES

3.1 PREPARATIONS

- Check oil level in tank.
- Any shut-off valves in the suction pipes must be fully opened.
- Prefill the housings of all piston pumps with clean hydraulic medium.
- Care shall be taken that the equipment can start up operationally reliable and that no operators are within hazard areas.

3.2 START-UP OF THE EQUIPMENT

- All time data, temperature values, and all other details worth knowing should be recorded for the future maintenance of the hydraulic equipment.
- The electric motor is briefly switched on and immediately switched off again. The sense of rotation of the pumps is checked.
- If automatic deaeration is not available, the screwed joint at the pressure side of the pump is loosened or deaeration valves opened so that the pump can start up more easily and air can leave the equipment better.
- The mobile parts of the equipment are to be run under low pressure so that air escapes. This shall be continued until the liquid emerging from the deaeration connections is completely free of bubbles.
- The equipment is to work against low pressure for some time. This is, so to speak, the last scavenging before the pump is started up under full load, always provided that oil flows through the filters of the equipment.
- Pressure valves are to be set to the lowest value the pump can satisfactorily work with.
- If the equipment is equipped with a compensator pump, an additional pressure control valve, then the value set at the pressure control valves must at least be 10 bar higher than the pump pressure.
- Once again the oil level in the pump shall be checked.
- As soon as operating temperature has reached its normal value all screwed pipe connections and mountings must be checked and retightened if necessary.

- Filters shall be cleaned frequently during the first running period and thereafter at intervals taught by experience. Fine filters with a high storage capacity are recommended for the first running period of hydraulic equipments that have to work with high precision.

3.3 HYDRAULIC FLUIDS

3.3.1 Hydraulic oil

The hydraulic media of a hydraulic equipment must have a series of properties in order to be able to guarantee operating safety for a long time. It is recommended to make the choice together with a well-known oil producer. Hydraulic oils contain a lot of additives, eg. with properties inhibiting corrosion and reducing wear. They must comply with DIN 51524. When hydraulic fluids are topped up, one should in principle use the identical fluid the system has been filled with. This should also be considered when using mineral oils, for oils that may comply with DIN 51524, part I or part II, are still differing in their use of additives and their basic oils.

Mixtures of various makes should in any case be avoided with hydraulic fluids of classes HFA, HFC, and HFD.

If fluids of the same type, but of different producers are mixed, any question of liability resulting from damages can no longer be allocated.

It can be said that producers of hydraulic fluids often decline any responsibility if any additives of different producers were mixed into the hydraulic fluid. The manufacturers of hydraulic devices adopt a similar attitude.

Even before using so called HEPG, HETG and HEES "Bio-Oils" manufacturer should be consulted.

In principle only relevantly qualified staff should carry out the start-up of hydraulic equipments. During initial operations the circuit diagram, list of parts, technical information of the installed devices and measuring equipment should be available. Start-up must be carried out with due care. Responsible staff of the manufacturer should be present. The same applies to the personnel of the operator. It is recommended to prepare a protocol of the start-up.

4. UPKEEP AND MAINTENANCE

4.1 MAINTENANCE

Maintenance measures must be planned and carried out in relation to value, action time and the required availability.

4.1.1 Inspection

Various inspection intervals will be required according to different action times eg. daily inspection, monthly inspection, and an inspection directly prior to a longer standstill of the equipment.

Important items are:

- Check liquid level in tank

Normally too low a level of liquid points to loss by outer leakages, after more intensive upkeep work the liquid level may continue to drop slowly for some time if the equipment is automatically deaerated. Too high a liquid level may indicate work without load in the upper parts of the equipment caused by penetration of air during standstill periods.

- Check heat exchanger for effectiveness -

Air - liquid heat exchanger -

Heat loss drops very quickly on account of dust deposits on the surface of the heat exchanger especially in locations with a high development of dust.

If cooling air then contains oil mist from outer leakages as well, heat loss is almost completely interrupted within a very short time.

Water - liquid heat exchanger -

Cooling water must be so clean that the transfer of heat is not hindered by insulating layers of mud. This is of particular importance if the temperature of the hydraulic fluid is kept constant by throttling the cooling water, thus not allowing for high water speeds to flush out deposits. It is important that water heat exchangers can drain neither at the side of the water nor the hydraulic fluid, because only partial surfaces will then be used for heat transmission.

- Check imperviousness of the equipment towards the outside

Pipelines, hosepipes (especially at the place of connection), pumps, control devices, hydraulic motors, and cylinders are to be checked.

- Check liquid temperature during operation

A rise in liquid temperature may have the following reasons:

The heat exchanger has too low an effect (surface soiled, ventilator out of action, cooling water supply insufficient or interrupted, cooling water temperature too high at entry, heat exchanger silted up).

Heat dissipation of the liquid tank, the pipeline system, and the hydraulic components has been reduced by dirt deposits.

Inner leakage of individual components has increased, pressure control valves respond to pressure that is too low. The equipment is operated outside the authorized data, etc.

- **Check pressures**

Check cutting-in pressures of pressure relief valves as well as set pressures of pressure reducing, unloading valves etc..

- **Check leakage quantities**

By measuring the quantities of leakage oil at hydraulic motors and partly at pumps, conclusions can be reached concerning the state of wear. Slow withdrawal (or extension) of a cylinder under outer load with all shut-off devices blocked may indicate defective piston packings.

- **Check noise development**

A hissing noise is heard when pressure valves open. Rattling or whistling noises indicate a defect.

Cylinders may create rattling or grating noises when extended or withdrawn. It may indicate worn guides, secondary bending (eg. rusted-in tilting bearings), unsuitable hydraulic fluid etc.

An increase in pump noises (applies to oil engines, too) under mounting pressure may indicate erosion or cavitation damages within the control surfaces or increasing play of the displacement elements or beginning roller bearing damages. Unpleasantly loud noise of the pumps, independent of the pressure load, point out too high negative pressure in the suction (or supply) pipe.

- **Filter check**

Whenever necessary, filter cartridges must be replaced in reflux filters equipped with contamination indicators. Dirty filters may lead to malfunctions of the hydraulics within a short time.

The same applies to pressure filters. In principle, filter elements are to be cleaned or replaced whenever the liquid is changed.

- **Performance and speed**

Check extension and withdrawal times of cylinders or the speed of oil engines, respectively.

Measure power consumption of electric motors of the pumps.

- **Pipeline system and hosepipes**

Sealing work at the pipeline system only to be carried out in pressureless state. Leakages in places sealed with soft sealings (O-rings, form ring seals) cannot be removed by re-tightening as these sealing elements are either destroyed or hardened.

Sealing can only be carried out by replacing the elements. Pipelines to be checked for firm seat at fixing points.

Hosepipes to be checked for chafe marks and hosepipe fittings for impermeability.

4.1.2 Cleaning the equipment

Hydraulic equipments are to be cleaned on the outside from time to time in order to find leakages, avoid taking dirt into the interior when refilling hydraulic fluid and replacing filter elements, protect cylinder rods against grooving during extension and withdrawal, not to reduce or completely interrupt head dissipation considered in the design.

When cleaning, it always has to be taken care that cleaning fluid does not enter the hydraulic system.

Exchange of hydraulic fluid should be carried out at regular intervals - but at least once a year. At the same time the medium contained in the pipelines and cylinders should also be removed. The hydraulic system is to be deaerated after the renewed start-up.

4.1.3 Maintenance of pressure vessels

The gas precharge pressure of accumulators is to be checked at intervals, always under almost identical ambient temperatures and with a pressureless hydraulic system. This applies to diaphragm accumulators, piston accumulators and bladder accumulators.

For further information see our information sheet 15-007.

With very few exceptions hydraulic accumulators are filled with nitrogen as far as gas is concerned.

For safety reasons a sign with the following, easily legible text must be so applied to all equipments that it cannot be overlooked.

**ATTENTION - PRESSURE
ACCUMULATOR EQUIPMENT!
Prior to starting repair work the
equipment must be rendered pressureless at the side of the liquid!**

4.1.4 Routine maintenance

The following list contains proposals for some checks to be made at regular intervals:

1. Machine cycle time.
2. Operating temperature of the hydraulic fluid.
3. Reading the individual manometers.
4. Unusual noises.
5. Regular measurements of leakages at the piston housing under identical pressure and temperature conditions to establish comparative values. Changes in these leakages will yield information about the general state of the equipment.

Essential deviations from the nominal values with regard to the above 5 items may lead to further investigations of the causes of these faults and to the implementation of maintenance or upkeep measures.

6. Clean all hydraulic devices on the outside and carry out leakage checks. If necessary, screwed connections must be retightened. If this is no longer feasible, they must be exchanged.
7. Check whether heat exchangers are clean and impervious.
8. Remove hydraulic fluid to check its chemical composition and degree of contamination.
9. Clean filter elements.
If necessary, replace with new ones.
10. Check ventilation filter for dirt deposits. Clean or replace, if necessary.
11. Check whether pre-charge pressure has the proper value when using gas-filled accumulators. Regulations of the manufacturer are to be precisely followed when refilling.

4.2 OVERHAUL OF THE EQUIPMENT

If routine maintenance and inspections show after some years of operation that the performance of the equipment no longer corresponds with the specified standard values, it should be time for a thorough overhaul of the hydraulic devices. This offers the opportunity to really thoroughly clean all the devices, namely the inside of the tank.

The use of a plain scraper of plastic is recommended for cleaning even surfaces, because fibres may be left behind when using cloths. These may lead to considerable breakdowns. It is advisable to get into contact with a firm of experts for hydraulics before starting overhaul work and obtain their advice for maintenance and the replacement of components suspected of faults or wear.

In this context a set of new sealings can also be ordered. This is recommendable as after several years of operation the sealing elements will no longer guarantee proper sealing on account of hardening.

Hoses in the supply lines and tubular feedthroughs that are worn or have become porous should also be checked.

The relevant provisions must be observed when reassembling the hydraulic devices. It has always to be taken care that the hydraulic devices are absolutely clean inside.

4.3 SAFETY MEASURES

Proper care must always be exercised when starting-up or maintaining the equipment. Let us remind you in particular of the following points:

1. Accumulators to be rendered absolutely pressureless when their pipelines are removed.
2. Check whether clutches slip or are loose.
3. Loads that may drop without control or may be moved in any other way during the disassembly of the hydraulic drive or work on the pipeline must be supported or wedged.
4. Before starting any work whatsoever on the equipment it must be guaranteed that the electrical current is switched off.

The best solution is to remove the main fuses.

5. GENERAL HAZARD INFORMATION

5.1. GENERAL REMARKS

Integral Hydraulic manufactures hydraulic components and complete hydraulic systems according to the acknowledged technical rules. They can be considered safe when used pursuant to provisions and properly maintained. On the other hand, hydraulic components and systems provide hazards that may lead to property damages and personal injuries if proper care is not taken.

5.2 HAZARDS

5.2.1 Caused by the pressure medium

Even without pressure the pressure medium may already cause damages by its chemical properties and in addition by the energy immanent under pressure.

5.2.1.1 Fire hazard

Mineral oil generally used is flammable in its liquid form and especially as an oil-air-mist. Therefore oil leakages, handling open flames as well as the presence of parts running hot (also during transport) are to be avoided. If need be, danger warnings are to be affixed.

5.2.1.2 Toxicity

Although hydraulic media are not expressly classified as toxic, any inhalation and prolonged contact with the skin should be avoided. Contaminated clothing should be changed as soon as possible. Affected skin is to be cleaned.

5.2.1.3 Temperature

At high temperatures emerging pressure medium may lead to scalding. If too high a temperature development is suspected, cooling should be provided.

5.2.1.4 Hazards for the environment

Freely leaking pressure medium may contaminate the soil, groundwater and other waters. Therefore free leakage should be avoided by collecting basins etc. Especially in cases of mobile deployment the use of pressure media environmentally compatible ("bio-oils") should be taken into consideration.

5.2.1.5 Hazards by kinetic energy

Under strong differences in pressure escaping jets of liquid may become extremely dangerous by their drilling (injecting) or separating effect. Prevention means the proper dimensioning of all devices, line connections, screwed connections, etc., proper tightening of all screwed connections, safe fixings for all devices, pre-scribed welded connection as well as reliable pressure control taking dynamic influences into consideration as well.

5.2.1.6 Oxygen displacement

This hazard may only occur when larger hydraulic accumulators filled with nitrogen are emptied at the side of the gas in narrow, unventilated rooms. If it should prove to be unavoidable, ventilation must be provided.

5.2.1.7 Consequential damages by lack of compatibility

Lacking compatibility towards sealing materials, bubbles, membranes, coats of paint, etc. may lead to leakages with the hazards described above or even to complete break-down. Therefore only pressure media recommended by the manufacturers should be used. The manufacturer should be consulted in case of other media or extraordinary operating conditions (eg. low minus temperatures).

5.2.2 Unintentional movement by pressure energy

Unintentional movements may be caused either by working devices in the sense of unexpected work-motions or by parts/devices flying away in case of loosening or bursting.

5.2.2.1 Breakdown of controls

Failing of controls should not lead to unintentional movements. Controls should be safe enough to stop or interrupt, in any case, hazardous movements when a control fails (fail-safe concept). Wherever this cannot be achieved, a different type of protection (eg. shields, covers, cut-offs, etc.) must be provided.

5.2.2.2 By accumulated energy

Hydraulic systems with an accumulator effect (hoses, hydraulic accumulators) store energy that may lead to hazardous movement even after the supply of primary energy has been cut. It is to be examined whether the system cannot automatically be rendered pressureless if primary energy fails. If this is not the case, eg. with accumulator-supported clamping fixtures, a clear hazard warning must be available.

5.2.3 By component failure

Bursting, tearing, detaching, loosening, opening, etc. may make fragments or entire devices fly off. It may be caused by static overloading, but even more so by dynamic vibration stresses. Finally outside forces may lead to damages.

6. ORDINANCES, PROVISIONS, TECHNICAL RULES

A small selection of publications and titles concerning safety:

- Directive of the Council of 14 June 1989 for the harmonization of legal provisions for machines of the member states.
- Pressure Vessel Ordinance (DruckbehV) of 27.02.1980 in its latest version (4.96).
- Technical Rules Pressure Vessels (TRB) in their relevant latest version.
- ISO 4413 Hydraulic Fluid Power - General rules for the application to transmission and control systems.
- DIN EN (Acc. to EN 982) Safety of machinery, Safety requirements for fluid power systems and their components, Hydraulics
- DIN 24 552 Hydraulic accumulators in hydraulic systems (with annex).
- Rules for the Prevention of Accidents - power-driven equipment VGB 5 April 1986.

For normal working conditions we recommend usage of HL and HLP Hydraulic oils (on Mineral oil basis). In special cases also HLPD and HVLP oils with same viscosity may be used. Please contact us if you wish to use HE-Liquids (HEPG, HETG and HEES oils) or HFC (not easily inflammable water – Glycole mixture).

The sequence of the listed suppliers is alphabetical and therefore no standard of value. We would also like to indicate that the list is not complete and that we cannot guarantee for it's context. Please take notice that some liquids cover a wide range of viscosity and therefore are only mentioned once (e.g. 32-68).

- 1) normally Hydraulic Oil acc. to DIN 51524 part 1
- 2) normally Hydraulic Oil acc. to DIN 51524 part 2
- 3) normally Hydraulic Oil acc. to DIN 51524 part 3
- 4) like 2) or 3) but detergent

Sheet 1 of 2

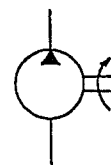
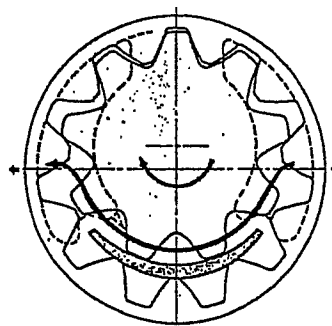
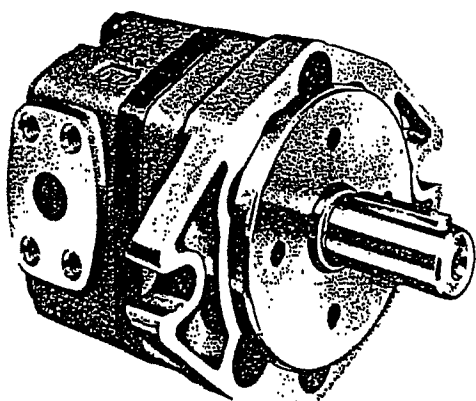
	VG 46		VG 68	
Oil company (alphabetical order)	a) HL ¹⁾ (mineral oil) b) HLP ²⁾ (mineral oil) c) HVLP ³⁾ (mineral oil) d) HLPD ⁴⁾ (mineral oil)	e) HEPG (polyglykol-basis) f) HETG (vegetable oil) g) HEES (synth. ester) h) HFC (water/glycol)	a) HL (mineral oil) b) HPL (mineral oil) c) HVLP (mineral oil) d) HLPD (mineral oil)	e) HEPG (polyglykol-basis) f) HETG (vegetable oil) g) HEES (synth. ester) h) HFC (water/glycol)
ARAL	a) Aral Vitam UF 46 b) Aral Vitam GF 46 c) Aral Vitam HF 46 Aral Vitam VF 46 d) Aral Vitam DE 46	e) Aral Vitam BAF 45 g) Aral Vitam EHF 45 h) Aral Montral 44	a) Aral Vitam UF 68 b) Aral Vitam GF 68 c) Aral Vitam VF 68 d) Aral Vitam DE 68	
BECEM	b) Staroil Nr. 46 c) Staroil HVI 46 d) Staroil H-LPD 46	e) Hydrostar UWF 45 f) UWS Hydraulik 32 g) Hydrostar HEP 45 h) Hydrostar HY 46	b) Staroil Nr. 68 c) Staroil HVI 68 d) Staroil H-LPD 68	e) Hydrostar UWF 68 g) Hydrostar HEP 68
BP	a) BP Energol HL 46 b) BP Energol HLP-HM 46 c) Bartran HV 46 d) BP Energol HLP-D 46	f) Biohyd 46 g) Biohyd SE 46 h) Enersyn SF-C 14	b) BP Energol HLP-HM 68 c) Bartran HV 68 d) BP Energol HLP-D 68	g) Biohyd SE 68
CASTROL	a) Magna 46 b) Hyspin AWS 46 Hyspin SP 46 c) Hyspin AWH-M 46 d) Varío HDX 46 Hydrauliköl HLP-D 46 SF	f) Carelube HTG 32 g) Castrol Produkt 695/13 Carelube HES 45 h) Anvol WG 46	a) Magna 68 b) Hyspin AWS 68 Hyspin SP 68 c) Hyspin AWH-M 68 d) Varío HDX 46 Hydrauliköl HLP-D 68 F	f) Carelube HTG 68 g) Castrol Produkt 695/14 Carelube HES 68
DEA	a) Astron HL 46 b) Astron HLP 46 Astron X HLP 46 c) Astron HVLP 46 Astron X HVLP 46 d) Actis HLPD 46 Actis X HLPD 46 Trion EP 46	e) Econa RG 46 f) (Econa R 32) g) Econa E 46 h) Tectro HF-C 46 S	a) Astron HL 68 b) Astron HLP 68 Astron X HLP 68 c) Astron HVLP 68 d) Actis HLPD 68 Actis X HLPD 68 Trion EP 68	
ELF	a) ELF POLYTELIS 68 b) ELFOLNA 68 ELFOLNA DS 68 ELFOLNA SP 68 c) HYDRELF DS 68 d) ELFOLNA HLPD 68 ELFOLNA HMD 68	f) ELF XTD 93031 g) HXDRELF BIO h) PYRELF HFC 46	a) ELF POLYTELIS 46 b) ELFOLNA 46 ELFOLNA DS 46 ELFOLNA SP 46 c) HYDRELF DS 46 d) ELFOLNA HLPD 46 ELFOLNA HMD 46	
ESSO	a) TERESSO 46 b) NUTO H 46 Hydrauliköl HLP 46 c) UNIVIS N 46 d) HLPD-OEL 46	e) Hydrauliköl PGK 45 f) Hydrauliköl PFL g) Hydrauliköl HE 46	a) TERESSO 68 b) NUTO H 68 Hydrauliköl HLP 68 c) UNIVIS N 68 d) HLPD-OEL 68	
FINA	a) CIRKAN 46 b) HYDRAN 46 c) HYDRAN HV 46 d) HYDRAN HLP-D 46 Hydrauliköl D3033	e) Hydrauliköl D3031-46 f) BIOHYDRAN RS 38 g) BIOHYDRAN SE 38 BIOHYDRAN TMP 46	a) CIRKAN 68 b) HYDRAN 68 c) HYDRAN HV 68 d) HYDRAN HLP-D 68	g) BIOHYDRAN TMP 68

Sheet 2 of 2

Oil company (alphabetical order)	VG 46		VG 68	
	a) HL ¹⁾ (mineral oil) b) HLP ²⁾ (mineral oil) c) HVLP ³⁾ (mineral oil) d) HLPD ⁴⁾ (mineral oil)	e) HEPG (polyglykol-basis) f) HETG (vegetable oil) g) HEES (synth. ester) h) HFC (water/glycol)	a) HL (mineral oil) b) HLP (mineral oil) c) HVLP (mineral oil) d) HLPD (mineral oil)	e) HEPG (polyglykol-basis) f) HETG (vegetable oil) g) HEES (synth. ester) h) HFC (water/glycol)
FUCHS	a) RENOLIN DTA 46 b) RENOLIN B15 VG 46 RENOLIN ZAF 46 B c) RENOLIN MR 46 MC RENOLIN ZAF 46 MC d) RENOLIN MR 15 VG 46 RENOLIN D 15 VG 46 RENOLIN ZAF 46 D	e) RENOLIN PG 46 f) PLANTOHYD 46 N g) PLANTOHYD 46 S PLANTOHYD Super S h) Hydrotherm 46 M Hydrotherm 46 NF3	a) RENOLIN DTA 68 b) RENOLIN B20 VG 68 RENOLIN ZAF 68 B c) RENOLIN MR 68 MC RENOLIN ZAF 68 MC d) RENOLIN MR 20 VG 68 RENOLIN ZAF 68 D	f) PLANTOHYD 68 N g) PLANTOHYD 68 S
MOBIL	a) Vactra Oil Medium DTE Oil Medium b) Mobil DTE 25 c) Mobil DTE 15 M d) Hydrauliköl HLPD 46	f) Mobil EAL 224 H g) Mobil EAL Syndraulic 46 Hydraulic Oil UF 46 h) Hydrofluid LT Nyvac FR 200 D Fluid	a) DTE Oil Heavy Medium Vactra Oil Heavy Medium b) Mobil DTE 26 c) Mobil DTE 16 M d) Hydrauliköl HLPD 68	
OEST	a) Hydrauliköl H-L 46 b) Hydrauliköl H-LP 46 c) Hydrauliköl HVI 46 d) Hydrauliköl 46 DD	f) (BIO HY-FLUID HV 34) g) BIO Synthetik HYD 46	a) Hydrauliköl H-L 68 b) Hydrauliköl H-LP 68 c) Hydrauliköl HVI 68 d) Hydrauliköl 68 DD	f) BIO HY-FLUID HV 68 g) BIO Synthetik HYD 68
PANOLIN	a) Panolin Indol ISO 46 b) Panolin HLP ISO 46 c) Pan. HLP Universal 37 d) Panolin HLP-D ISO 46	g) Panolin HLP Synth 46	a) Panolin Indol ISO 68 b) Panolin HLP ISO 68 c) Panolin GP 55 d) Panolin HLP-D ISO 68	g) Panolin HLP Synth 68
PETROFER	a) Isolubric VG 46 L b) Isolubric VG 46 c) Isolubric VG 46 HV d) Isolubric VG 46 D	e) Syntolubric 46 f) Tribolubric 32 g) Envolubric HE 46 h) Ultra-Safe 620	a) Isolubric VG 68 L b) Isolubric VG 68 c) Isolubric VG 68 HV d) Isolubric VG 68 D	g) Envolubric HE 68 h) Ultra-Safe 360
QUAKER		f) GREENSAVE N 30 g) GREENSAVE N 40 h) QUINTOLUBRIC 730		
SHELL	a) Morlina Oil 46 b) Shell Tellus Oil 46 c) Shell Tellus Oil TD 46 d) Shell Tellus Oil DO 46	e) Shell Fluid BD 46 f) Shell Naturelle HF-R g) Shell Naturelle HF-E 46 h)	b) Shell Tellus Oil 68 d) Shell Tellus Oil DO 68	g) Shell Naturelle HF-E 68
STUART		e) ISOCOR E 46 g) ISOCOR HF 46 h) HYDROCOR CC 44		g) ISOCOR HF 68
TEBIOL		f) Florahyd HVI 46 g) Esterhyd HE 46		f) Florahyd HVI 68
TRIBOL	a) Tribol 772 b) Tribol 943 AW 46		a) Tribol 773 b) Tribol 943 AW 68	
WISURA	a) Dynex 46 b) Tempo 46 c) Hydroma 46 d) HLPD 46	f) Hydroma NAT 40 g) Hydrofluid SE 46	a) Dynex 68 b) Tempo 68 c) Hydroma 68 d) HLPD 68	

2,10
2,20

$p_{max} = 175 \text{ bar (250 bar)}$
 $Q_{max} = 176 \text{ l/min}$



Integral Hydraulic Internal gear pumps have the following features:

Low noise level

The special Trochocentric gear profile virtually eliminates the trapped volume between the gear tips of the pinion and the root of the gearing on the ring gear. This ensures very quiet running of the pump with low pulsation. The design reduces gear tooth pressure to a minimum promoting long life.

High power handling

The long sealing zone between suction and pressure produces high volumetric efficiency. The small outside diameter of the ring gear and the large volume provided for the pinion minimises mechanical losses. The small number of teeth, wide gearing and large flow cross sectional area ensures low filling resistance and promotes good suction behaviour.

Compactness

The long sealing zone of the crescent and small tooth difference between pinion and ring gear ensures that the maximum displacement is achieved thereby keeping size small. Casing parts are few due to the method of fitting the bearings into the pump housing. Single and double pumps are available.

Characteristics

Operating pressure

Continuous pressure

: 175 bar (250 bar on request)

Peak pressure

: 200 bar (300 bar)

(3 s max. or 10 % of duty cycle)

Suction conditions

: min 0.6 bar absolute

: max 0.5 bar

Delivery

: 7,5 — 176 l/min (1500 rpm)

Effective displacement*

: 5 — 125 cm³/rev

Speed

: see size table

Fluid temperature range

: 0 to 70°C

Viscosity range

: 20 to 300 mm²/s (cSt)

Recommended filtration

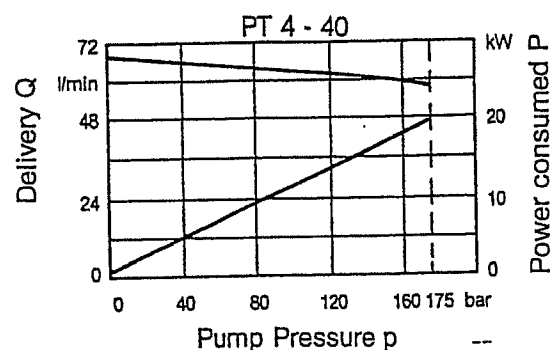
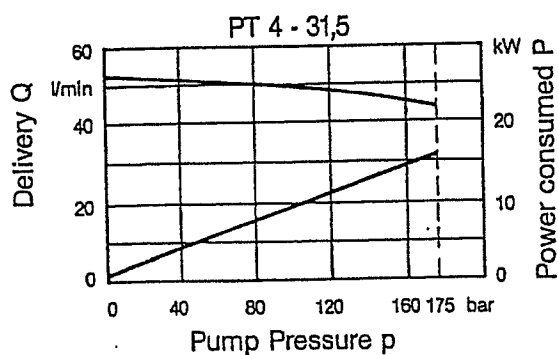
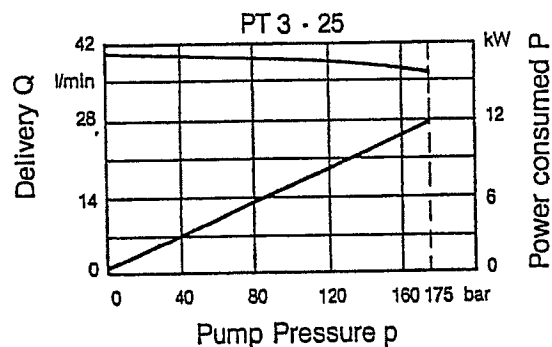
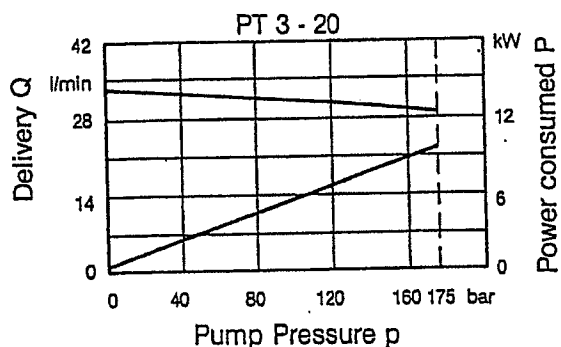
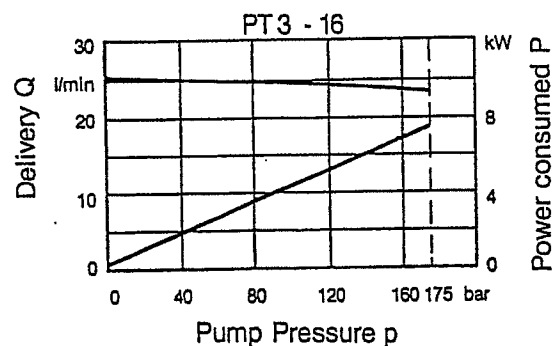
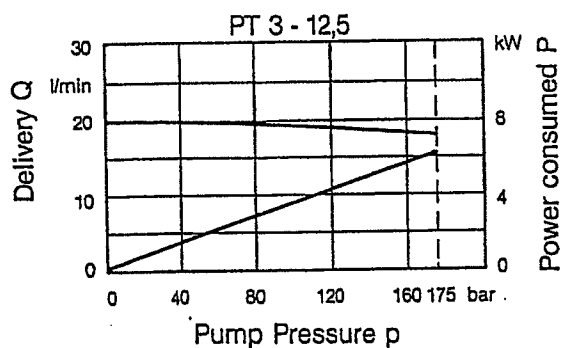
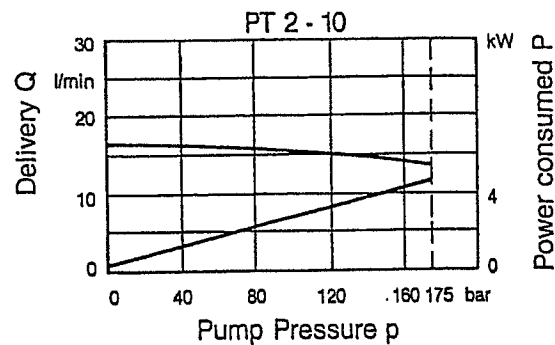
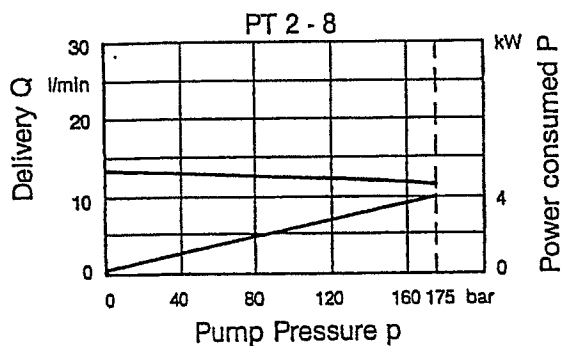
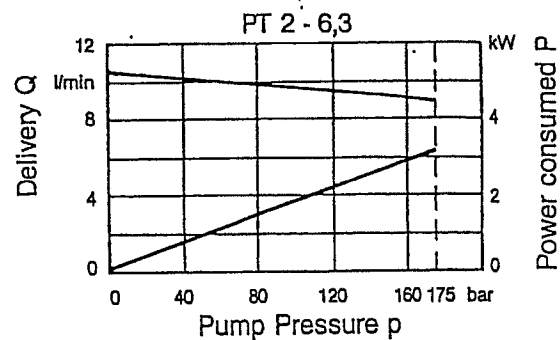
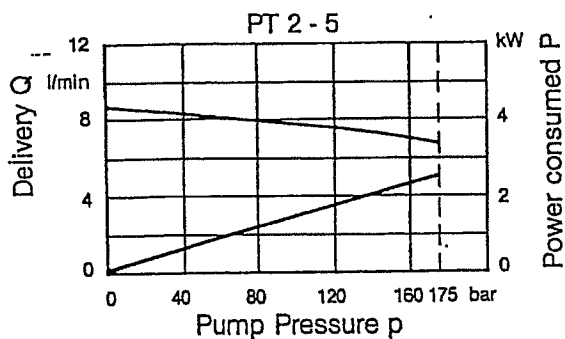
: 25µm absolute

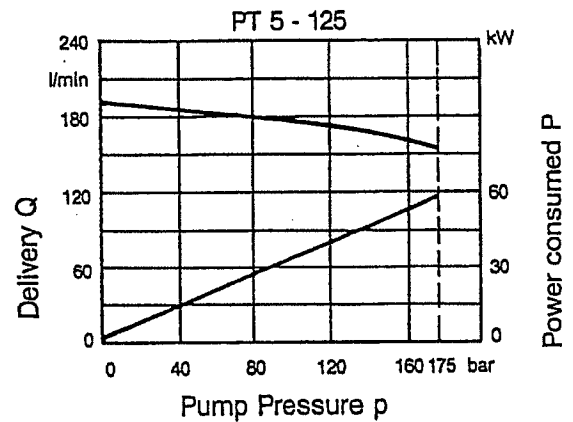
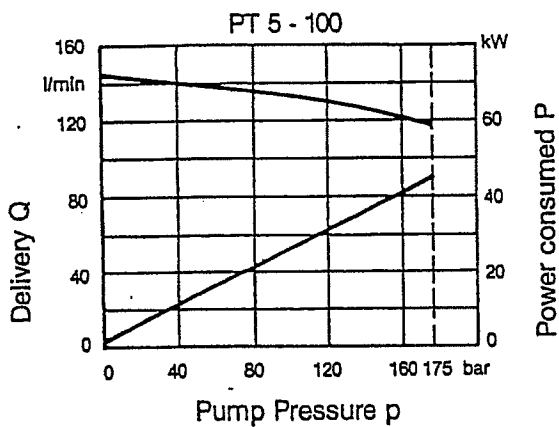
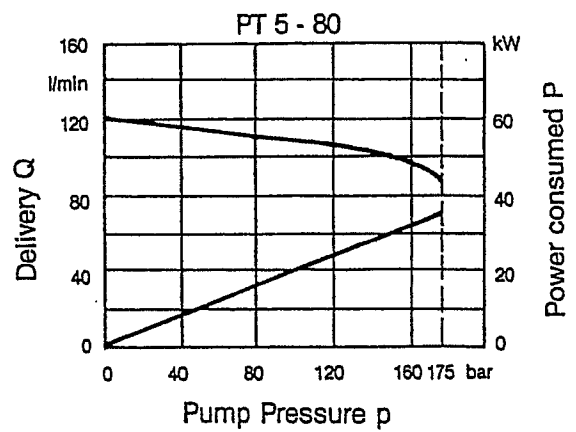
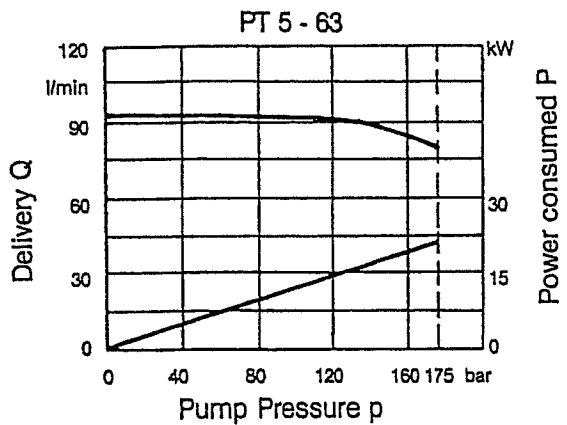
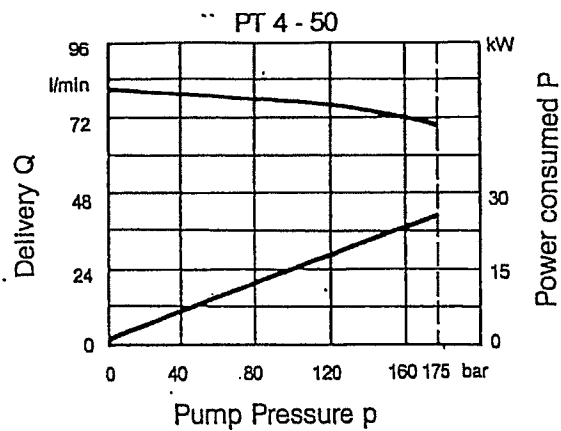
(Suction filter not required)

Size table

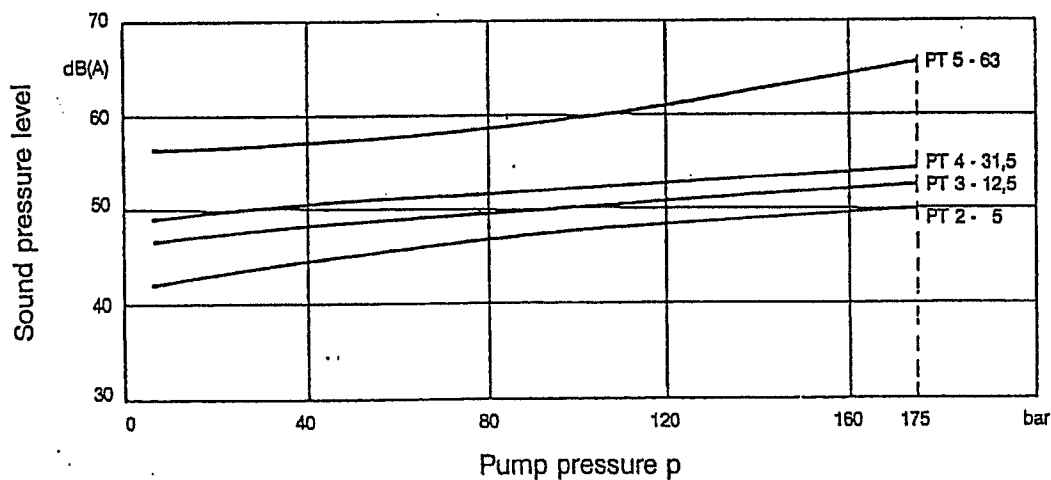
Type	Size	Delivery* l/min	Speed rpm min max		Order code
PT 2 - 5	2	7,5	800	4000	211 - 1319 - 014 - 005
PT 2 - 6,3		9,4			211 - 1319 - 014 - 006
PT 2 - 8		12			211 - 1319 - 014 - 008
PT 2 - 10		15			211 - 1319 - 014 - 010
PT 3 - 12,5	3	18,7	800	3000	311 - 1319 - 014 - 012
PT 3 - 16		24			311 - 1319 - 014 - 016
PT 3 - 20		30			311 - 1319 - 014 - 020
PT 3 - 25		37,5			311 - 1319 - 014 - 025
PT 4 - 31,5	4	47,2	800	3000	411 - 1319 - 014 - 031
PT 4 - 40		60		2800	411 - 1319 - 014 - 040
PT 4 - 50		75		2500	411 - 1319 - 014 - 050
PT 5 - 63	5	86	600	2200	511 - 1319 - 014 - 063
PT 5 - 80		110		2400	511 - 1319 - 024 - 080
PT 5 - 100		138		2100	511 - 1319 - 024 - 100
PT 5 - 125		176		1800	511 - 1319 - 024 - 125

*The quoted deliveries and effective displacements are at a pump pressure of 100 bar at 1500 rpm and 36 mm²/s viscosity

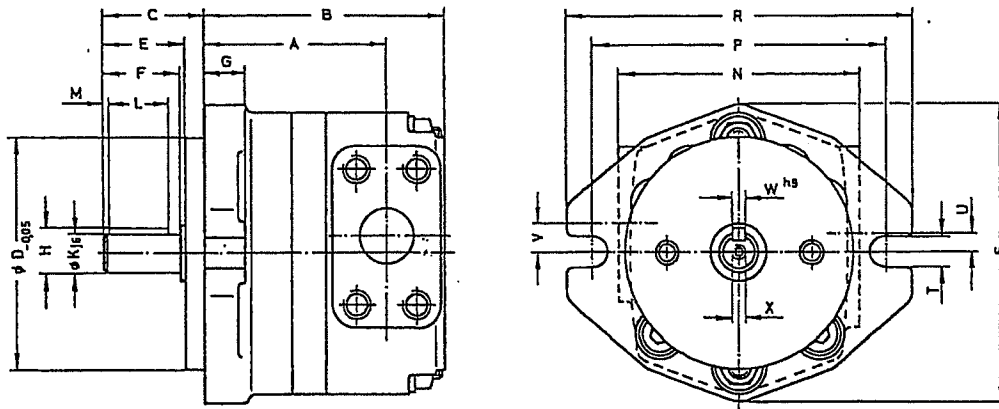




Sound pressure (measured in quiet room)



Dimensions

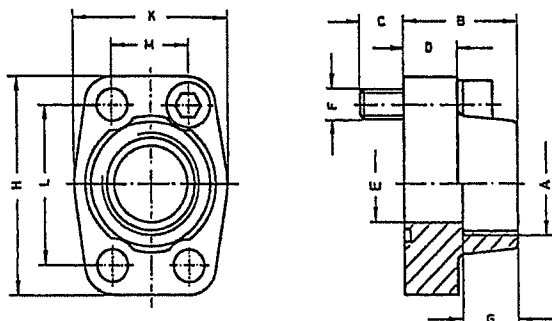


Size	PT 2 - ...				PT 3 - ...				PT 4 - ...			PT 5 - ...			
cm³/rev	5	6,3	8	10	12,5	16	20	25	31,5	40	50	63	80	100	125
A	64	66,5	70	74	81	84,5	89	94	109	114,5	120	123	129	136	146
B	85	87,5	91	95	106	109,5	114	119	146	151,5	157	162	168	175	185
C	37				61,5				86			94			
D	82,55				101,6				127			152,4			
E	29,5				53,5				76			82			
F	28				49,5				73			80			
G	14,5				18,5				20			34			
H	16				27				35			41			
K	14				24				32			38			
L	22				40				61			61			
M	1,5				3,5				5			8			
N	87				115				155			200			
P	106				146				181			229			
R	125				168				205			268			
S	106				136				186			233			
T	11				14				18			22			
U	6				6				0			0			
V	10				8				8			0			
W	5				8				10			10			
X	M5				M8				M12			M12			
Weight kg	2,4	2,5	2,6	2,8	4,9	5,2	5,5	5,9	12,3	13,1	13,9	22,2	23,9	25,6	27,8
Suction pipe flanges	SAE G 3/4				SAE G 1				SAE G 1 1/2			SAE G 2			
Pressure pipe flanges	SAE G 1/2				SAE G 3/4				SAE G 1			SAE G 1 1/2			

Socket weld flanges on request

Pipe flanges

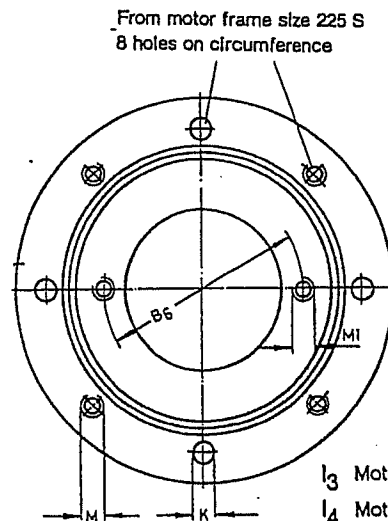
(O-ring and screws included)



	A	B	C	D	E	F	G	H	K	L	M
SAE G 1/2	1/2	36	14	16	14	M 8	19	56	46	38	17,5
SAE G 3/4	3/4	36	14	18	19	M10	19	65	48	47,6	22,2
SAE G 1	1	38	15	18	25	M10	19	70	52	52,4	26,2
SAE G 1 1/2	1 1/2	44	18	25	38	M12	24	93	76	70	35,7
SAE G 2	2	45	18	25	51	M12	26	102	88	77,8	42,9
SAE G 3	3	50	23	27	73	M16	34	134	123	106,4	62

	O-ring	Order code
SAE G 1/2	18,6 x 3,5	011 - 1319 - 014 - 012
SAE G 3/4	25 x 3,5	011 - 1319 - 014 - 034
SAE G 1	32,9 x 3,5	011 - 1319 - 014 - 100
SAE G 1 1/2	47,2 x 3,5	011 - 1319 - 014 - 112
SAE G 2	56,7 x 3,5	011 - 1319 - 014 - 200
SAE G 3	85,3 x 3,5	011 - 1319 - 014 - 300

Accessories



- l₃ Motor shaft connection is binding
- l₄ Motor shaft connection is' longer
- l₅ Motor shaft connection is' shorter

Bell housings

Type	Order code	Ø A1	Ø B	Ø B1	Ø B3	Ø B4	Ø B5	B6	L1	L2	L3	L5	L6	Ø K	M	M1	h	
PK 200/3/3	200-1361-074-027	200	165	130	145	82,55	124	106,4	99	28	16	12	7	11,5	M10	M12	4	1)
PFL 200/5/2	200-1361-094-027					101,6	166	146	124				13					2) 3)
PL 200/3/11	200-1361-104-027					82,55	124	106,4	110				11					2)
PK 250/6/30	250-1361-024-027	250	215	180	190	82,55	180	106,4	120	33	18		8,5	14		M10	5	1)
PK 250/13/6	250-1361-064-027					101,6	186	146	159				14,5			M12		2)
PFL 250/18/7	250-1361-044-027					127,0	250	181	175				18,5			M16		2) 3)
PL 300/3/2	300-1361-024-027	300	265	230	234	82,55	124	106,4	144	35	20	15	8	14	M12	M10	1)	
PL 300/5/9	300-1361-034-027					101,6	230	146	160				16			M12		2)
PL 300/13/16	300-1361-074-027					127,0	231	181	210				8			M16		1)
PFK300/6/13	300-1361-054-027					152,4	260	228,6	196				29			M20		1) 3)
PL 350/5/9	350-1361-024-027	350	300	250	260	101,6	230	146	190	40	25	15	18			M12	6	2)
PK 350/15/9	350-1361-034-027					127,0	252	181	222				10			M16		1)
PL 350/15/20	350-1361-044-027					152,4	252	228,6	237				18			M20		2)
PL 400/4/3	400-1361-024-027	400	350	300	300	127,0	230	181	214	40	25	20	10	18	M16	M16	1)	
PK 400/5/6	400-1361-034-027	400	350	300	300	152,4	290	228,6	228				10			M20		1)
PK 450/2/3	450-1361-014-027	450	400	350	350	127,0	230	181	234				30			M16		2)
PK 450/3/7	450-1361-024-027	450	400	350	350	152,4	325	228,6	262				10					1)
PL 550/8/17	550-1361-014-027	550	500	450	450	152,4	340	228,6	248	41	26	25	25			M20		2)
PK 550/3/8	550-1361-024-027	550	500	450	450	152,4	340	228,6	275	41	26	25	20					2)

- 1) Countersunk for screws (Dim. M) to DIN 6912
- 2) Countersunk for screws (Dim. M) to DIN 912
- 3) These bell housings are only for horizontal mounting where B_5 is bigger than B_3 .

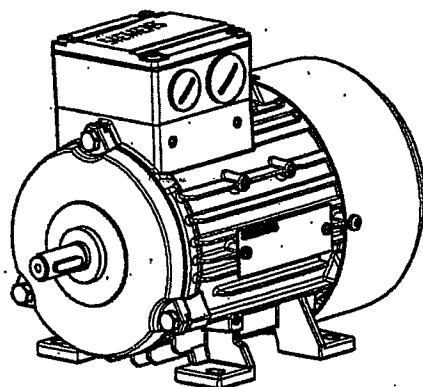
Couplings

Type		E-Motor		Coupling		A	Ø Dh	E	Ø d	Ø d1	L	l1	l2	l3	l4	l5
Bell housing	Pump	Size	kW	Rotex	Order code											
PK 200/3/3	PT 2..	80	0,55	R-19.19-14	190-2295-014-195	12	40	16	14	19	66	25	25	40	-	6
PFL 200/5/2	PT 3..		0,75	R-19/24.19-24	192-2295-124-190	37			24						-	6
PL 200/3/11	PT 2..		90 S	1,1	R-19/24.24-14	192-2295-414-245			10						14	24
PFL 200/5/2	PT 3..	90 L	1,5	R-19/24.24-24	192-2295-424-240	35			24					2	-	
PK 250/6/30	PT 2..	100 L	2,2	R-24/28.28-14	242-2295-814-286	9	55	18	14	28	78	30	30	60	-	3
PK 250/13/6	PT 3..	112 M	3,0	R-24/28.28x50-24	242-2295-828-524	29,5			24		98	50	-		19	
PFL 250/18/7	PT 4..		4,0	R-28/38.28-32	283-2295-832-280	52			32			-	8			
PL 300/3/2	PT 2..	132 S	5,5	R-28/38.38-14	283-2295-814-388	9	65	20	14	38	90	35	35	80	-	-
PL 300/5/9	PT 3..			132 M	7,5	R-28/38.38-24			283-2295-824-380		25	24			-	-
PL 300/13/16	PT 4..				R-28/38.38x60-32	283-2295-638-032			51		32	115	60		-	24
PFL 300/6/13	PT 5..			R-28/38.38-38	283-2295-838-388	5			38	90	35			-	2	
PL 350/5/9	PT 3..	160 M	11,0	R-38/45.42-24	384-2295-524-420	12	80	24	24	42	114	45	45	110	2	-
PK 350/15/9	PT 4..	160 L	15,0	R-38/45.42-32	384-2295-532-420	41			32						-	2
PL 350/15/20	PT 5..		R-38/45.42-38	384-2295-538-420	53	38			-						5	
PK 350/15/9	PT 4..	180 M	18,5	R-42/55.48-32	425-2295-532-480	36	95	26	32	48	126	50	50	110	-	-
PL 350/15/20	PT 5..	180 L	22,0	R-42/55.48-38	425-2295-538-480	47			38						-	4
PL 400/4/3	PT 4..	200 L	30,0	R-42/55.55-32	425-2295-532-551	28			32						55	-
PK 400/5/6	PT 5..			R-42/55.55-38	425-2295-538-550	42			32					-	-	
PK 450/2/3	PT 4..	225 S	37,0	R-48/60.60-32	486-2295-032-600	13	105	28	38	60	140	56	56	140	3	-
PK 450/3/7	PT 5..	225 M	45,0	R-48/60.60-38	486-2295-038-600	38									-	-
PK 550/8/17	PT 5..	250 M	55,0	R-57/70.65-38	557-2295-038-650	14			120						30	38
PK 550/3/8	PT 5..	280 S/M	75/90	R-75.75-38	750-2295-038-750	14	160	40		75	210	85	85		4	-

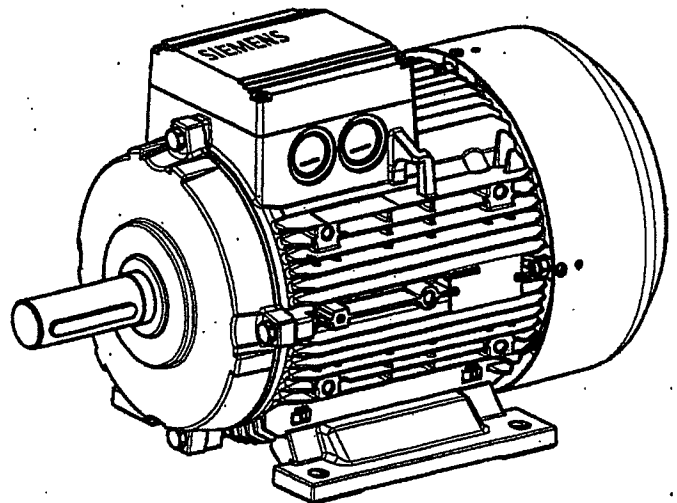
Drehstrommotoren Three-phase AC Motors Moteurs triphasés Motores trifásicos Motori trifasi Trefasmotorer

Betriebsanleitung
Instructions

1LA7^{CE}, 1LP7^{CE}, 1PP7^{CE}
1MA7, 1MF7



BG 71... 90 L - IM B3



BG 100... 160 L - IM B3

Baugrößen (BG) / Frame sizes (BG) / Désignation de carcasse (BG)
Tamaños constructivos (BG) / Grandezza costruttiva (BG) / Konstruktionsstorlekar (BG)

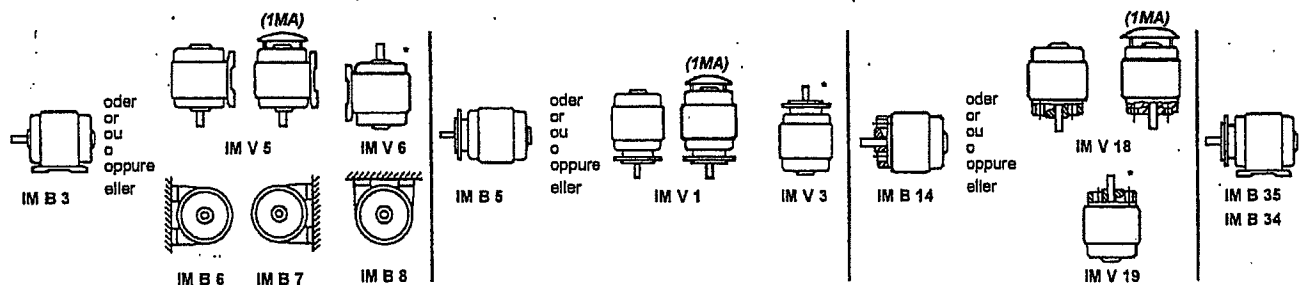


Fig. 1 Bauformen / Types of construction / Formes de construction
Formas constructivas / Forme costruttive / Monteringsätt

* s. "4 EEx e Motoren"/see "4 EEx e motors"/voir "4 Moteurs EEx e"/
véase "4 Motores EEx e"/cfr. "4 Motori EEx e"/se "EEx e-Motoren"

General note



The data and recommendations specified in all the instructions supplied ("Information on safety and commissioning"), and in all other related instructions, must always be observed in order to avoid hazardous situations and the risk of possible injury or damage.

Furthermore, the pertinent national, local and plant-specific regulations and requirements should be kept in mind!

Special designs and other versions may vary in technical details! If in doubt, be sure to contact the manufacturer, quoting the type designation and serial number (No. E ..., see rating plate), or have maintenance work done by one of the SIEMENS Service Centres.

1 Description

1.1 Application

Normal use of the standard motors:

The motors are designed to comply with degree of protection IP55 (degree of protection indicated on rating plate). They are suitable for installation in dusty and damp environments. The insulation is tropicalized. If they are properly stored or installed outdoors, special weatherproofing measures are not usually required for these motors. They must however be protected against intense sunlight, e.g. by means of a canopy.

Ambient temperature -30°C ... +60°C

Site altitude ≤ 1000 m

1.2 Construction and mode of operation

Motor types:

1LA- and 1MA motors are self-ventilated (with fans). 1PP motors are self-ventilated either by separate fans arranged on the rotors, such as for service as fan motors, or by external fans.

1LP and 1MF motors are non-ventilated (without fans).

The feet on foot-mounted motors are bolted onto the frame (see Fig. 2).

For this reason, rearranging the feet (such as for changing the position of the terminal box - see Fig. 5) is only possible with frame sizes 100L to 160L. In this case, threads will have to be cut in the existing drilled holes in the frame to receive the feet. However, the support faces of the feet may have to be remachined or provided with shims to ensure that the motor is level.

Where brake motors are concerned, please also observe the brake operating instructions.

2 Operation



Before starting any work on the machine, be sure to isolate it from the power supply.

2.1 Transport, storage

The motors should always be lifted at all lifting eyes during transport.



For lifting machine sets (such as built-on gearboxes, fan units), always use the lifting eyes or lifting pegs provided! Machine sets may not be lifted by suspending the individual machines! Check the lifting capacity of the hoist!

If, after delivery, the motors are stored for more than 4 years under favourable conditions (kept in a dry place free from dust and vibration) prior to commissioning, the rolling-contact bearings should be regreased. Under unfavourable conditions, this period is considerably shorter.

If necessary, any unprotected, machined surfaces (flanging surface, shaft end, etc.) should be treated with an anti-corrosion agent.

If necessary, the insulation resistance of the winding should be checked, see Section 2.5.

2.2 Installation

After installation, screwed-in lifting eyes should either be removed or tightened down.

In the case of motors installed vertically, additional measures must be taken to ensure that no water can penetrate along the shaft.

Quiet running: Exact alignment of the coupling and a well-balanced transmission element (coupling, pulleys, fans, etc.) are essential for quiet vibration-free running. If necessary, the whole motor and transmission element should be balanced.

BG90S/L: The rear feet of motors of type of construction IM B3 have double holes in order to ensure that these machines adhere to the standard foot dimensions. On the other hand, all feet of motors of type of construction IM B3 with side-mounted terminal box (special designs) have double holes. This enables the standard hole clearances to be maintained even when the terminal box is arranged on the left-hand side (see Fig. 10).

BG71...90L: In the case of 1LA, 1LP and 1LP motors, the top part of the terminal box can be turned through 4x90 degrees.

BG100...160L: The terminal box moulded onto the motor frame cannot be turned. If the top part of the terminal box is retrofitted (Fig. 2 - 5.90), the box can be turned through 4 x 90 degrees.

In the case of motors with a bolted-on terminal box, the top part of the box can be turned through 4 x 90 degrees.

2.3 Balancing, transmission elements

A suitable device should always be used for fitting and removing the transmission elements (couplings, pulleys, pinions, etc.) (Fig. 7).

As standard, the rotors are dynamically balanced with the half featherkey.

When fitting the transmission element, keep the type of balance in mind! (Option: F = balanced with full featherkey)

The transmission elements must be balanced in accordance with ISO 1940!

If the transmission elements are balanced with a half featherkey, the visible, protruding part of the featherkey T_p must be cut back (see Fig. 9).



The usual measures should be taken to guard transmission elements from touch. If a motor is started up without the transmission element attached, the featherkey should be secured to prevent it being thrown out.

2.4 Electrical connection

NOTE: If the openings for cables and leads in the terminal box are sealed with a "skin" of cast iron, it must be shaken out using a suitable tool (see Fig. 4)!

Care must be taken not to damage the terminal box, the terminal board, the cable connections, etc. inside the terminal box!

See Fig. 4 for details of the screwed connections for cables and leads. An adapter must be screwed in for PTC thermistor connections!

The terminal box must be sealed so that it is dust and water-tight.

The system voltage and the frequency must agree with the data given on the rating plate. Voltage or frequency deviations of $\pm 5\%$ and $\pm 2\%$ respectively from the rated voltage and frequency values are permitted without needing to derate the output. The connection and arrangement of the terminal links must agree with the diagram provided in the terminal box.

Connect the earthing conductor to the terminal with the $\oplus$ marking.

Wherever terminal clips are used (for example, to DIN 46282), arrange the conductors so that the clips are virtually level on both sides. This method of connection means that the ends of single conductors must be bent in the shape of a U or be fitted with a cable lug (see Fig. 6.1). This also applies to the green-yellow protective earthing conductor and the outer earthing conductor (see Fig. 6.2).

Please refer to Fig. 5 for the tightening torques for the screwed electrical connections - terminal board connections (except for terminal strips).

The anti-condensation heater must not be switched on during operation.

2.5 Checking the insulation resistance

The insulation resistance of the windings must be measured prior to initial startup of the motor, or after long periods of storage or standstill (approx. 6 months).

While the measurement is being taken and immediately afterwards, some of the terminals carry dangerous voltages and must not be touched.

Insulation resistance

- The minimum insulation resistance of new, cleaned or repaired windings with respect to ground is 10 Mohms.

- The critical insulation resistance R_{crit} is calculated first by multiplying the rated voltage U_N , e.g. 0.69 kVAC, with the constant factor (0.5 Mohms/kV):

$$R_{crit} = 0.69 \text{ kV} \cdot 0.5 \text{ Mohms/kV} = 0.345 \text{ Mohms.}$$

Measurement

The minimum insulation resistance of the windings to ground is measured with 500 V DC. The winding temperature should then be $25^\circ\text{C} \pm 15^\circ\text{C}$.

The critical insulation resistance should be measured with 500 V DC with the winding at operating temperature.

Checking

If the minimum insulation resistance of a new, cleaned or repaired motor, which has been stored or at standstill for a prolonged period of time, is less than 10 Mohms, this may be due to humidity. The windings must then be dried.

After long periods of operation, the minimum insulation resistance may drop to the critical insulation resistance. As long as the measured value does not fall below the calculated value of the critical insulation resistance, the motor may continue in operation. If it does, the motor must be stopped immediately.

The cause must be determined, and the windings or winding sections repaired, cleaned or dried as necessary.

2.6 Commissioning

NOTE: Electromagnetic compatibility

Emitted interference: Where the torque is very uneven (the drive of a piston-type compressor, for example), the inevitable result is a non-sinusoidal motor current, whose harmonics can lead to excessive system perturbation and thus excessive emitted interference.

In the case of converter-fed motors, interference is emitted to a greater or lesser degree, depending on the converter version concerned (type, interference suppression measures, manufacturer). The instructions of the converter manufacturer regarding electromagnetic compatibility must be heeded at all times. If the use of a shielded motor cable is recommended, the shield will have the greatest effect if it is conductively connected over a large area on the metal terminal box of the motor (with a screwed metal conduit thread). Noise voltages may occur on the sensor leads of motors with integrated sensors (e.g. PTC thermistors) as a result of the converter.

Noise immunity: If the motor has an integrated sensor (e.g. a PTC thermistor), the owner is responsible for ensuring adequate noise immunity by choosing a suitable sensor signal lead (possibly with shielding, connected like the motor supply lead) and evaluator.

The data and recommendations specified in all the instructions supplied ("Information on safety and commissioning"), and in all other related instructions, must always be observed prior to commissioning!

After motor installation, the brake, if fitted, should be checked for proper functioning.

3 Maintenance

Safety precautions

Before starting any work on the motor or other equipment, particularly before opening covers over live parts, the motor must be properly isolated from the power supply. Besides the main circuits, any additional or auxiliary circuits that may be present must also be isolated.

The usual "5 safety rules" (as set forth in DIN VDE 0105) are:

- Isolate the equipment
- Take effective measures to prevent reconnection
- Verify equipment is dead
- Earth and short-circuit
- Cover or fence off adjacent live parts

The precautions listed above should remain in force until all maintenance work is finished and the motor has been fully assembled.

NOTE: Where motors are fitted with closed condensed water openings, these should be opened from time to time to allow any accumulated condensed water to drain away.

Condensed water openings should always be at the lowest point of the motor!

Fitting new bearings, type of grease

Under normal operating conditions, with horizontally mounted motors and the following coolant temperatures and motor speeds, the bearings should be changed at the intervals [h] specified below:

Frame sizes	25°C	40°C
71...160L:	... 1800 rev/min approx. 40,000 h	approx. 20,000 h
	... 3600 rev/min approx. 20,000 h	approx. 10,000 h

NOTE: The permissible axial and transverse forces (see Catalog) must not be exceeded!

Irrespective of the number of operating hours, the rolling-contact bearing should be renewed every 3 years because of grease ageing.

In the case of motors operating under special conditions, such as a vertical motor position, heavy vibration, sudden load changes, frequent reversing operation, etc., the bearing should be changed at considerably more frequent intervals than the operating hours stated above.

The motors feature deep-groove ball bearings which are provided with cover plates (2ZC3 version). The cover plate material should withstand temperatures from -30 °C to +150 °C, e.g. polyacryl-rubber (ACM).

Type of grease for standard machines: UNIREX N3 (Esso); synthetic greases must conform to DIN 51825-K3N.

Special greases should be indicated on the rating plate or on a separate plate.

Dismantle the motor to the extent necessary. Pull off the rolling-contact bearing with a suitable device (see Fig. 7). Clean the journal!

Heat the rolling-contact bearing evenly to about 80-100 °C and press on. Heavy blows (such as with a hammer, etc.) should be avoided.

Any worn sealing elements (such as the shaft sealing ring, etc.) should also be renewed.

If springless radial shaft sealing rings are used, the replacement sealing rings must also be of the springless type.

Regreasing device

In the case of motors with a regreasing device, take note of the information given on the rating plate or the lubrication instruction plate! The bearings should be relubricated while the motor is running!

4 1MA and 1MF motors with increased EExe protection



The information in italics is intended to serve as supplementary or special information on these types of motors.

The increased hazards in areas which are exposed to the danger of explosion or firedamp necessitate that the general notes on safety and commissioning are carefully complied with.

Electrical machines which are protected against explosion are in line with the standards EN 60034 (VDE 0530) and EN 50014 to 50020. It is permitted to use these machines in areas exposed to the danger of explosion only in accordance with the stipulations of the responsible authority which also determines whether a danger of explosion exists (division into zones).

If the relevant certification is supplemented by an X, any special stipulations in the certificate of conformity are to be complied with.

The cable entries must be approved for the explosion-endangered area and be secured to prevent accidental loosening. Unused openings are to be closed with approved plugs.

In the case of 1MA-motors, a cover to prevent foreign bodies from falling into the motor-fan cowl (see DIN EN 50014) is to be located on the end of the shaft at the top, e.g. types of construction IMV3, IMV6 and IMV19 (see Fig. 1). This cover must not hinder cooling of the motor by its fan.

Unless other information regarding type of operation and tolerances is given on the test certificate or on the rating plate, electrical machines are designed for continuous operation and normal, rarely recurring starting procedures during which no excessive temperature increases occur. The motors must only be used for the type of operation indicated on the rating plate.

Section A in EN 60034-1 (VDE 0530, Part 1) - voltage $\pm 5\%$, frequency $\pm 2\%$, waveform, power-system symmetry - must be complied with in order to ensure that any increase in temperature remains within the permitted limits. Greater deviations from the rated values can

lead to non-permissible increases in the temperature of the machine and must be indicated on the rating plate.

The temperature class of the motor given on the rating plate must agree with the temperature class of the inflammable gas which may occur.

Each machine is to be protected in all phases against non-permissible temperature increases by means of a current-dependent, time-delayed circuit-breaker with phase-failure protection to VDE 0660 or an equivalent device. The protective device is to be set to the rated current. In the case of delta-connected windings, the trips are to be connected in series with the winding phases and set to 0.58 times the rated current. If such a circuit is not possible, additional protective measures are necessary (e.g. thermal machine protection).

If the rotor is blocked, the protection device must switch off the machine within the time t_E indicated for the respective temperature class.

Electrical machines for heavy starting (power-up time $> 1.7 \times t_E$) are to be protected according to the stipulations of the certificate of conformity by means of a starting-cycle monitoring circuit.

Thermal machine protection by means of direct temperature monitoring of the winding is permissible if this is certified and indicated on the rating plate.

In the case of pole-changing motors, separate and reciprocally locked protective devices are necessary for each speed step. Devices with the test report of an approved testing agency are recommended.

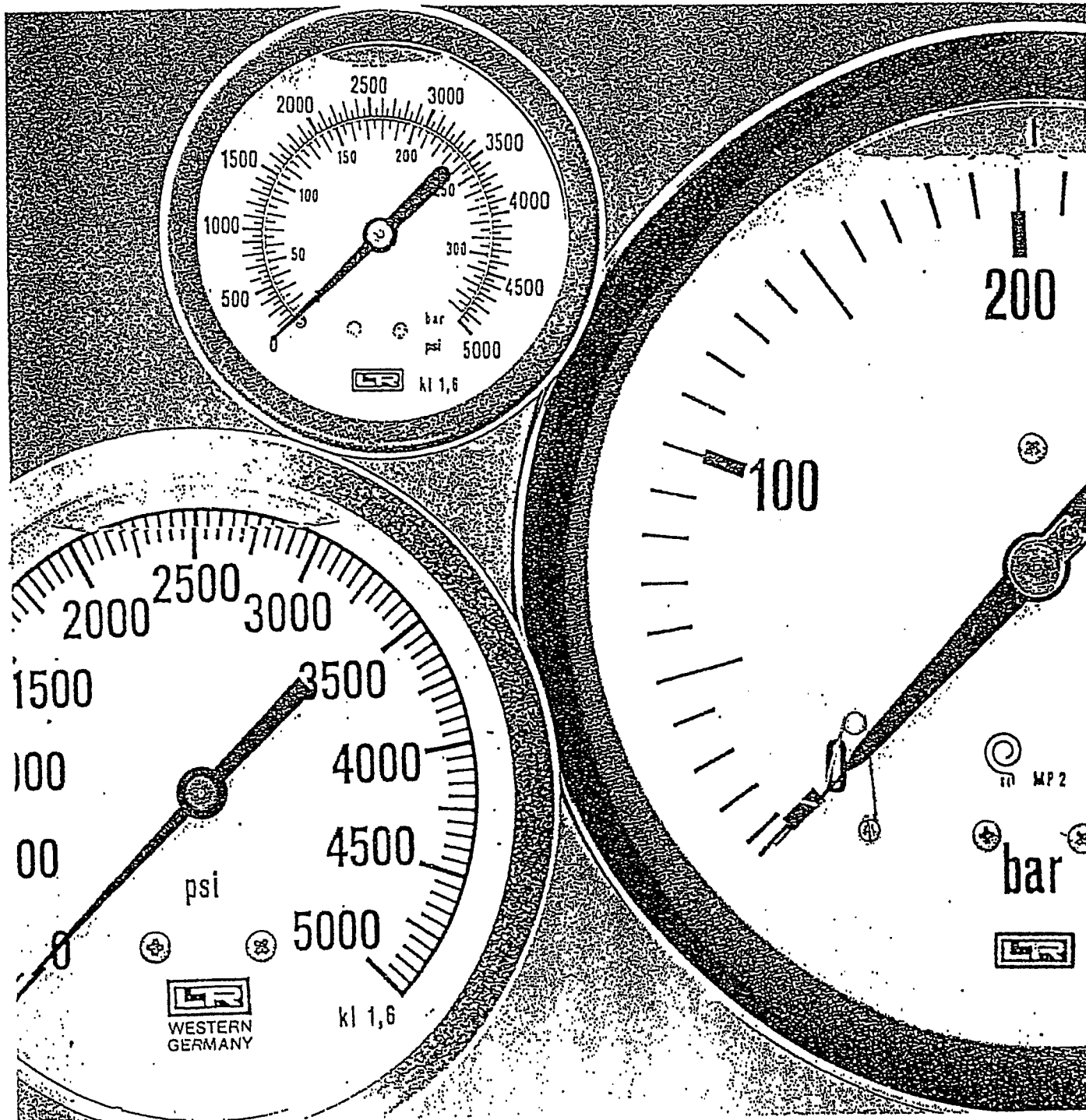
In Germany, please refer to DIN 57165/VDE 0165 and ElexV when erecting electrical installations in areas exposed to the danger of explosion. In countries other than Germany, the relevant national regulations are to be complied with in each case!

Operation on a converter must be certified. The separate instructions of the manufacturer must be complied with. For EExe type of protection, the motor, converter and protective devices must be marked to show they belong together and the permissible operating data must be specified in the shared test certificate (VDE 0165).

MGS 10/3

4.1
4.2
4.3

liquid-filled vibration-damping pressure gauges



MATERIAL SPECIFICATION

TYPE S10/3

STANDARD VERSION

Socket material: OT 58 brass.

Elastic element: phosphor bronze or AISI 316L stainless steel.

Movement: gearing, OT 59 brass - plates AISI 304 stainless steel.

Dial: white aluminium, with black markings.

Special Dial: ranges different from standards, or custom artwork, available on application.

Pointer: black oxidized aluminium.

Case: AISI 304 stainless steel.

Gasket: polychloroprene.

Window: plexiglas.

Ring: AISI 304 stainless steel polished, crimped.

Protection: IP 65.

TYPE S18/3

STAINLESS STEEL VERSION TYPE S4

Socket material: AISI 316 stainless steel.

Elastic element: AISI 316L stainless steel.

Movement: reinforced of stainless steel our standard MP2R for DS 100 (4 in.) and DS 150 (6 in.); stainless steel for DS 50-63 (2 - 2 1/2 in.).

Dial: white aluminium, with black markings.

Special Dial: ranges different from standards, or custom artwork, available on application.

Pointer: balanced aluminium or stainless steel; micrometric adjustable only for DS 100 (4 in.) and DS 150 (6 in.).

Case: AISI 304 stainless steel.

Gasket: polychloroprene.

Window: plexiglas; optional: shatterproof glass for DS 100 (4 in.) and 150 (6 in.).

Ring: AISI 304 polished stainless steel; with bayonet connection for DS 100 (4 in.) and DS 150 (6 in.); crimped for DS 50-63 (2 - 2 1/2 in.).

Protection: IP 65.

TYPE S6

Socket material: K Monel.

Elastic element: K Monel.

Window: plexiglas.

Other specifications identical type S4.

DAMPING LIQUIDS

TAB. 1

Damping liquids	Limit ambient temperature	
Glycerine filled 98%	15/65 °C	60/150 °F
Silicone oil filled	-45/65 °C	-50/150 °F
Fluorolube filled	-60/65 °C	-76/150 °F

Glycerine or silicone should not be used with highly oxidizing agents such as oxygen, chlorine, nitric acid or hydrogen peroxide, because of danger of spontaneous

chemical reaction, inflammability or explosion. The use of fluorinated fluid is recommended in these cases.

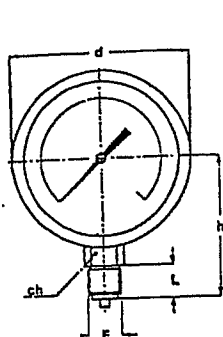
DIAMETERS, ORDERING CODE AND CONNECTING THREADS

TAB. 2

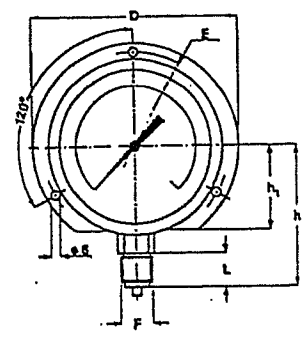
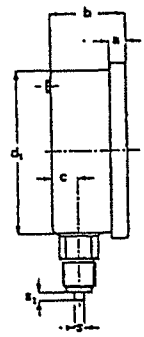
DS	Model	Pressure range (bar)	Ordering Code	Connections	Socket material	C-shaped tubular element	Spiral tubular element	Welding
50	S10/3	2,5/40	N1	1/4"-1/8" Gas-NPT	OT58brass	phosphorbronze	—	Tin alloy
50	S10/3	60/400	S1	1/4"-1/8" Gas-NPT	OT58brass	—	phosphorbronze	Tin/Silver alloy
50	S18/3	2,5/40	S4	1/4"-1/8" Gas-NPT	AISI316L	AISI316L	—	AISI316 TIG
63	S10/3	≤ 40	N1	1/4" Gas-NPT	OT58brass	phosphorbronze	—	Tin alloy
63	S10/3	60/400	S1	1/4" Gas-NPT	OT58brass	—	phosphorbronze	Silver alloy
100	S10/3	≤ 40	N1	1/4" NPT, 1/2" Gas-NPT	OT58brass	phosphorbronze	—	Tin alloy
100	S10/3	60	S2	1/4" NPT, 1/2" Gas-NPT	OT58brass	AISI316L	—	Silver alloy
100	S10/3	100/1000	S2	1/4" NPT, 1/2" Gas-NPT	OT58brass	—	AISI316L	Silver alloy
63	S18/3	≤ 40	S4	1/4" Gas-NPT	AISI316	AISI316L	—	AISI316 TIG
63	S18/3	60/400	S4	1/4" Gas-NPT	AISI316	—	AISI316L	AISI316 TIG
100 + 150	S18/3	≤ 60	S4	1/4" NPT, 1/2" Gas-NPT	AISI316	AISI316L	—	AISI316 TIG
100 + 150	S18/3	100/1000	S4	1/4" NPT, 1/2" Gas-NPT	AISI316	—	AISI316L	AISI316 TIG



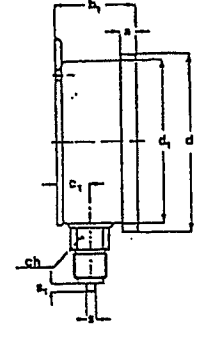
DS 100 S10/3 - MOUNTING STYLES AND DIMENSIONS



TYPE A
Lower Connection - Stem Mounting.

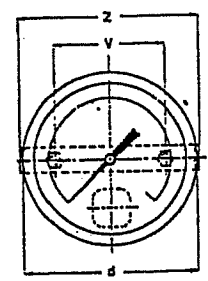


TYPE C
Lower Connection - Surface Mounting Back Flange.

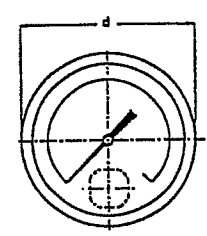
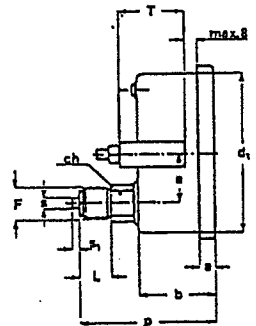


TAB. 16 - DIMENSIONS AND WEIGHTS TYPE A-C

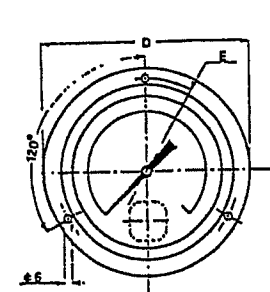
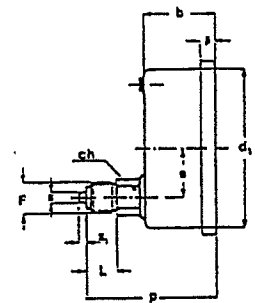
F	a	b	b ₁	c	c ₁	d	d ₁	D	E	h	h ₁	L	ch	Kg
1/2" Gas / NPT	10	47	51	16	20	110	101	130	116	86	52	20	□ 24	= 0.900
1/2" NPT	10	47	51	16	20	110	101	130	116	81	52	15	□ 24	



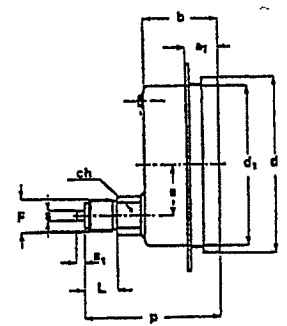
TYPE B
Back Connection - Flush Mounting, U-Clamp.



TYPE D
Back Connection - Stem Mounting.

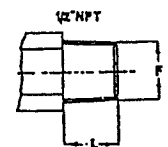
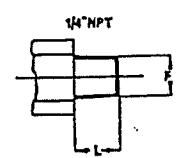
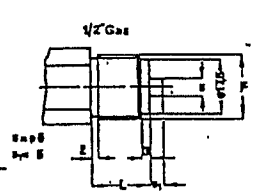


TYPE E
Back Connection - Flush Mounting Front Flange.



TAB. 17 - DIMENSIONS AND WEIGHTS TYPE B-D-E

F	a	a ₂	b	d	d ₁	D	E	e	L	p	T	V	Z	ch	Kg.
1/2" Gas / NPT	10	18	47	110	101	130	116	31	20	86	41,5	70	112	□ 24	0,900
1/4" NPT	10	18	47	110	101	130	116	31	15	81	41,5	70	112	□ 24	

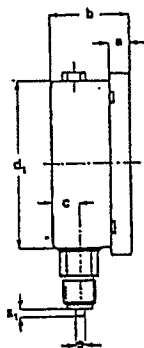
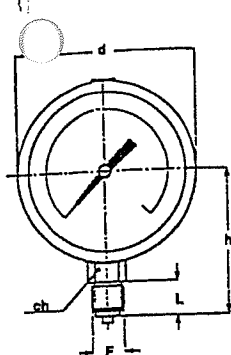


S10/3

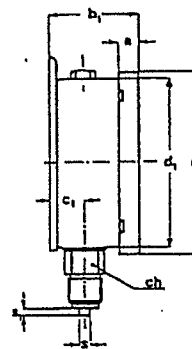
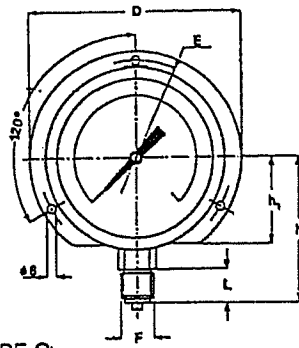
S18/3

DS 100 AND 150

S18/3 - MOUNTING STYLES AND DIMENSIONS



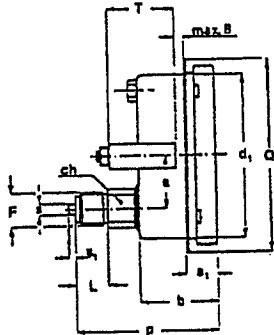
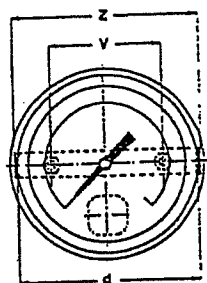
TYPE A
Lower Connection - Stem Mounting.



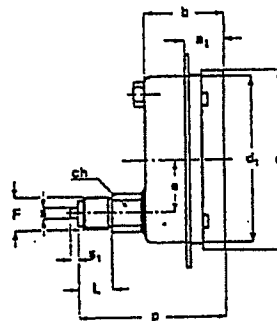
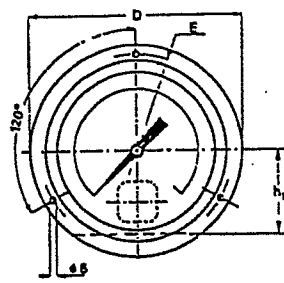
TYPE C
Lower Connection - Surface Mounting Back Flange.

TAB.18-DIMENSIONS AND WEIGHTS TYPE A-C

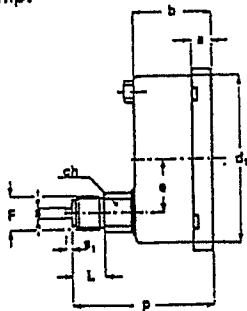
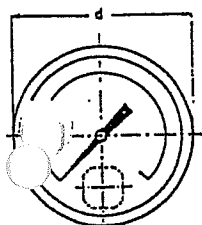
DS	F	a	b	b ₁	c	c ₁	d	d ₁	D	E	h	h ₁	L	ch	Kg
100	1/4" NPT	13	48	52	16	20	110,5	101	130	116	81	52	15	□ 24	≈ 1.000
100	1/2" Gas/NPT	13	48	52	16	20	110,5	101	130	116	86	52	20	□ 24	≈ 1.000
150	1/4" NPT	15	50,5	54	16,5	20	161,5	150	190	175	113	85	15	□ 24	≈ 1.800
150	1/2" Gas/NPT	15	50,5	54	16,5	20	161,5	150	190	175	118	85	20	□ 24	≈ 1.800



TYPE B
Back Connection -
Flush Mounting, U-Clamp.



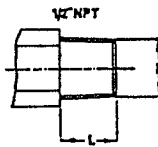
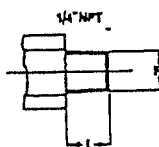
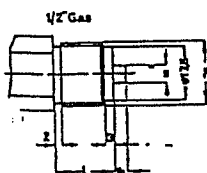
TYPE E
Back Connection - Flush Mounting Front Flange.



TYPE D
Back Connection - Stem Mounting.

TAB.19-DIMENSIONS AND WEIGHTS TYPE B-D-E

DS	F	a	a ₁	a ₂	b	d	d ₁	D	E	e	L	p	T	V	Z	Q	ch	Kg
100	1/4" NPT	13	20	20	48	110,5	101	130	116	31	15	82	41,5	70	112	112	□ 24	≈ 1.000
100	1/2" Gas/NPT	13	20	20	48	110,5	101	130	116	31	20	87	41,5	70	112	112	□ 24	≈ 1.000
150	1/4" NPT	15	21	25	50,5	161,5	150	190	175	48	15	80	41,5	106	155	164	□ 24	≈ 1.800
150	1/2" Gas/NPT	15	21	25	50,5	161,5	150	190	175	48	20	85	41,5	106	155	164	□ 24	≈ 1.800



D-7402 Kirchentellinsfurt, Postfach 64
Bahnhofstrasse 33, Tel. 0 71 21 / 6 02 91
Telex 729 663, Telefax 0 71 21 / 60 09 59

RESERVE THE RIGHT TO THEMSELVES TO MAKE ALL THE MODIFICATIONS THAT THEY DEEM INDISPENSABLE AT ANY TIME.



PRODUCTION.

IN ORDER TO IMPROVE

High-Pressure 2-Way Ball Valves

BKH Steel

PN 500 bar max. temperature 100°C

BKH Stainless Steel

PN 500 bar max. temperature 220°C



ZENTGRAF

Merziger
Hochdruck-
Armaturen
GmbH

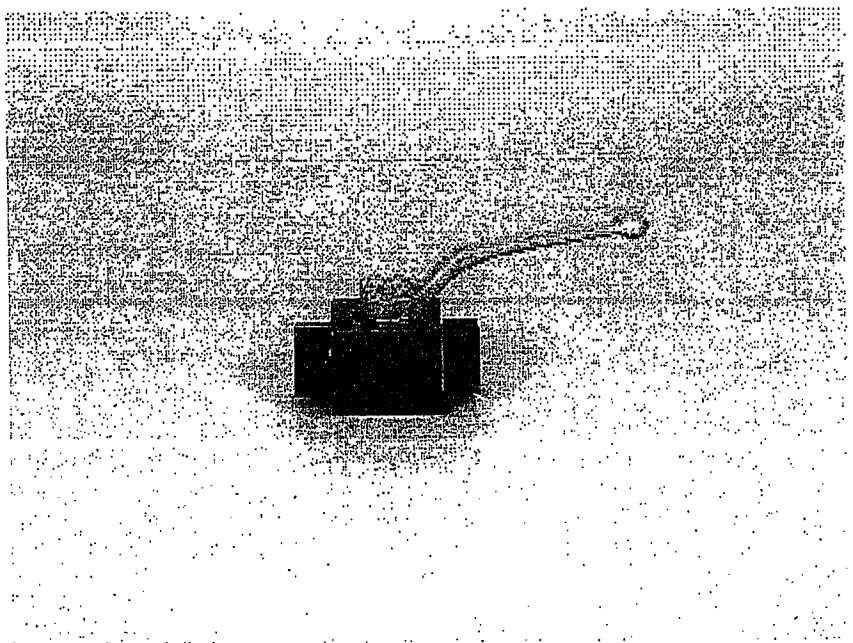
Ballerner Straße 8
D-66663 Merzig

Phone
0 68 61 / 70 00-0
Fax
0 68 61 / 70 00 77

e-mail
info@mha-zentgraf.com

5.2
5.5

MHA-high-pressure ball valves type BKH are our well-established ball valves for the high-pressure range up to 500 bar Nominal pressure. They can be delivered in Nominal diameters from DN 4 to DN 25. Due to the different possible seal materials, the ball valves are suitable for various applications. Our technical sales department will be pleased to assist you in finding the suitable material combination.



Dimensions

DIN 2353 · Light Series

Please note the pressure ratings of the tube connections !

Type	DN ⁽¹⁾	PN ⁽²⁾	RA	LW	L	I	B	H	h±0,5	m	V	SW	K ⁽³⁾	i	d	Weight _{kg}
BKH- 6L	4	500	6	5	67	40	26	47	33	13,5	11	9	115	10	M 12x1,5	0,36
BKH- 8L	6	500	8	6	67	40	26	47	33	13,5	11	9	115	10	M 14x1,5	0,37
BKH-10L	8	500	10	8	74	40	26	47	33	13,5	11	9	115	11	M 16x1,5	0,38
BKH-12L	10	500	12	10	74	43	32	52	38	17,5	11	9	115	11	M 18x1,5	0,5
BKH-15L-SW9	13	500	15	13	82	48	35	54	40	19	11	9	115	12	M 22x1,5	0,61
BKH-15L-SW12	16	400	15	15	82	48	38	63	46	19	13	12	160	12	M 22x1,5	0,7
BKH-18L-SW9	13	500	18	13	82	48	35	54	40	19	11	9	115	12	M 26x1,5	0,6
BKH-18L-SW12	16	400	18	15	82	48	38	63	46	19	13	12	160	12	M 26x1,5	0,75
BKH-22L	20	315	22	20	101	62	49	75	57	24,5	14	14	200	14	M 30x2	1,49
BKH-28L	25	315	28	25	108	66	58	83	65	29,5	14	14	200	14	M 36x2	2,0
BKH-35L	25/32	315	35	25	112	66	58	83	65	29,5	14	14	200	16	M 45x2	2,12

DIN 2353 · Heavy Series

Please note the pressure ratings of the tube connections !

Type	DN ⁽¹⁾	PN ⁽²⁾	RA	LW	L	I	B	H	h±0,5	m	V	SW	K ⁽³⁾	i	d	Weight _{kg}
BKH- 8S	4	500	8	5	73	40	26	47	33	13,5	11	9	115	12	M 16x1,5	0,38
BKH-10S	6	500	10	6	73	40	26	47	33	13,5	11	9	115	12	M 18x1,5	0,39
BKH-12S	8	500	12	8	76	40	26	47	33	13,5	11	9	115	12	M 20x1,5	0,39
BKH-14S	10	500	14	10	80	43	32	52	38	17,5	11	9	115	14	M 22x1,5	0,5
BKH-16S-SW9	13	500	16	13	86	48	35	54	40	19	11	9	115	14	M 24x1,5	0,6
BKH-16S-SW12	16	400	16	15	86	48	38	63	46	19	13	12	160	14	M 24x1,5	0,75
BKH-20S-SW9	13	500	20	13	90	48	35	54	40	19	11	9	115	16	M 30x2	0,6
BKH-20S-SW12	16	400	20	15	90	48	38	63	46	19	13	12	160	16	M 30x2	0,8
BKH-25S	20	315	25	20	109	62	49	75	57	24,5	14	14	200	18	M 36x2	1,55
BKH-30S	25	315	30	25	120	66	58	83	65	29,5	14	14	200	20	M 42x2	2,1
BKH-38S	25/32	315	38	25	124	66	58	83	65	29,5	14	14	200	22	M 52x2	2,3

Female Thread DIN/ISO 228

Type	DN ⁽¹⁾	PN ⁽²⁾	LW	L	I	B	H	h±0,5	m	V	SW	K ⁽³⁾	i	d	Weight _{kg}
BKH-G1/8	4	500	5	69	40	26	47	33	13,5	11	9	115	10	G1/8	0,41
BKH-G1/4	6	500	6	69	40	26	47	33	13,5	11	9	115	14	G1/4	0,4
BKH-G3/8	10	500	10	72	43	32	52	38	17,5	11	9	115	14	G3/8	0,54
BKH-G1/2-SW9	13	500	13	83	48	35	54	40	19	11	9	115	16,3	G1/2	0,65
BKH-G1/2-SW12	16	400	15	83	48	38	63	46	19	13	12	160	16,3	G1/2	0,7
BKH-G5/8-SW9	13	500	13	83	48	35	54	40	19	11	9	115	16	G5/8	0,61
BKH-G5/8-SW12	16	400	15	83	48	38	63	46	19	13	12	160	16	G5/8	0,7
BKH-G3/4	20	315	20	95	62	49	75	57	24,5	14	14	200	18	G3/4	1,5
BKH-G1	25	315	25	113	66	58	83	65	29,5	14	14	200	20	G1	2,2
BKH-G1 1/4	25/32	315	25	120	66	58	83	65	29,5	14	14	200	22	G1 1/4	2,3

NPT Female Thread ANSI B1.20.1

Type	DN ⁽¹⁾	PN ⁽²⁾	LW	L	I	B	H	h±0,5	m	V	SW	K ⁽³⁾	i	d	Weight _{kg}
BKH-1/8"NPT	4	500	5	69	40	26	47	33	13,5	11	9	115	13	1/8 "NPT	0,3
BKH-1/4"NPT	6	500	6	69	40	26	47	33	13,5	11	9	115	17	1/4 "NPT	0,4
BKH-3/8"NPT	10	500	10	78	43	32	52	38	17,5	11	9	115	17,4	3/8 "NPT	0,55
BKH-1/2"NPT	13	500	13	104	48	35	54	40	19	11	9	115	25	1/2 "NPT	0,75
BKH-3/4"NPT	20	315	20	102	62	49	75	57	24,5	14	14	200	23,1	3/4 "NPT	1,63
BKH-1"NPT	25	315	25	119	66	58	83	65	29,5	14	14	200	27,8	1 "NPT	2,3
BKH-1 1/4"NPT	25/32	315	25	130	66	58	83	65	29,5	14	14	200	23	1 1/4 "NPT	2,51

(1) DN in mm

(2) PN in bar (1 bar=14,503 psi) depending on the seal material

(3) The ball valve will be delivered without the handle being mounted.

Standard handles:

SW 9 - Zn bent

SW 12 - Al bent

SW 14 - Zn straight

Other designs and materials on request.

General Information

Admissible working temperature for body material

11SMn30: - 10 °C to + 100 °C

St52-3: - 60 °C to + 300 °C

Stainless steel: - 60 °C to + 400 °C

Attention: Please note lower pressures at higher temperatures.

Admissible working temperature for seal material

Buna-N - 35 °C to + 100 °C

Viton - 25 °C to + 200 °C

POM-MoS₂ - 30 °C to + 100 °C

Teflon - 200 °C to + 220 °C

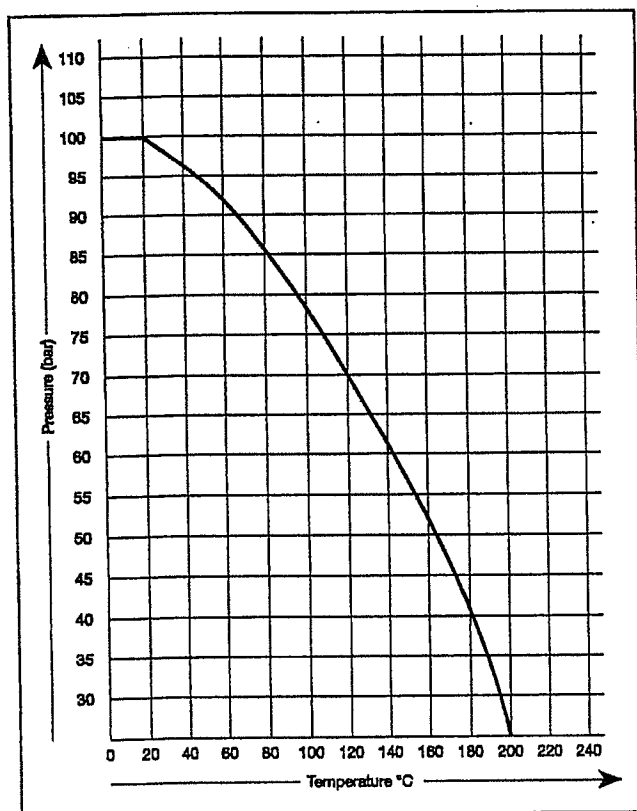
EPDM - 50 °C to + 130 °C

Attention: Please note lower pressures at higher temperatures.

Storing and Assembling

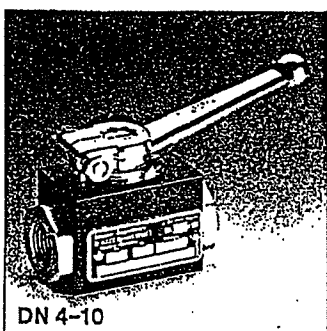
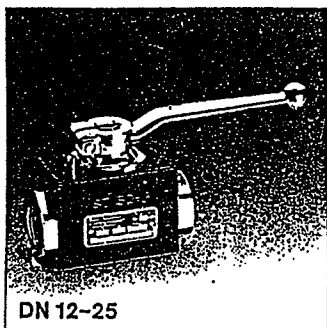
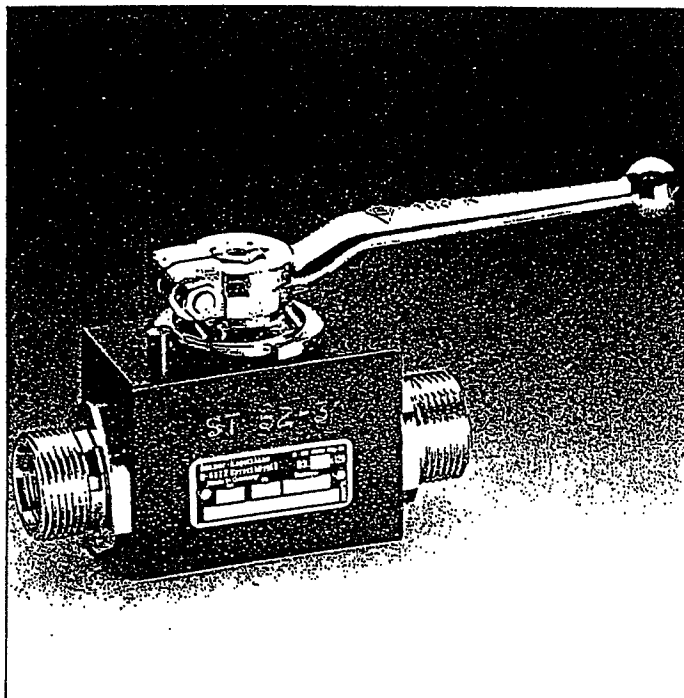
1. Do not remove the dust caps during storing.
2. Pipes into which ball valves will be mounted have to be rinsed thoroughly before the assembly.
3. All pressure and temperature values in the catalogue are maximum values under normal conditions. Do not hesitate to enquire in special cases. MHA-ball valves with metal seals may also be used as throttling valves.

Pressure-temperature diagramme for ball seal consisting of teflon



BHG

Block-Hochdruck-Kugelhahn mit Gewindeanschluß
Block type high pressure ball valve with threaded ports
Robinet monobloc haute pression à boisseau sphérique, raccords mâles ou femelles



● Der Schaltgriff gehört zum Lieferumfang.

● Bei Kugelhähnen mit Dichtungen aus Teflon werden im Normalfall O-Ringe aus Viton eingebaut.

● Sonderausführungen auf Anfrage.

● The handle is included.

● Ball valves with teflon seals are normally delivered with viton O-rings.

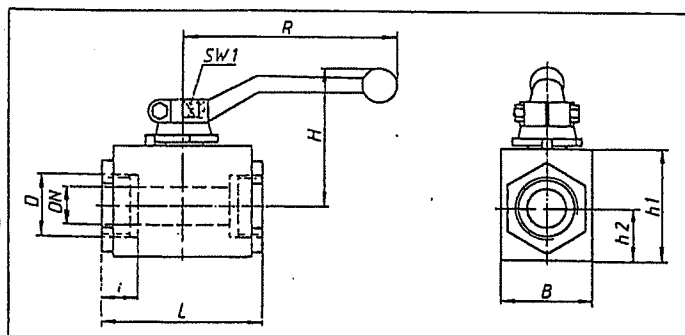
● Special designs upon request.

● La manette est comprise dans la livraison.

● Les robinets à boisseau sphérique avec joints teflon, sont prévus normalement avec joints torique en viton.

● Exécutions spéciales sur demande.

	PN bar	DN mm	Normalausführungen Standard versions Exécutions standards	
			Typ 1	Typ 2
Gehäuse Body Corps	150-500	4-25	Stahl Steel Acier	Stahl Steel Acier
Kugel Ball Bille	150-500	4-25	Stahl, hartverchromt Steel, hard chromed Acier, chromé dur	Stahl, hartverchromt Steel, hard chromed Acier, chromé dur
Dichtungen Seals Joints	150-500	4-25	Polyamid	Teflon



Bestell-Nr. und Typ
bitte angeben.

Please quote order-no.
and typ.

Mentionner référence
et type s.v.p.

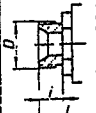
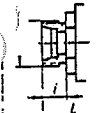
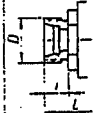
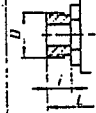
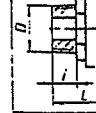
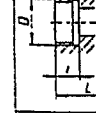
Typ 1
Typ 2

400
BHG V500.104/...

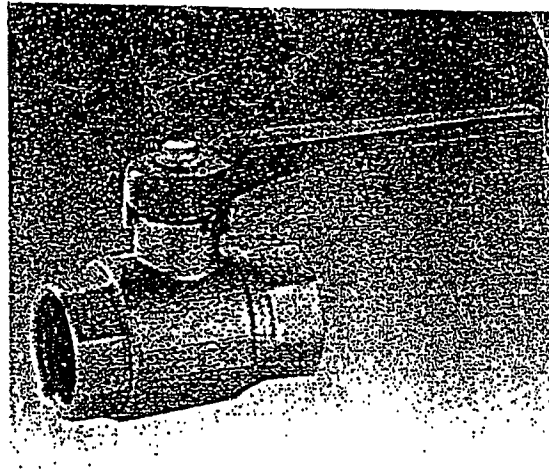
Golddurchgang
Längs DIN 3202 V1, M3
DN 4- 25
PN 150-500

Full bore
Length, face-to-face DIN 3202 V1, M3
DN 4- 25
PN 150-500

Passage intégral
Longueur hors tout DIN 3202 V1, M3
DN 4- 25
PN 150-500

Anschlußart Type of connection Type de raccordement	DN	Rohr AD pipe OD ø ext. du tube	PN		D	i	L	B	h ₁	h ₂	H	R	SW1	Gew. kg Weight kg Poids kg	Bestell-Nr. Order-No. Référence
			Poly- amid	Teflon											
1 Für Rohrverschraubung mit Kugelbuchse, DIN 7601  For compression fitting, with ball-type nipple, DIN 7601 <i>Pour raccord avec cône, DIN 7601</i>	4	-	500	350	M 12x1,5	10	76	25	30	12	45	90	8	0,3	BHG V 500.104
	6	-	500	350	M 14x1,5	10	76	25	30	12	45	90	8	0,3	BHG V 500.106
	8	-	500	350	M 16x1,5	11	76	25	30	12	45	90	8	0,3	BHG V 500.108
	10	-	500	350	M 18x1,5	11	80	30	35	15	51	90	8	0,4	BHG V 500.110
	12	-	500	350	M 22x1,5	12	96	35	45	18	66	130	10	0,8	BHG V 500.112
	16	-	400	250	M 26x1,5	12	105	40	50	21,5	68	130	10	1,0	BHG V 400.116
	20	-	315	175	M 30x1,5	14	110	45	55	24	77	180	10	1,3	BHG V 315.120
	25	-	315	150	M 38x1,5	14	120	55	65	30,5	82	180	12	2,2	BHG V 315.125
2 Für Rohrverschraubung leichte Reihe, DIN 2353  For compression fitting, light series, DIN 2353 <i>Pour raccord avec cône DIN 2353, série légère</i>	4	6	250*	250*	M 12x1,5	10	76	25	30	12	45	90	8	0,3	BHG V 250.204
	6	8	250*	250*	M 14x1,5	10	76	25	30	12	45	90	8	0,3	BHG V 250.206
	8	10	250*	250*	M 16x1,5	11	76	25	30	12	45	90	8	0,3	BHG V 250.208
	10	12	250*	250*	M 18x1,5	11	80	30	35	15	51	90	8	0,4	BHG V 250.210
	12	15	250*	250*	M 22x1,5	12	96	35	45	18	66	130	10	0,8	BHG V 250.212
	16	18	160*	160*	M 26x1,5	12	105	40	50	21,5	68	130	10	1,0	BHG V 160.216
	20	22	160*	160*	M 30x2	14	110	45	55	24	77	180	10	1,3	BHG V 160.220
	25	28	100*	100*	M 36x2	14	120	55	65	30,5	82	180	12	2,2	BHG V 100.225
3 Für Rohrverschraubung schwere Reihe, DIN 2353  For compression fitting, heavy series, DIN 2353 <i>Pour raccord avec cône DIN 2353, série lourde</i>	4	8	500	350	M 16x1,5	12	76	25	30	12	45	90	8	0,3	BHG V 500.304
	6	10	500	350	M 18x1,5	12	76	25	30	12	45	90	8	0,3	BHG V 500.306
	8	12	500	350	M 20x1,5	12	76	25	30	12	45	90	8	0,3	BHG V 500.308
	10	14	500	350	M 22x1,5	14	80	30	35	15	51	90	8	0,4	BHG V 500.310
	12	16	400*	350	M 24x1,5	14	96	35	45	18	66	130	10	0,8	BHG V 400.312
	16	20	400	250	M 30x2	16	105	40	50	21,5	68	130	10	1,0	BHG V 400.316
	20	25	315	175	M 36x2	18	110	45	55	24	77	180	10	1,3	BHG V 315.320
	25	30	250*	150	M 42x2	20	120	55	65	30,5	82	180	12	2,2	BHG V 250.325
4 Flachdichtend, metrisches Außengewinde, DIN 13  Flat sealing metric male thread, DIN 13 <i>Filet métrique mâle avec étanchéité, DIN 13</i>	4	-	500	350	M 12x1,5	10	76	25	30	12	45	90	8	0,3	BHG V 500.404
	6	-	500	350	M 14x1,5	10	76	25	30	12	45	90	8	0,3	BHG V 500.406
	8	-	500	350	M 16x1,5	11	76	25	30	12	45	90	8	0,3	BHG V 500.408
	10	-	500	350	M 18x1,5	11	80	30	35	15	51	90	8	0,4	BHG V 500.410
	12	-	500	350	M 22x1,5	12	96	35	45	18	66	130	10	0,8	BHG V 500.412
	16	-	400	250	M 26x1,5	12	105	40	50	21,5	68	130	10	1,0	BHG V 400.416
	20	-	315	175	M 30x1,5	14	110	45	55	24	77	180	10	1,3	BHG V 315.420
	25	-	315	150	M 38x1,5	14	120	55	65	30,5	82	180	12	2,2	BHG V 315.425
5 Flachdichtend, Rohraußen- gewinde, DIN ISO 228/1  Flat sealing BSP male thread, DIN ISO 228/1. <i>Filet whitworth mâle avec étanchéité, DIN ISO 228/1</i>	4	-	500	350	G 1/8 A	10	76	25	30	12	45	90	8	0,3	BHG V 500.504
	6	-	500	350	G 1/4 A	11	76	25	30	12	45	90	8	0,3	BHG V 500.506
	8	-	500	350	G 3/8 A	12	76	25	30	12	45	90	8	0,3	BHG V 500.508
	10	-	500	350	G 1/2 A	12	80	30	35	15	51	90	8	0,4	BHG V 500.510
	12	-	500	350	G 5/8 A	12	96	35	45	18	66	130	10	0,8	BHG V 500.512
	16	-	400	250	G 3/4 A	14	105	40	50	21,5	68	130	10	1,0	BHG V 400.516
	20	-	315	175	G 1 A	16	110	45	55	24	77	180	10	1,3	BHG V 315.520
	25	-	315	150	G 1 1/4 A	18	120	55	65	30,5	82	180	12	2,2	BHG V 315.525
6 NPT-Außengewinde, ANSI/ASME B 1.20.1  NPT male thread, ANSI/ASME B 1.20.1 <i>Filet mâle NPT, ANSI/ASME B 1.20.1</i>	4	-	500	350	NPT 1/8	9,5	69	25	30	12	45	90	8	0,3	BHG V 500.604
	6	-	500	350	NPT 1/4	14,5	79	25	30	12	45	90	8	0,3	BHG V 500.606
	8	-	500	350	NPT 3/8	14,5	79	25	30	12	45	90	8	0,3	BHG V 500.608
	10	-	500	350	NPT 1/2	19	90	30	35	15	51	90	8	0,4	BHG V 500.610
	12	-	500	350	NPT 5/8	19	110	35	45	18	66	130	10	0,8	BHG V 500.612
	16	-	400	250	NPT 3/4	19	112	40	50	21,5	68	130	10	1,0	BHG V 400.616
	20	-	315	175	NPT 1	24	125	45	55	24	77	180	10	1,4	BHG V 315.620
	25	-	315	150	NPT 1 1/4	24,5	141	55	65	30,5	82	180	12	2,3	BHG V 315.625
7 Rohrinngewinde, DIN ISO 228/1  BSP female thread, DIN ISO 228/1 <i>Taradage, whitworth DIN ISO 228/1</i>	4	-	500	350	G 1/8	11	50	25	30	12	45	90	8	0,3	BHG V 500.704
	6	-	500	350	G 1/4	12,5	50	25	30	12	45	90	8	0,3	BHG V 500.706
	8	-	500	350	G 3/8	12,5	70	25	30	12	45	90	8	0,3	BHG V 500.708
	10	-	500	350	G 1/2	12,5	60	30	35	15	51	90	8	0,4	BHG V 500.710
	12	-	500	350	G 5/8	15	75	35	45	18	66	130	10	0,7	BHG V 500.712
	16	-	400	250	G 3/4	15	75	40	50	21,5	68	130	10	0,9	BHG V 400.716
	20	-	315	175	G 1	18	80	45	55	24	77	180	10	1,2	BHG V 315.720
	25	-	315	150	G 1 1/4	20	90	55	65	30,5	82	180	12	2,1	BHG V 315.725
8 NPT-Innengewinde, ANSI/ASME B 1.20.1  NPT female thread, ANSI/ASME B 1.20.1 <i>Taradage NPT, ANSI/ASME B 1.20.1</i>	4	-	500	350	NPT 1/8	11	50	25	30	12	45	90	8	0,3	BHG V 500.804
	6	-	500	350	NPT 1/4	12,5	50	25	30	12	45	90	8	0,3	BHG V 500.806
	8	-	500	350	NPT 3/8	12,5	70	25	30	12	45	90	8	0,3	BHG V 500.808
	10	-	500	350	NPT 1/2	12,5	60	30	35	15	51	90	8	0,4	BHG V 500.810
	12	-	500	350	NPT 5/8	15	75	35	45	18	66	130	10	0,7	BHG V 500.812
	16	-	400	250	NPT 3/4	15	75	40	50	21,5	68	130	10	0,9	BHG V 400.816
	20	-	315	175	NPT 1	18	80	45	55	24	77	180	10	1,2	BHG V 315.820
	25	-	315	150	NPT 1 1/4	20	90	55	65	30,5	82	180	12	2,1	BHG V 315.825

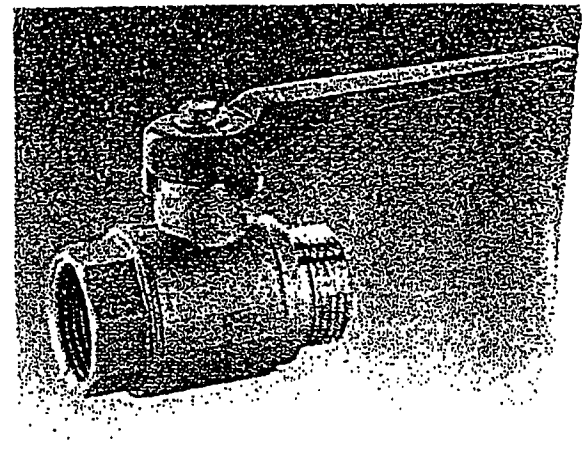
* Druckstufe der Rohrverschraubung.
Do not exceed nominal pipe pressure. Veuillez vous reporter à la pression nominale du tube.



Ball Valve PN 40 with lever

*both ends female threads full flow
brass nickel-plated
Type 105*

Dimension	Best.-Nr.
1/4"	101001
3/8"	101002
1/2"	101003
3/4"	101004
1"	101005
1 1/4"	101006
1 1/2"	101007
2"	101008
2 1/2"	101009
3"	101010



Ball Valve PN 40 with lever

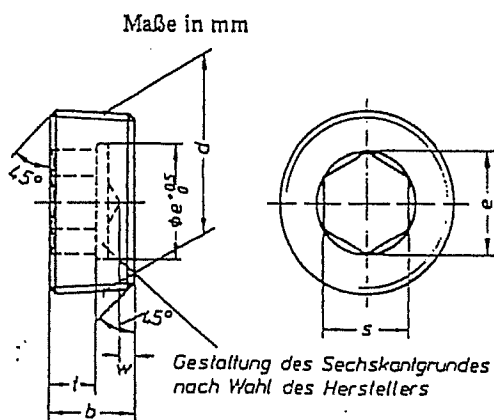
*female/male thread full flow
brass nickel-plated
Type 120*

Dimension	Best.-Nr.
1/4"	101016
3/8"	101017
1/2"	101018
3/4"	101019
1"	101020
1 1/4"	101021
1 1/2"	101022
2"	101023

5.7.1

Hexagonal head screw plug
Taper thread

DIN
906



G 3/4

6.1
6.4

Direct Operated

$p_{\max} = 315 \text{ bar}$
 $Q_{\max} = 80 \text{ l/min}$

Design

Pressure relief valves are designed as seat valves with cushion piston. The seat valve provides a drainless sealing between the pressure port and the return port below the adjusted setting pressure. The cushion piston prevents uncontrolled oscillations of the poppet when opened. Due to improved streamlining the valve achieves high flow rates.

Function

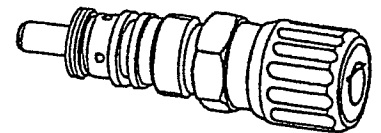
The pressure relief valve meets the requirements of "AD-Merkblatt A2" and is officially approved for the safety of hydraulic accumulators. The valve has been type tested by TÜV and can be supplied with TÜV test certificate upon request.

Safety valves with TÜV certificate are recommended if the built-in valves cannot be positioned to respond and there is no separate test bench available. The pressure is set by the manufacturer in stages of 10 bar. Higher pressure settings are avoided by means of a seal, however it is possible to relieve the hydraulic accumulator when the pump is turned off.

The valve is available as screw in type DBE 6 or as threaded connection type DBG 6.

Cartridge

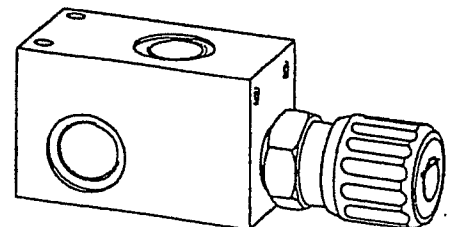
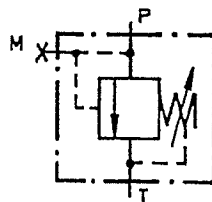
Type DBE 6



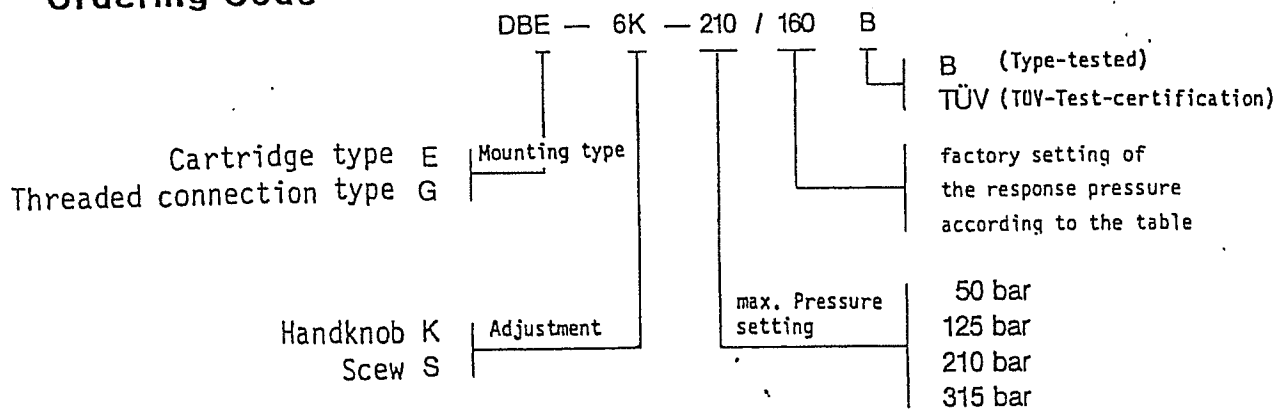
Threaded Connection

Type DBG 6

G 1/2 ISO 228



Ordering Code



Ordering No.

Standard Types

The third figure of the ordering No. is a 3 instead of 2 for valves with TÜV certificate

relief setting p in bar	permissible flow in l/min	TÜV-component mark	Ordering No.			
			Screw-in Valve DBE - 6 operated by hand knob	operated by inner hexagon	Pipe Connection Valve DBG - 6 operated by hand knob	operated by inner hexagon
20 30 40 50	20 30 40 40	TÜV. SV.88-711.5.F. 20.20 30.30 40.40 40.50	352 - 1330 - 012 - 120	352 - 1330 - 012 - 140	302 - 1330 - 013 - 120	302 - 1330 - 013 - 140
60 70 80 90 100 110 120 125	40 45 50 55 60 60 60 60	TÜV. SV.88-711.5.F. 40.60 45.70 50.80 55.90 60.100 60.110 60.120 60.125	352 - 1330 - 012 - 420	352 - 1330 - 012 - 440	302 - 1330 - 013 - 420	302 - 1330 - 013 - 440
130 140 150 160 170 180 190 200 210	55 65 70 75 80 80 80 80 80	TÜV. SV.88-711.5.F. 55.130 65.140 70.150 75.160 80.170 80.180 80.190 80.200 80.210	352 - 1330 - 012 - 620	352 - 1330 - 012 - 640	302 - 1330 - 013 - 620	302 - 1330 - 013 - 640
220 230 240 250 260 270 280 290 300 310 315	80	TÜV. SV.88-711.5.F. 220 230 240 250 260 80.270 280 290 300 310 315	352 - 1330 - 012 - 820	352 - 1330 - 012 - 840	302 - 1330 - 013 - 820	302 - 1330 - 013 - 840

Sealing Set: Ordering No. 351-1330-019-009

Housing for flange mounting and sandwich mounting with mounting surface A6 as per DIN 24340 is available.
Please see Data Sheet 30-00109-84E for dimension drawings.
Valves with VITON seals available upon request.

Technical Data

Pressure setting range

Port P : up to 315 bar

Port T : up to 50 bar

Recommended flow : see characteristics

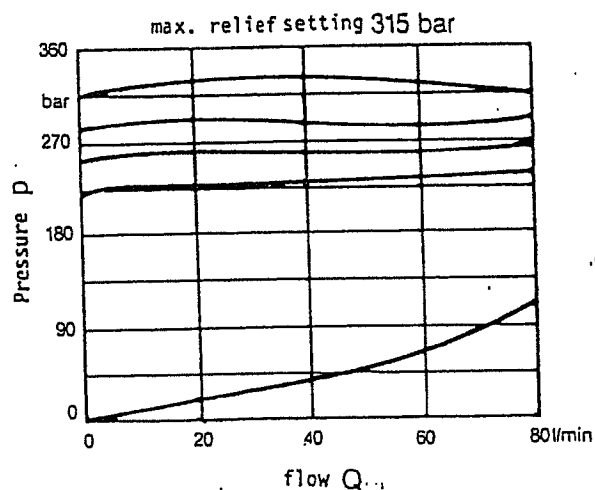
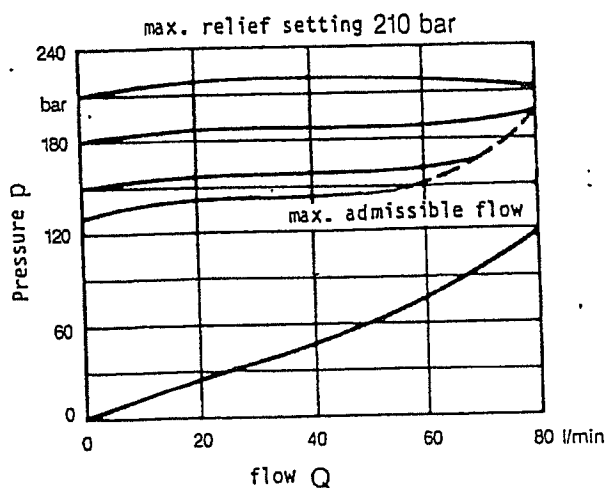
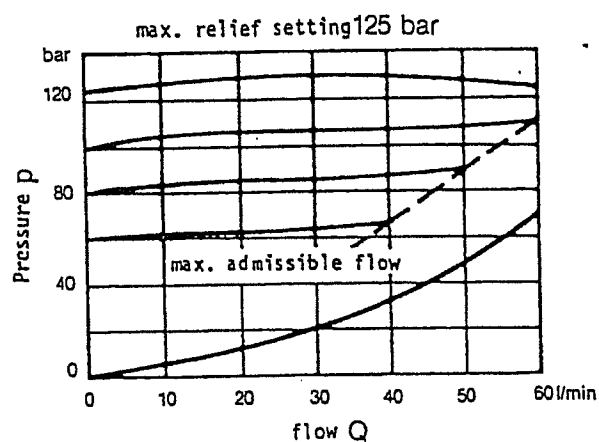
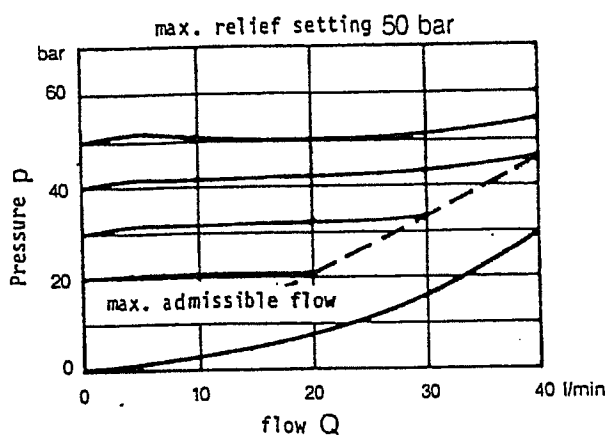
Temperature range : -20 to +70°C

Viscosity range : 12 to 300 mm²/s (cSt)

Recommended filtration : 25 µm abs.

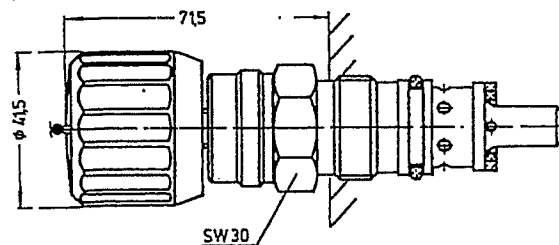
Characteristics (for 50°C oil temperature and 36 cSt viscosity)

Threaded Connection

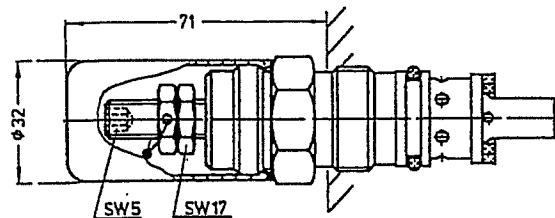


Cartridge Type

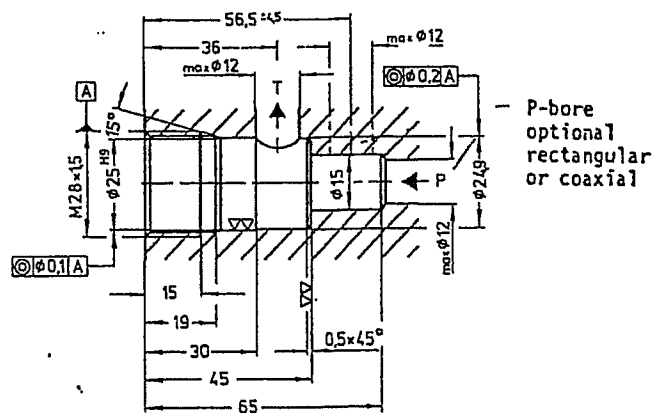
Handknob Adjustment



Screw Adjustment



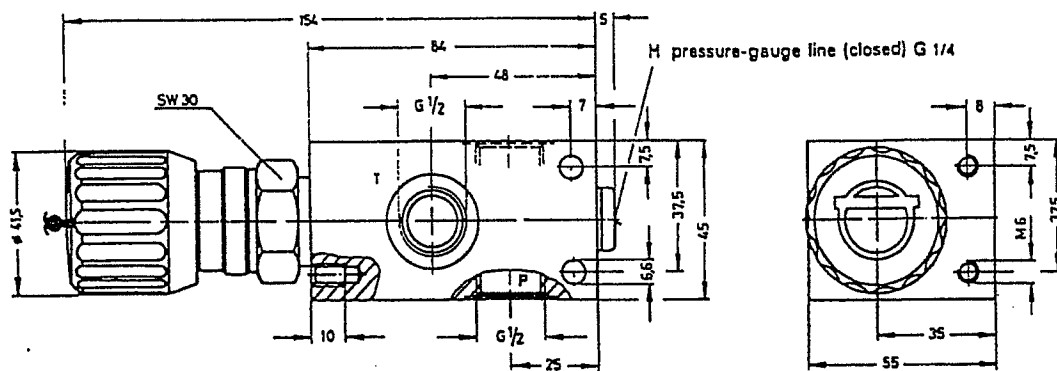
cavity details



Torque 50^{+5} Nm

Weight ab. 0,35 kg

Threaded Connection Type



Weight ab. 1,5 kg

6.2 6.10
6.5 6.20

Direct Operated

$p_{max} = 315 \text{ bar}$
 $Q_{max} = 80 \text{ l/min}$

Design

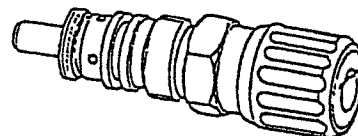
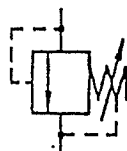
Pressure relief valves are designed as seat valves with cushion piston. The seat valve provides a drainless sealing between the pressure port and the return port below the adjusted setting pressure. The cushion piston prevents uncontrolled oscillations of the poppet when opened.

Cartridge Type

Type DBE 6

Function

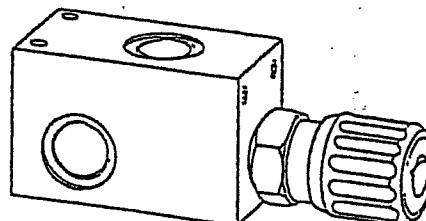
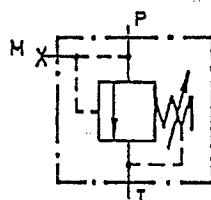
Pressure relief valves limit the pressure in hydraulic circuits and thus prevent overloading of the system. Thanks to their very good characteristics pressure relief valves are ideal as safety valves for accumulator circuits. Type-tested valves are available on request with TÜV test certificate (see Data Sheet 30-10009-84).



Threaded Connection

Type DBG 6

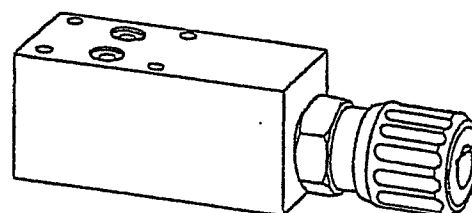
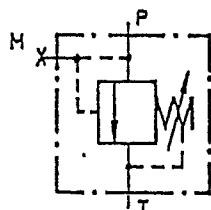
G 1/2 ISO 228



Subplate Mounting

Type DBA 6

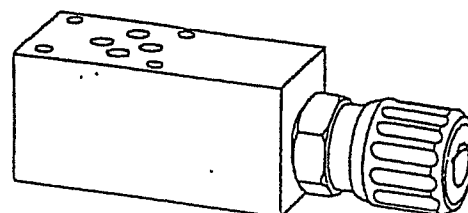
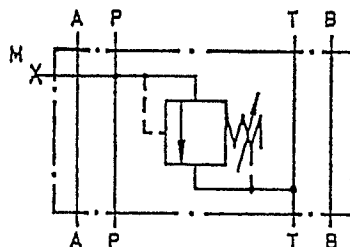
Mounting surface A6 DIN 24340



Sandwich

Type DBZ 6

Mounting surface A6 DIN 24340



Technical Data

Pressure setting range

Port P : up to 315 bar

Port T : up to 50 bar

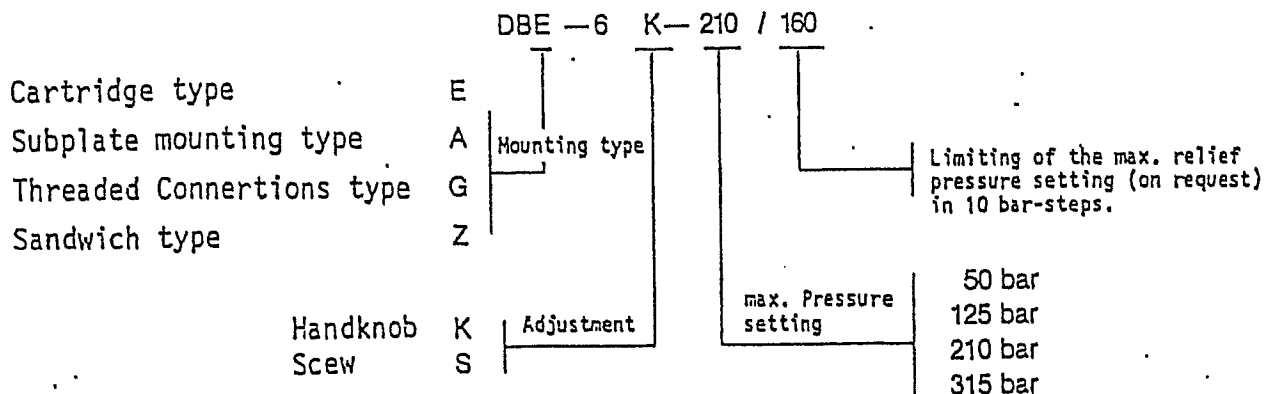
Recommended flow : see characteristics

Temperature range : -20 to +70°C

Viscosity range : 12 to 300 mm²/s (cSt)

Recommended Filtration : 25 µm abs.

Ordering Code



Ordering No.

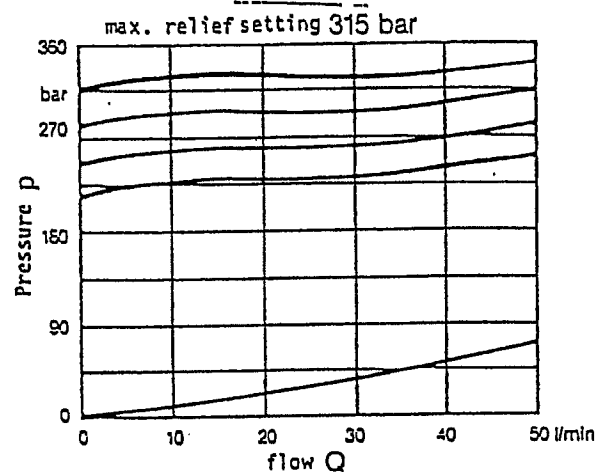
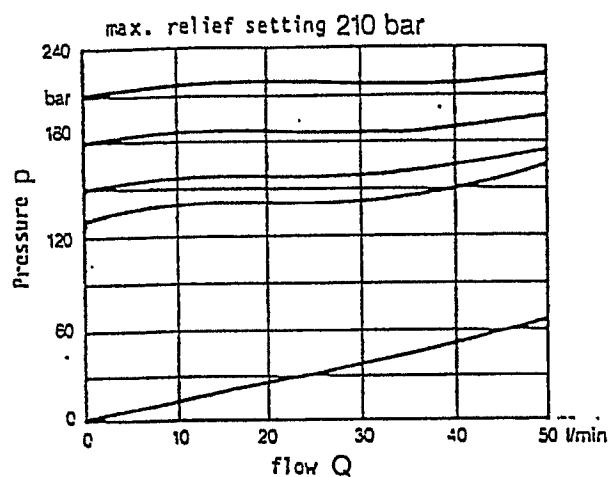
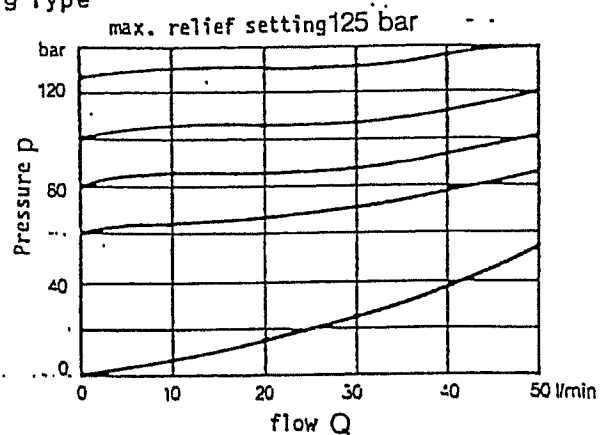
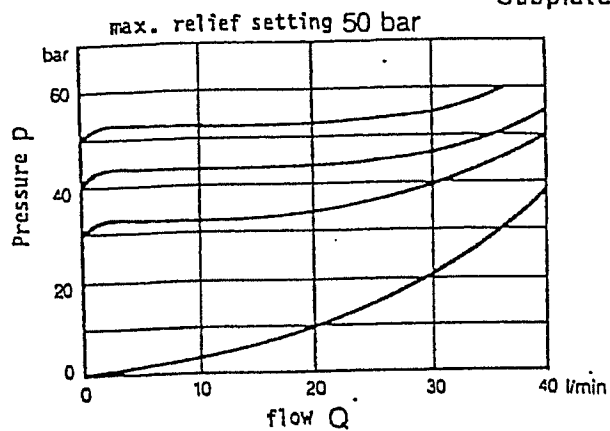
Standard Types

Type		Ord. number		Ord. number	Accessories
DBE - 6 - 50	Handknob Adjustm.	351 - 1330 - 012 - 120	Screw Adjustm.	351 - 1330 - 012 - 140	Mounting screws for suplate mount.types M 5 x 45 DIN 912 Ord. no. 405 - 1328 - 019 - 045
DBE - 6 - 125		351 - 1330 - 012 - 420		351 - 1330 - 012 - 440	
DBE - 6 - 210		351 - 1330 - 012 - 620		351 - 1330 - 012 - 640	
DBE - 6 - 315		351 - 1330 - 012 - 820		351 - 1330 - 012 - 840	
DBA - 6 - 50		321 - 1330 - 012 - 120		321 - 1330 - 012 - 140	
DBA - 6 - 125		321 - 1330 - 012 - 420		321 - 1330 - 012 - 440	
DBA - 6 - 210		321 - 1330 - 012 - 620		321 - 1330 - 012 - 640	
DBA - 6 - 315		321 - 1330 - 012 - 820		321 - 1330 - 012 - 840	
DBG - 6 - 50		301 - 1330 - 013 - 120		301 - 1330 - 013 - 140	sealing set Ord. no. 351 - 1330 - 019 - 009
DBG - 6 - 125		301 - 1330 - 013 - 420		301 - 1330 - 013 - 440	
DBG - 6 - 210		301 - 1330 - 013 - 620		301 - 1330 - 013 - 640	
DBG - 6 - 315		301 - 1330 - 013 - 820		301 - 1330 - 013 - 840	
DBZ - 6 - 50		341 - 1330 - 013 - 120		341 - 1330 - 013 - 140	
DBZ - 6 - 125		341 - 1330 - 013 - 420		341 - 1330 - 013 - 440	
DBZ - 6 - 210		341 - 1330 - 013 - 620		341 - 1330 - 013 - 640	
DBZ - 6 - 315		341 - 1330 - 013 - 820		341 - 1330 - 013 - 840	

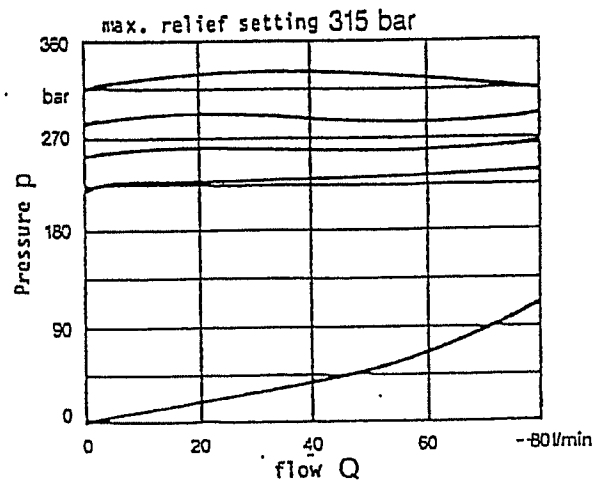
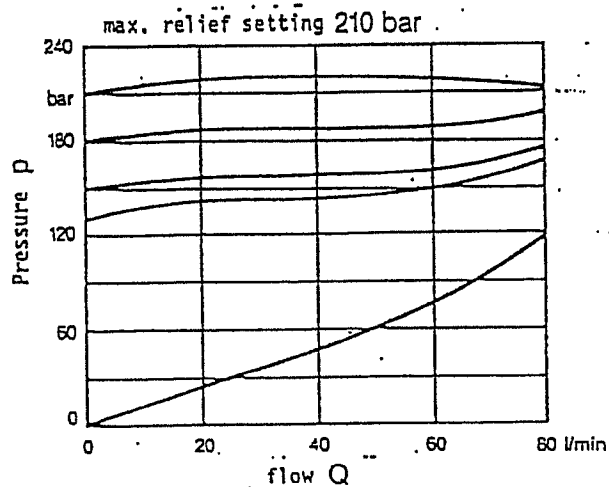
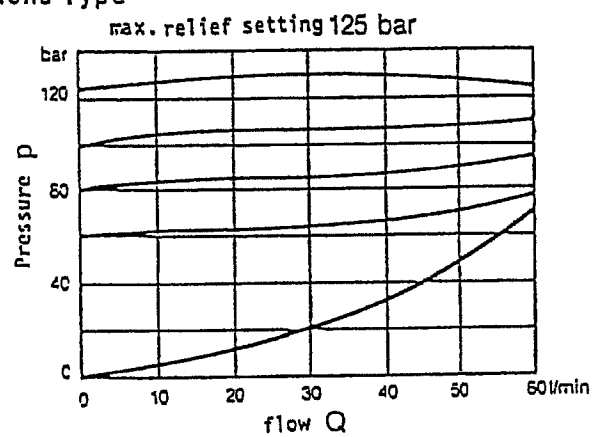
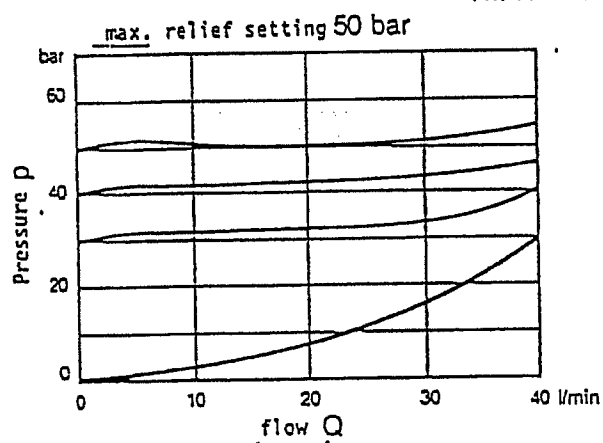
Valves with VITON seals upon request

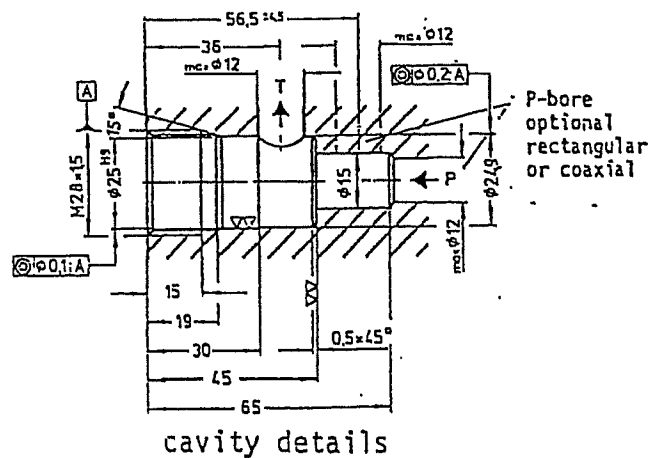
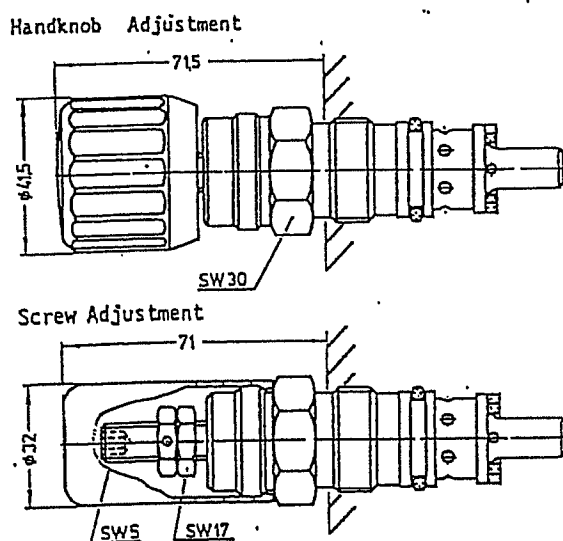
Characteristics (for 50°C Oil-temperature and 36 cSt viscosity)

Subplate Mounting Type

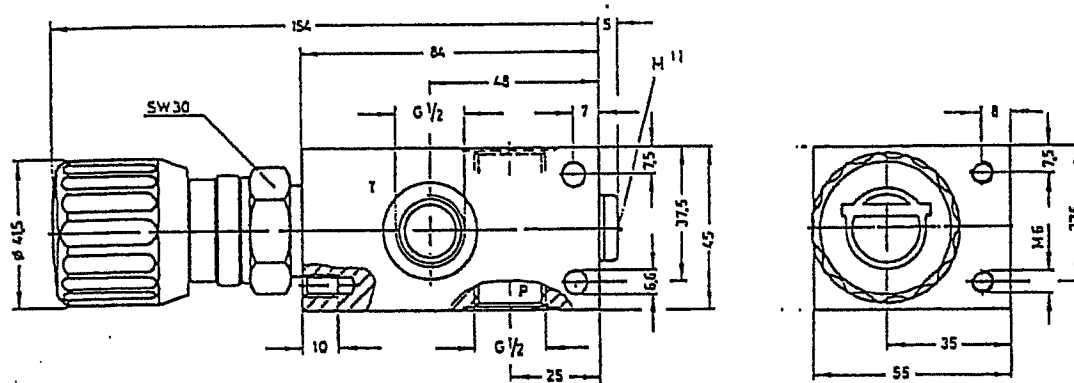


Threaded Connections Type



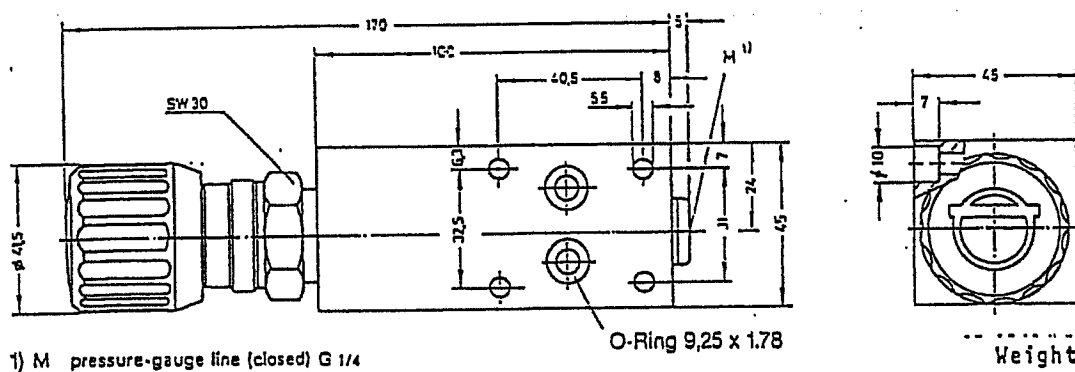
Torque 50^{+5} Nm

Weight ab. 0,35 kg



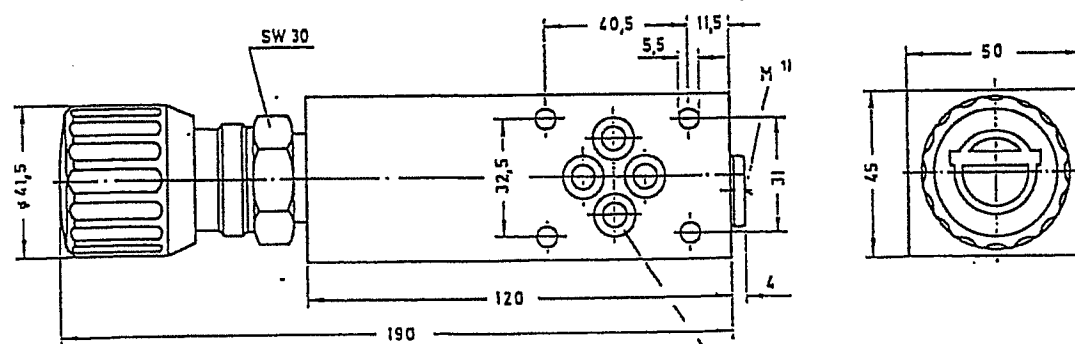
1) M pressure-gauge line (closed) G 1/4

Weight ab. 1,5 kg



1) M pressure-gauge line (closed) G 1/4

Weight ab. 1,6 kg



1) M pressure-gauge line (closed) G 1/4

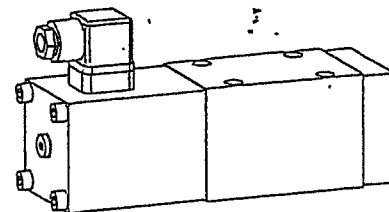
O-Ring 9,25 x 1.78

Weight ab. 1,8 kg

Solenoid poppet valve

- 2/2-, 3/2- and 3/4-way construction
- $Q_{\max} = 80 \text{ l/min}$
- $p_{\max} = 350 \text{ bar.}$

NG10
ISO 4401-05

**DESCRIPTION**

Poppet valve, flanged design NG10, available as a 2/2 or 3/2-way valve (normally open or closed) and as a 3/4-way valve (normally closed). The central functioning element of all directly controlled poppet valves in the NG10 series is the poppet valve cartridge NG10. See data sheet 1.11-2040. The solenoids correspond to VDE standard 0580.

Important: When commissioning, the valve must be vented under pressure (max. 2 revolutions of screw E).

FUNCTION

The valve is direct operated by a wet pin push type solenoid which in turn either opens or closes the poppet. The design of the poppet spool, which is equal in surface area on both sides and thus pressure balanced, means there are no undue opening and closing hydraulic forces. Due to this the oil flow through the poppet valve is possible in both directions. The valve is tight in both flow directions.

APPLICATION

Wandfluh poppet valves can be used anywhere absolutely leak tight closing functions are important. Completely sealed loading, gripping and clamping operations are all important functions which Wandfluh poppet valves can perform. Cartridge type poppet valves can be neatly accommodated in valve blocks. From a mechanical and functional point of view, poppet valves can replace slide valves at any time. NG10 valves are used where a light, compact unit is needed.

CONTENT

TYPE CODE

GENERAL SPECIFICATIONS	1.11-2160/1
HYDRAULIC SPECIFICATIONS	1.11-2160/1
ELECTRICAL CONTROL	1.11-2160/1
SYMBOLS	1.11-2160/1
CHARACTERISTICS	1.11-2160/2
DIMENSIONS	1.11-2160/2
PARTS LIST	1.11-2160/2
ACCESSORIES	1.11-2160/2

2/2- or 3/2-way construction	A	☐	☐	2	10	☐	-	☐	#	☐
3/4-way construction	A	☐	3	4	10	☐	-	☐	#	☐
International mounting interface ISO										
Medium-solenoid	M									
Super-solenoid	S									
2-way (connections)	2									
3-way (connections)	3									
2 position										
4 position										
Nominal size 10										
Normally closed,	solenoid on A-side					1a				
Normally open,	solenoid on B-side					0b				
Standard nominal voltage	12VDC	G12		110VAC		R110				
	24VDC	G24		115VAC		R115				
				230VAC		R230				

Design-Index (Subject to change)

GENERAL SPECIFICATIONS

Description	2/2-, 3/2- and 3/4-way poppet valve
Nominal size	NG10 to ISO 4401
Construction	Direct operated poppet valve
Operations	Solenoid
Mounting	Flange, 4 mounting holes for socket head screws M6x65
Connections	Threaded connection plates Multi-flange subplates Longitudinal stacking system
Ambient temperature	-20 ... +50°C
Mounting position	any, preferable horizontal
Fastening torque	$M_0 = 9,5 \text{ Nm}$ (Quality 8,8)
Weight	2/2-, 3/2-way 3/4-way $m = 4,6 \text{ kg}$ $m = 6,4 \text{ kg}$
Volume flow direction	any (see characteristics)

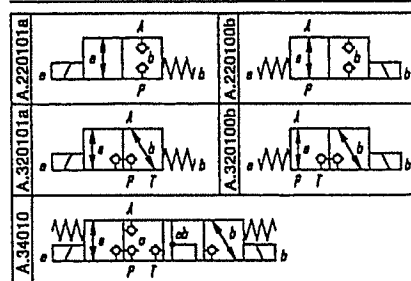
HYDRAULIC SPECIFICATIONS

Fluid	Mineral oil, other fluid on request
Contamination efficiency	ISO 4406, classe 18/14 (Required filtration grade $\beta_{10...16275}$) refer to data sheet 1.0-50/2
Viscosity range	12 mm ² /s bis 320 mm ² /s
Fluid temperature	-20 ... +70°C
Working pressure	Medium: $p_{\max} = 160$ bar Super: $p_{\max} = 350$ bar
Max. volume flow	$Q_{\max} = 80$ l/min see characteristics

ELECTRICAL CONTROL

Construction	Solenoid, wet pin push type, pressure tight	
Standard-nominal voltage	$U_N = 12 \text{ VDC}, 24 \text{ VDC}$ $U_N = 110 \text{ VAC}^*, 115 \text{ VAC}^*, 230 \text{ VAC}^*$ $AC = 50 \text{ to } 60 \text{ Hz}$ * Rectifier integrated in the plug Other nominal voltages and nominal performances on request	
Voltage tolerance	+/- 10% of nominal voltage	
Protection class	IP 65 to DIN 40050	
Relative duty factor	100% ED (see data sheet 1.1-430)	
Switching cycles	15'000/h	
Operating life (number of switching cycles)	10 ⁷	
Connection/Power supply	Over device plug connection to ISO 4400 / DIN 43650, (2P+E), other connections on request	
Solenoid:	- Medium	SIN60 (data sheet 1.1-145)
	- Super	SIS60 (data sheet 1.1-150)

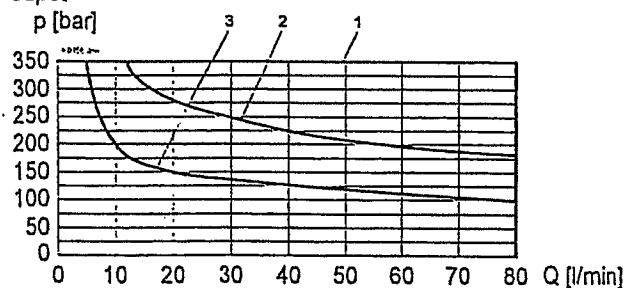
SYMBOLS



CHARACTERISTICS Oil viscosity $\nu = 30 \text{ mm}^2/\text{s}$

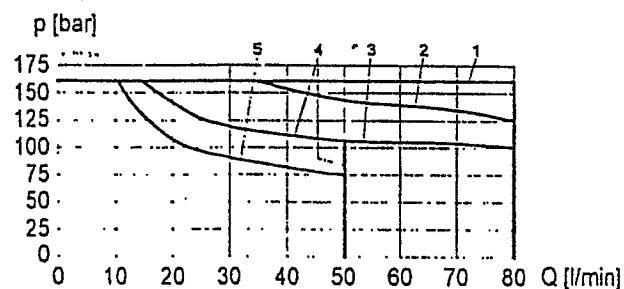
$p = f(Q)$ Performance limit by standard voltage at -10 %

Super

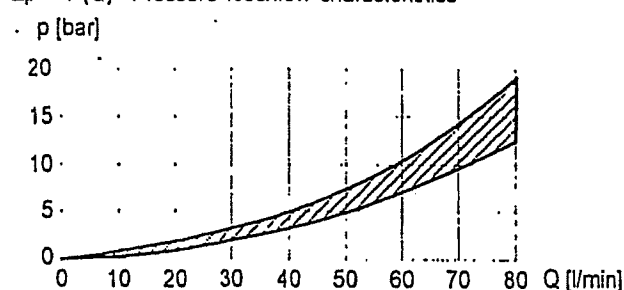


$p = f(Q)$ Performance limit by standard voltage at -10 %

Medium



$\Delta p = f(Q)$ Pressure loss/flow characteristics

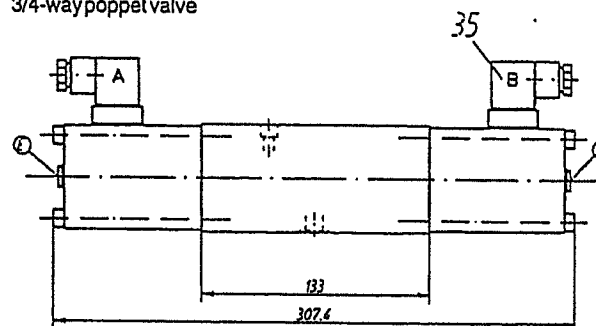


Type	Flow direction			
	P - A	A - T	A - P	T - A
AS22101a	1	-	2	-
AS22100b	1	-	2	-
AS32101a	1	2	3	1
AS32100b	1	2	3	1
AS3410	1	1	2	2

Type	Flow direction			
	P - A	A - T	A - P	T - A
AM22101a	1	-	4	-
AM22100b	1	-	2	-
AM32101a	1	3	5	1
AM32100b	1	3	3	1
AM3410	1	1	4	4

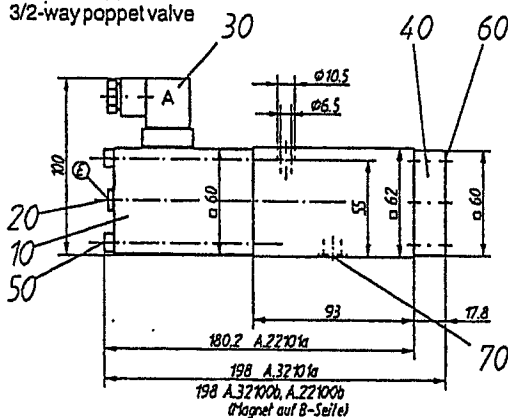
DIMENSIONS

3/4-way poppet valve

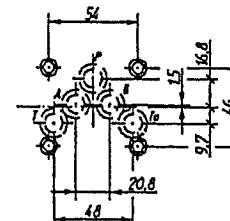


2/2-way poppet valve

3/2-way poppet valve



E = air bleed screw



PARTS LIST

Position	Article	Description
10	260.8...	Medium-solenoid SIN60
	260.9...	Super-solenoid SIS60
20	239.2024	Plug (incl. seal ring) HB0
30	219.2001	Plug A (grey)
35	219.2002	Plug B (black)
40	59.2200	Lid
50	246.3190	Zyl. screw M6x90 DIN912
60	246.3120	Zyl. screw M6x20 DIN912
70	160.2140	O-Ring ID 14,00x1,78

ACCESSORIES

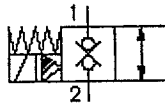
Threaded connection plates, Multi-flange subplates and Longitudinal stacking system see Register 2.9

**2 way, 2 pos. sol. valve, poppet
(normally closed)
SBV11-12.-C-0-.....H**

Symbol:

Nom. pressure:
350 bar

Nominal flow:
114 l/min.



7.3

IMAV
HYDRAULIK GMBH

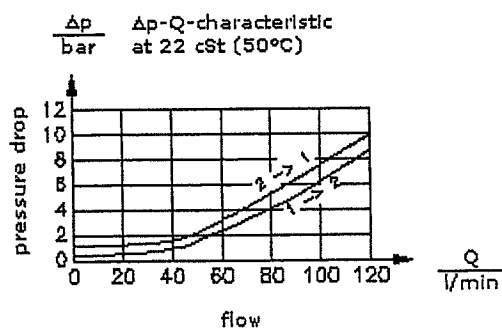
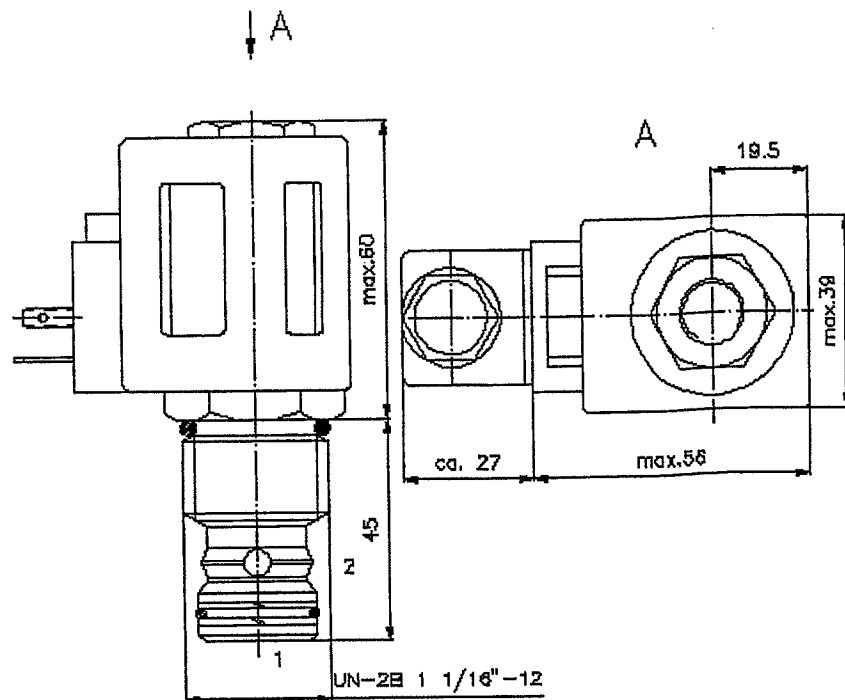
IMAV-Hydraulik GmbH
Breite Strasse 10
D-40670 Meerbusch (Osterath)
Telefon +49 (0)2159 - 9188.0
Telefax +49 (0)2159 - 4242
E-mail: info@imav.com
Internet: <http://www.imav.com>

MS-260-121-00

HEX 3/4"
(torque
cartridge:
5 - 8 Nm)

HEX 1 1/4"
(torque
cartridge:
85 - 94 Nm)

cavity:
C-12-2
see
data-sheet no.:
IK-100-129-00



**2 way, 2 pos. sol. valve, poppet
(normally closed)
SBV11-12.-C-0-.....H**



IMAV-Hydraulik GmbH
Breite Strasse 10
D-40670 Meerbusch (Osterath)
Telefon +49 (0)2159 - 9188.0
Telefax +49 (0)2159 - 4242
E-mail: info@imav.com
Internet: <http://www.imav.com>

MS-260-121-00

Ordering example:

SBV11 - 12N - C - 0 - 024D - G H
 | | | | | |
A **B** **C** **D** **E** **F** **G**

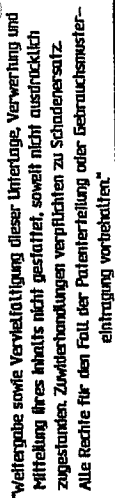
Keytype:

- | | |
|--|---|
| <p>A design
solenoid valve
double poppet valve
pilot operated</p> <p>B size and sealing
12N = Buna (NBR)
12V = Viton (FPM)</p> <p>C symbol function mode
C = normally closed</p> <p>D construction
0 = screw-in valve</p> <p>E voltage
DC current:
012D = 12V DC
024D = 24V DC
AC current:
120A = 120V AC
240A = 240V AC
other voltage on request</p> | <p>F power supply
G = DIN 43650 connector, other
connectors see data sheet no.: MS-010-000-00</p> <p>G coil
H = high pressure variant</p> |
|--|---|

Technical Data:

nominal pressure:	350 bar
nominal flow:	114 l/min
weight:	0.6 kg
temperature range:	-40°C to 100°C
fluids:	hydraulic oil as per DIN 51524 T1 (ISO TC 131)
max. leakage:	5 drops/min at 22 cSt (50°C) and 350 bar
seals:	optional Buna (NBR) or Viton (FPM), as well as Teflon wascher (PTFE)
power consumption:	26 W
response times:	with 100% change of current, nominal pressure and approx. 80% of the flow open: .. ms close: .. Ms
cavity:	C-12-2 see datasheet no. <u>IK-100-129-00</u>
housing:	2LH-12.-B.. see datasheet no. <u>IH-100-000-00</u>
seal kit:	SP-MDS-12N-22 (NBR), SP-MDS-12V-22 (FPM)

9.3



INTEGRAL HYDRAULIK											
Anlagen + Systeme											
Werkstoff X											
Halbzeug-Nr. X											
Allgemeintoleranzen ISO 2768-m				Ersatz für				Zeichen Änderungen Ä.-M.-Nr Datum			
über 0,5 6 30 120 400				.				Datum Name Zeichnungs Nr.			
bis 6 30 120 400 1000				Ersetzt durch				gez. 24.09.03 Marek			
±0,1 ±0,2 ±0,3 ±0,5 ±0,8				.				gepr. N. gepr. 020-1531-014-014			
Oberflächenangaben nach DIN ISO 1302											
ID-Nr.				Gewicht		Maßstab		Benennung			
								Check valve			

IN LINE NEEDLE VALVES CHECK VALVES IN CARBON STEEL

FT 257/2

They allow flow control in both directions. Needle adjustments give:

- full shut off (via metal seat);
- accurate control for a wide range of flowrates.

The valve has a graduated adjustment scale below the handle to indicate accurately the valve position.

There is a locking screw in the handle to allow the handle to be fixed. (Preventing accidental adjustment or movement due to vibration).

A panel mounting nut (G) can be supplied on request.

On request

- Versions in AISI 316 stainless steel code FT 2257/2
- Viton seals (V)
- NPT threads
- Complete with panel mounting nut (G)
- Handwheel in ABS (mp) - plastic

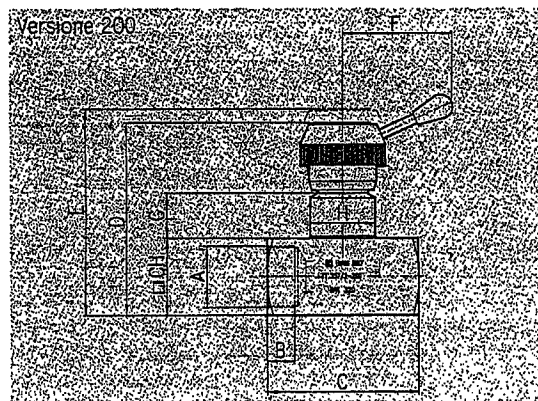
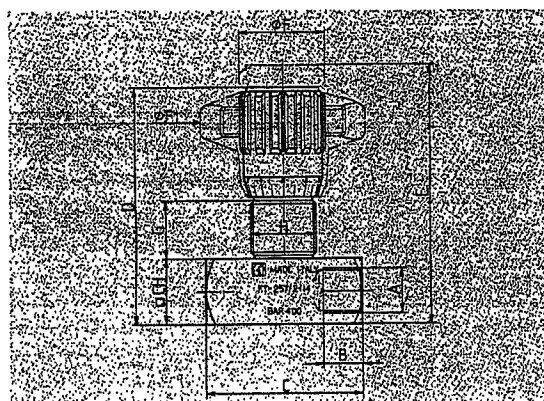


MATERIALS

Body	9 s Mn Pb 23 - UNI 5105
Needle	1 C 40 - UNI 8373
Or	Nitrile
Antiextrusion ring	PTFE
Handwheel	GD - Al Si 12 - UNI 5706 aluminium
Handwheel (mp)	ABS - plastic

EXAMPLE FOR ORDERING

Accessories on request					
	Code	Type	Panel ring	Viton seal	Handwheel in plastic
Steel	FT 257/2	18	G	V	mp
Stainl. steel	FT 2257/2	38	G	-	mp



CODE FT 257/2

Type	A UNI 338	B	C	D	E	F	F1	G	H	CH	Weight kg
18	1/8"G	8.5	38	59	64	22	40	13.5	M17x1	16	0.110
14	1/4"G	12.5	49	71	78	27	50	17	M20x1	20	0.200
38	3/8"G	12.5	59	84	93	33	70	19.5	M25x1.5	25	0.375
12	1/2"G	15.5	68	97	107	38	80	21	M30x1.5	30	0.600
34	3/4"G	17	86	120.5	132.5	47	100	26.5	M40x1.5	40	1.250
100	1"G	20	105	151.5	167.5	58	120	35	M50x1.5	50	2.550
114	1 1/4"G	22	120	156.5	172.5	58	120	35	M50x1.5	55	3.000
112	1 1/2"G	24	134	167	181	58	120	35	M55x2	65	4.217
200	2"G	27	150	188	202	85	/	44	M65x2	75	7.300

Note: Instructions for panel mounting on page 17

Specification

Type	air throughput [l/min]*			fig.	connection size ***	part number			
	Mic	Sm-L	Mol			Mic	Mic-UM	Sm-L	Sm-L-UM
Pi 0145	2100	1800	**	1	DIN	610.210.7	610.211.5	610.212.3	610.213.1
Pi 0146	1800	1500	**	1+1a	DIN	610.214.9	610.215.6	610.216.4	610.217.2
Pi 0147	2000	1700	**	3 **	40	610.218.0	610.219.8	610.220.6	610.221.4
Pi 0148	2400	2000	**	2 **	52	610.222.2	610.223.0	610.224.8	610.225.5
Pi 0149	1600	1300	**	4 **	1 1/2	610.226.3	610.227.1	610.228.9	610.229.7

* throughput at delta-p of 10 mbar

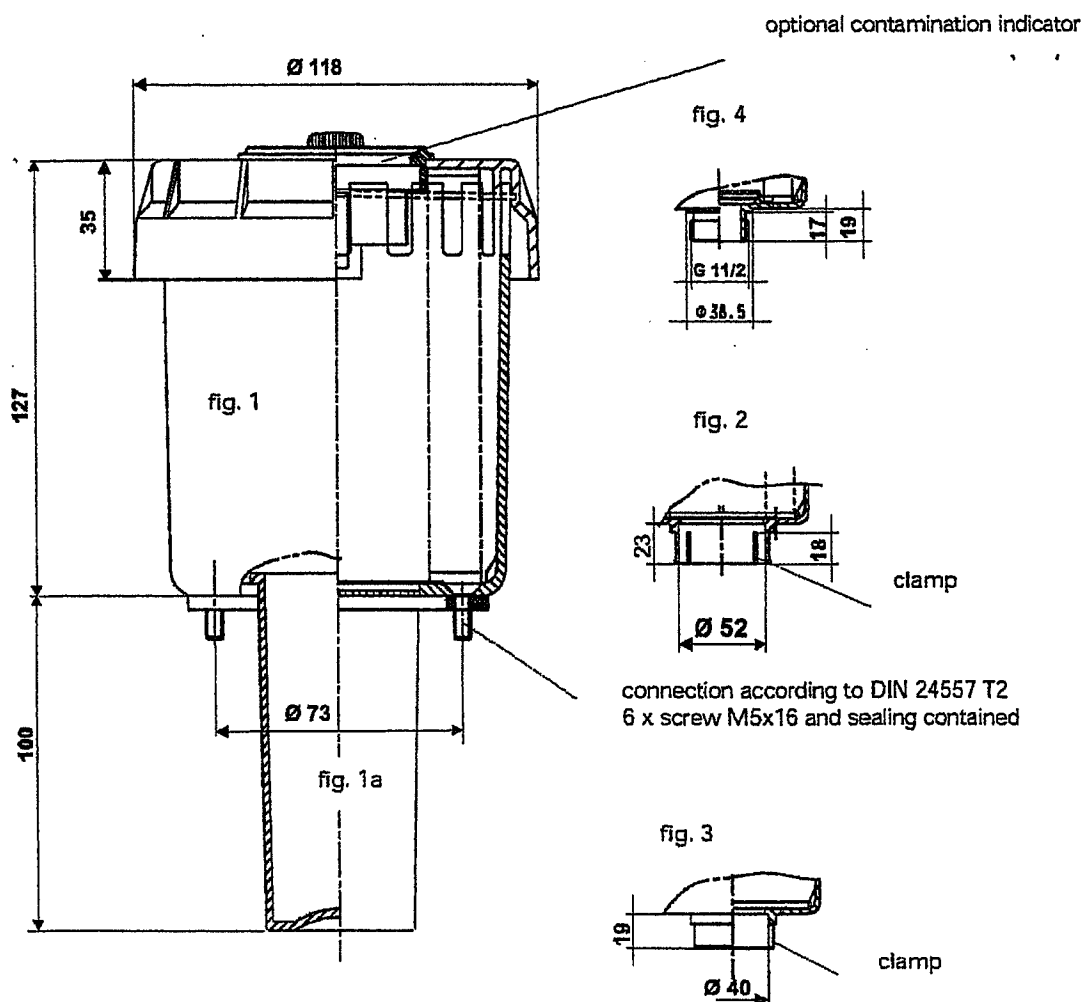
** in preparation

*** fit in the range H11

Retention rate	
Mic	10 µm > 99 %
Sm-L	3 µm > 99 %

Temperature range	-40 °C up to +100 °C
Housing material	black polyamide
Sealing material	rubberized cork
Compatibility	all hydraulic oils

Special variations on request



MAHLE Filtersysteme GmbH
 Bereich Industriefilter
 Schleifbachweg 45 D-74613 Öhringen
 P.O. Box 13 09 D-74603 Öhringen
 Tel./Phone +49 79 41 67-0
 Telefax +49 79 41 67-4 29
 Internet: www.mahle.com
 E-mail: ub2.Industrie@mahle.com

Subject to technical alteration!

20.1

Medium Pressure Filter

Pi 3000

Operating pressure 200 (315) bar, Nominal size up to 400,
according to DIN 24550

1. Features

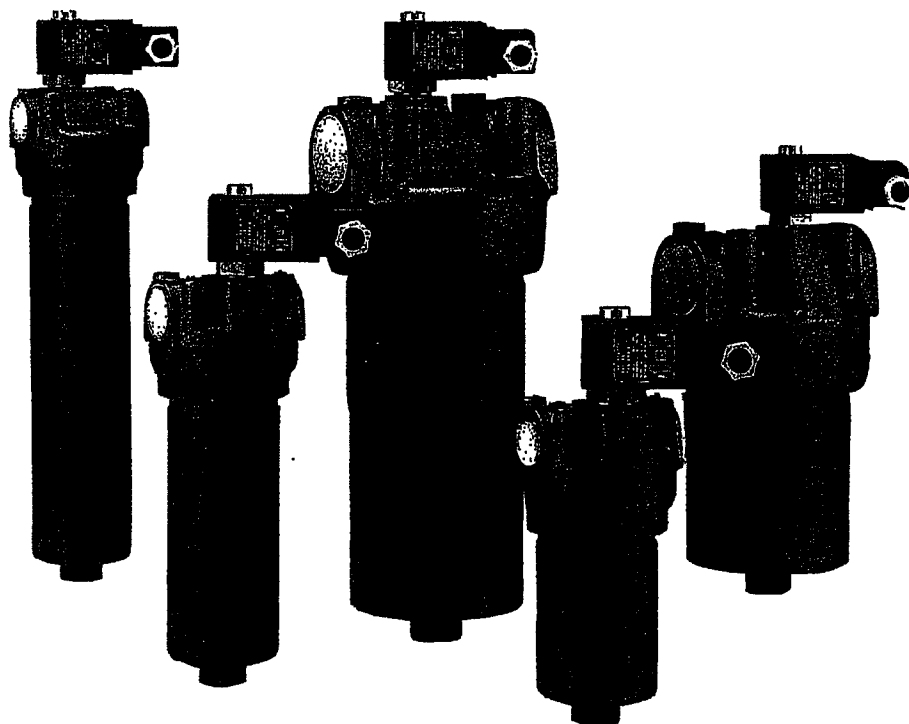
Efficient filters for modern hydraulic systems

- Modular design
- Minimal pressure loss
- Compact design
- Visual / electrical / electronical differential pressure indication
- Threaded ports

Quality filters, easy to service

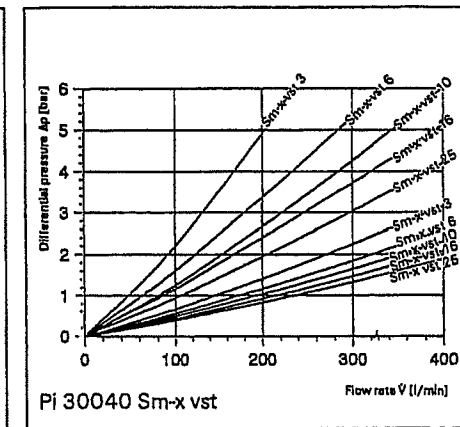
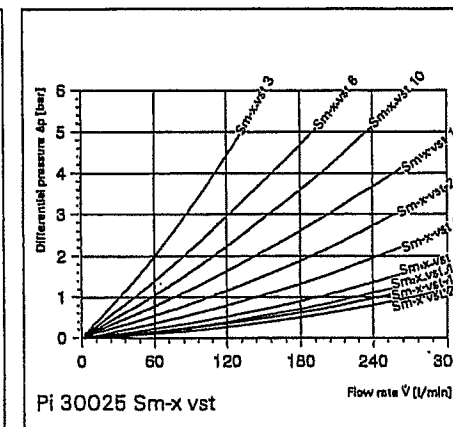
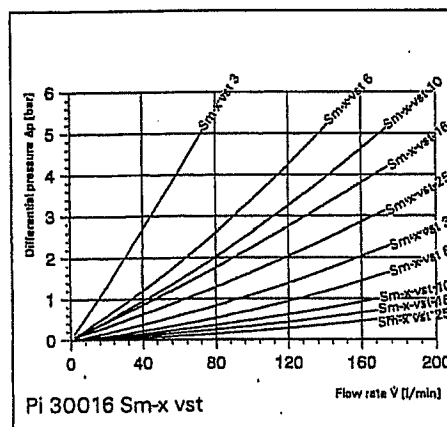
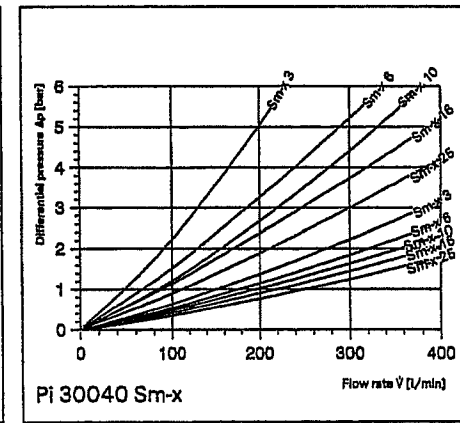
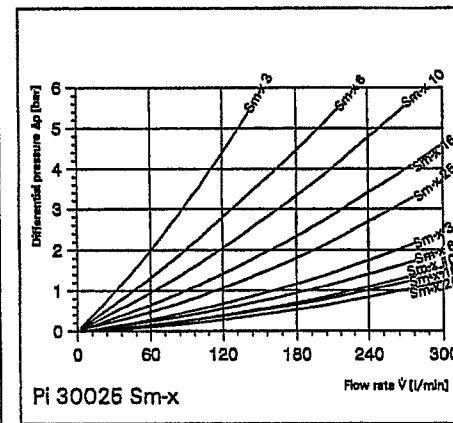
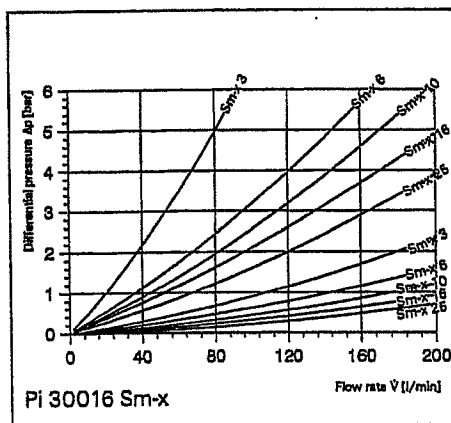
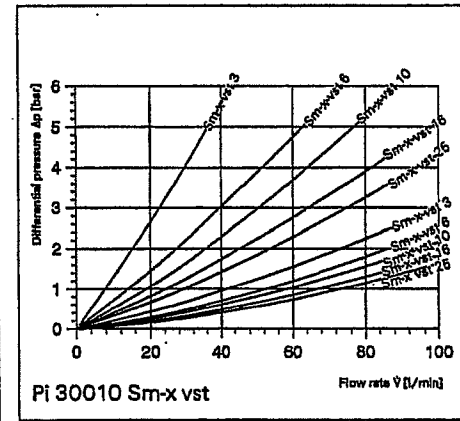
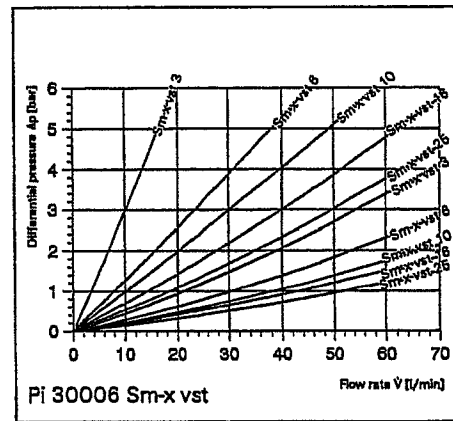
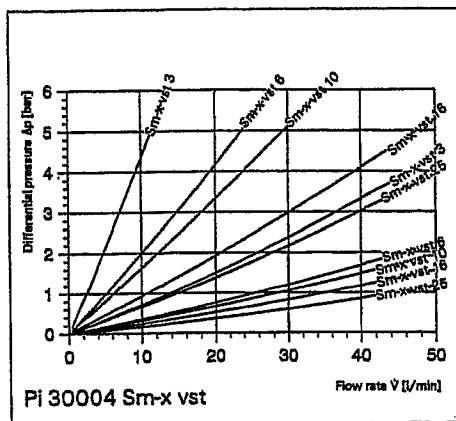
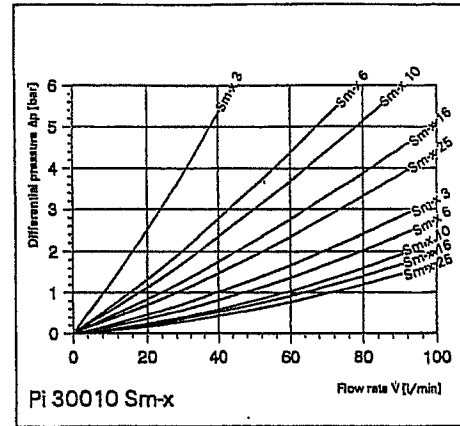
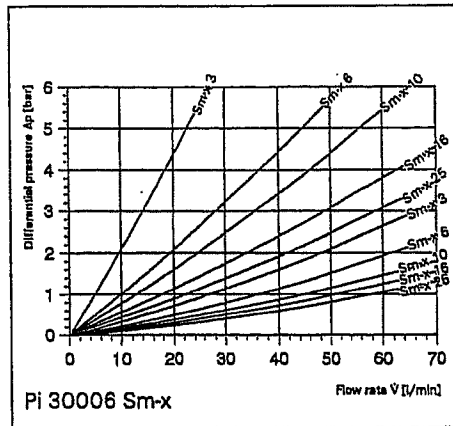
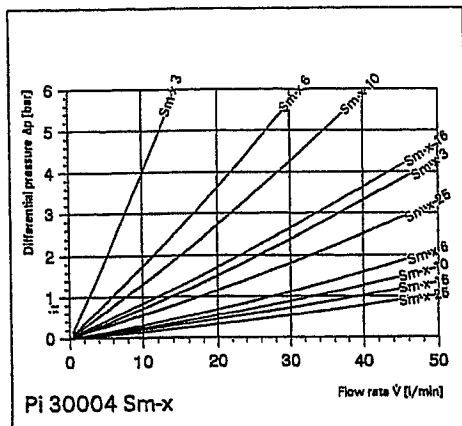
- Highly efficient Sm-x filter elements
- β -rated elements per ISO 4572
- Large dirt holding capacity and high differential pressure stability providing optimal element service life
- 100% bubble-point tested elements

World-wide sales

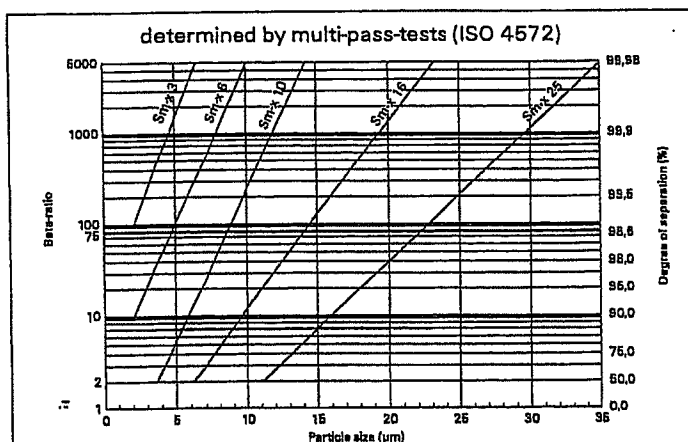
**MAHLE**[®]
Industriefilter

2. Flow rate / pressure drop curve compl. filter

190 mm²/s (25° E)
33 mm²/s (4,5° E)



3. Separation characteristics



4. Filter performance data

tested according to ISO 4572 (Multi-Pass-Test)

Sm-x elements
with Δp 20 bar

Sm-x 3 $\beta_{0.5} \geq 75$
Sm-x 6 $\beta_{0.5} \geq 75$
Sm-x 10 $\beta_{1.0} \geq 75$
Sm-x 16 $\beta_{1.5} \geq 75$
Sm-x 25 $\beta_{2.5} \geq 75$

at 7 bar differential pressure

Sm-x vst elements
with Δp 210 bar

Sm-x vst 3 $\beta_{0.5} \geq 75$
Sm-x vst 6 $\beta_{0.5} \geq 75$
Sm-x vst 10 $\beta_{1.0} \geq 75$
Sm-x vst 16 $\beta_{1.5} \geq 75$
Sm-x vst 25 $\beta_{2.5} \geq 75$

at 16 bar differential pressure

Example for ordering filters:

1. Housing design V = 100 l/min, electrical indicator
Type no. Pi 30 010-15 Order no. 820.808.4
2. Filter element Sm-x vst 3
Type no. Pi 71 010 DN Order no. 822.748.0

7. Order numbers

7.1 Housing design								
Part number	Type number	Nominal size (NG)	① With indicator cavity	② With bypass valve and indicator cavity	③ With bypass valve and visual indicator	④ With bypass valve and electrical indicator	⑤ With visual indicator	⑥ With electrical indicator
		40						
820.789.6	Pi 30 004-010							
820.790.4	Pi 30 004-011							
833.738.8	Pi 30 004-012							
830.420.6	Pi 30 004-013							
820.793.8	Pi 30 004-014							
820.794.6	Pi 30 004-015							
		63						
820.796.1	Pi 30 006-010							
820.797.9	Pi 30 006-011							
820.798.7	Pi 30 006-012							
830.421.4	Pi 30 006-013							
820.800.1	Pi 30 006-014							
820.801.9	Pi 30 006-015							
		100						
820.803.5	Pi 30 010-010							
820.804.3	Pi 30 010-011							
820.805.0	Pi 30 010-012							
830.422.2	Pi 30 010-013							
820.807.6	Pi 30 010-014							
820.808.4	Pi 30 010-015							
		160						
820.810.0	Pi 30 016-010							
820.811.8	Pi 30 016-011							
820.812.6	Pi 30 016-012							
825.997.0	Pi 30 016-013							
820.814.2	Pi 30 016-014							
820.815.9	Pi 30 016-015							
		250						
820.816.7	Pi 30 025-010							
820.817.5	Pi 30 025-011							
820.818.3	Pi 30 025-012							
825.998.8	Pi 30 025-013							
820.820.9	Pi 30 025-014							
820.821.7	Pi 30 025-015							
		400						
820.822.5	Pi 30 040-010							
820.823.3	Pi 30 040-011							
820.824.1	Pi 30 040-012							
825.999.6	Pi 30 040-013							
820.826.6	Pi 30 040-014							
820.827.4	Pi 30 040-015							

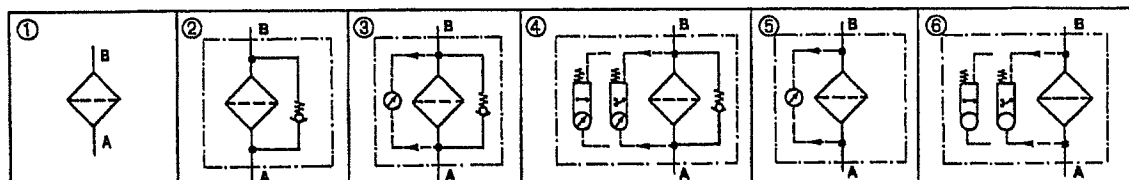
When filter with non bypass configuration is selected the collapse pressure of the element may not be exceeded.

5. Test regulations

MAHLE filter elements are manufactured respectively, tested in accordance with the following international standards:

No.	Designation
ISO 2941	Hydraulic-filter elements: Verification of burst resistance
ISO 2942	Hydraulic-filter elements: Determination of fabrication integrity
ISO 2943	Hydraulic-filter elements: Verification of material compatibility with hydraulic fluids
ISO 3723	Hydraulic-filter elements: Method for testing end-cap load
ISO 3724	Hydraulic-filter elements: Verification of flow fatigue characteristics
ISO 3968.2	Hydraulic-filters: Evaluation of pressure drop versus flow
ISO 4572	Hydraulic-filter elements: Testing of filter performance (multi-pass-test)

6. Symbols



7.2 Filter elements

(A) = Filter surface area; (L) = Type number

	Sm-x 3 Δp 20 bar (540 cm²)	Sm-x 6 Δp 20 bar (540 cm²)	Sm-x 10 Δp 20 bar (540 cm²)	Sm-x 16 Δp 20 bar (540 cm²)	Sm-x 25 Δp 20 bar (540 cm²)	Sm-x vst 3 Δp 210 bar (440 cm²)	Sm-x vst 6 Δp 210 bar (440 cm²)	Sm-x vst 10 Δp 210 bar (440 cm²)	Sm-x vst 16 Δp 210 bar (440 cm²)	Sm-x vst 25 Δp 210 bar (440 cm²)
						[PI 71 004 DN] 821.607.9	[PI 72 004 DN] 796.015.6	[PI 73 004 DN] 792.565.4	[PI 74 004 DN] 821.608.7	[PI 75 004 DN] 821.609.6
	[PI 21 004 DN] 826.092.9 (910 cm²)	[PI 22 004 DN] 796.085.9 (910 cm²)	[PI 23 004 DN] 792.557.1 (910 cm²)	[PI 24 004 DN] 826.093.7 (910 cm²)	[PI 25 004 DN] 826.094.5 (910 cm²)					
						[PI 71 006 DN] 821.613.7	[PI 72 006 DN] 796.014.9	[PI 73 006 DN] 792.566.2	[PI 74 006 DN] 821.614.5	[PI 75 006 DN] 821.615.2
	[PI 21 006 DN] 826.096.0 (1490 cm²)	[PI 22 006 DN] 796.086.7 (1490 cm²)	[PI 23 006 DN] 792.558.9 (1490 cm²)	[PI 24 006 DN] 826.097.8 (1490 cm²)	[PI 25 006 DN] 826.098.8 (1490 cm²)					
						[PI 71 010 DN] 822.748.0	[PI 72 010 DN] 796.013.1	[PI 73 010 DN] 792.567.0	[PI 74 010 DN] 826.128.1	[PI 75 010 DN] 821.616.0
	[PI 21 010 DN] 822.747.2 (2420 cm²)	[PI 22 010 DN] 796.087.5 (2420 cm²)	[PI 23 010 DN] 792.559.7 (2420 cm²)	[PI 24 010 DN] 826.100.0 (2420 cm²)	[PI 25 010 DN] 826.101.8 (2420 cm²)					
						[PI 71 016 DN] 794.063.8	[PI 72 016 DN] 796.012.3	[PI 73 016 DN] 792.568.8	[PI 74 016 DN] 826.979.7	[PI 75 016 DN] 821.617.8
	[PI 21 016 DN] 826.103.4 (4020 cm²)	[PI 22 016 DN] 796.082.6 (4020 cm²)	[PI 23 016 DN] 792.560.5 (4020 cm²)	[PI 24 016 DN] 826.104.2 (4020 cm²)	[PI 25 016 DN] 826.105.9 (4020 cm²)					
						[PI 71 025 DN] 794.064.6	[PI 72 025 DN] 796.011.6	[PI 73 025 DN] 792.569.6	[PI 74 025 DN] 826.981.3	[PI 75 025 DN] 821.618.6
	[PI 21 025 DN] 822.751.4 (6700 cm²)	[PI 22 025 DN] 796.083.4 (6700 cm²)	[PI 23 025 DN] 792.561.3 (6700 cm²)	[PI 24 025 DN] 826.107.5 (6700 cm²)	[PI 25 025 DN] 826.108.3 (6700 cm²)					
						[PI 71 040 DN] 794.065.3	[PI 72 040 DN] 796.010.7	[PI 73 040 DN] 793.082.9	[PI 74 040 DN] 826.982.1	[PI 75 040 DN] 826.090.0
	[PI 21 040 DN] 822.752.2	[PI 22 040 DN] 796.084.2	[PI 23 040 DN] 792.562.1	[PI 24 040 DN] 826.110.9	[PI 25 040 DN] 826.111.7					

* Further elements available upon request

8. Specifications

Design:	line mounting filter
Operating pressure:	200 bar*
Static test pressure:	260 bar*
Temperature range:	-10 °C to +120 °C (other temperature ranges on request)
Bypass opening pressure:	Δp 7 bar $\pm$ 10 %
Filter head material:	GGG
Filter bowl material:	St
Material of seals:	NBR / PTFE / Cu
Activating pressure of visual / electrical differential pressure indicator:	Δp 5 bar $\pm$ 10 %
Electrical data of contamination indicator:	
Maximum voltage:	230 V $\sim$ / =
Maximum current on contact:	2,5 A
Maximum contact load:	60 VA / 40 W
Inrush current:	70 VA
Type of protection:	IP 65 when inserted and secured
Contact:	bistable
Cable connection:	PG 11 $\varnothing$ 6-10

The electrical indicator function can be changed from the Normally Open position to the Normally Closed position or visa versa by inverting the electrical section.

With the inrush current of 70 VA the indicator can trigger small contactors or contactor relays.

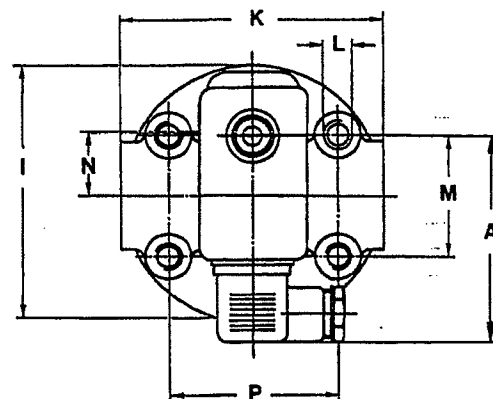
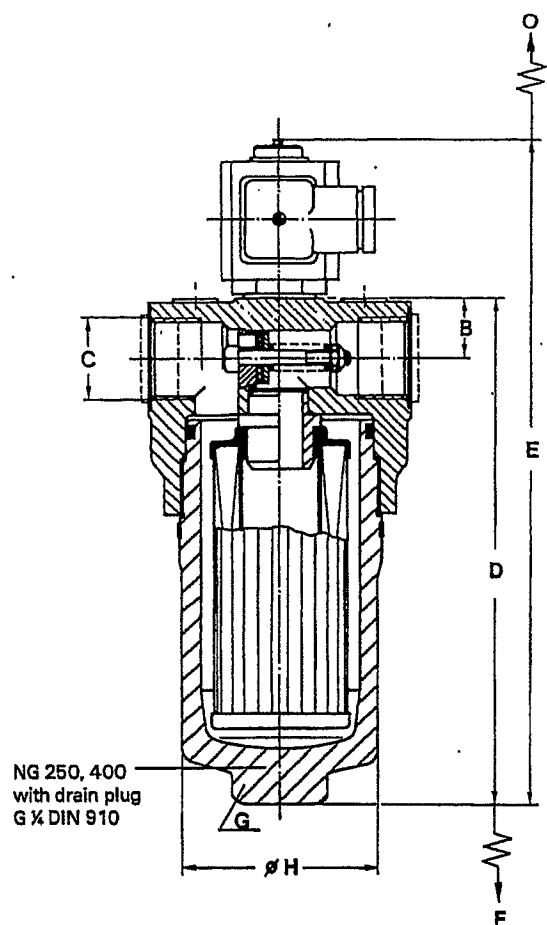
Inductivity in the direct current may require the use of a signal eraser.

For further information and executions please see our leaflet "contamination indicators".

Filters compatible with standard mineral oils.

Please contact us in case of using other media.

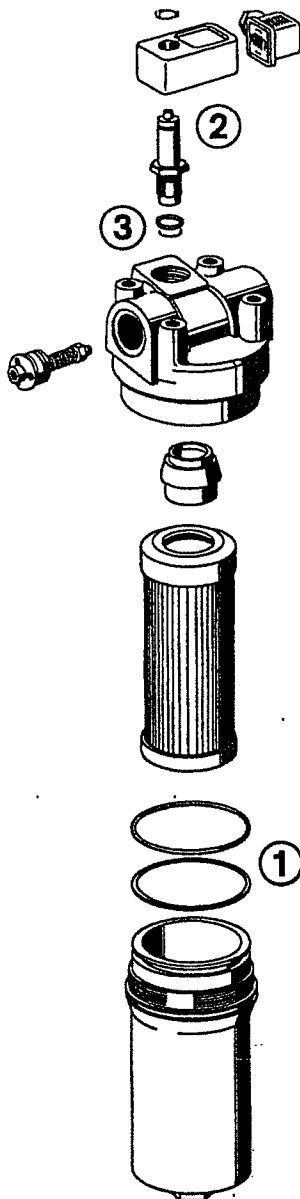
* Types Pi 30 004 – Pi 30 010 without bypass have an operating pressure of 315 bar/test pressure 410 bar, with bypass 250 bar/325 bar.



9. Dimensions

All dimensions (except "C") in mm

Dimension Type	A	B	C	D	E	F	G (SW)	H	I	K	L	M	N	O	P	Weight [kg]
Pi 30 004	78	23	G ½	182	234	80	27	66	84	85	M 8x15	40	23	45	55	3,6
Pi 30 006	78	23	G ½	242	294	80	27	66	84	85	M 8x15	40	23	45	55	3,8
Pi 30 010	78	23	G ¾	325	377	80	27	66	84	85	M 8x15	40	23	45	55	4,4
Pi 30 016	78	32	G 1 ¼	267	319	110	30	109	137	142	M 12x16	60	28	45	80	11,6
Pi 30 025	78	32	G 1 ¼	368	420	110	30	109	137	142	M 12x16	60	28	45	80	14,0
Pi 30 040	78	32	G 1 ¼	518	570	110	30	109	137	142	M 12x16	60	28	45	80	16,6



11. Spare parts list

Pos.	part number / housing	
	Pi 30 004-Pi 30 010	Pi 30 016-Pi 30 040
①	Seal kit	Seal kit
	NBR 838.374.7	NBR 838.377.0
	FPM 838.375.4	FPM 838.378.8
	EPDM 838.376.2	EPDM 838.379.6
②	Contamination indicator	
	visual	electrical
	766.991.4	766.986.4
	Pis 3093/5 bar	Pis 3092/5 bar
③	Seal kit for contamination indicator	
	NBR 776.027.5	
	FPM 776.028.3	
	EPDM 776.029.1	

10. Installation, operating and maintenance instructions

10.1 Filter installation

When installing the filter make sure that sufficient space is available to remove filter element and filter bowl. Preferable the filter should be installed with the filter bowl pointing downwards. The contamination indicator must be visible.

10.2 Connecting the electrical contamination indicator

The electrical indicator is connected via a 2-pole appliance plug according DIN 43650 with poles marked 1 and 2. The electrical section can be inverted to change from Normally Open position to Normally Closed position or visa versa.

10.3 When must the filter element be replaced?

- Filters equipped with visual and electrical contamination indicator:
During cold starts, the indicator may give a warning signal. Depress the red button of the visual indicator once again only after operating temperature has been reached. If the red button immediately pops out again and/or the electrical signal has not switched off after reaching operating temperature, the filter element must be replaced after the end of the shift.
- Filters without contamination indicator:
The filter element should be replaced after the trial run or flushing of the system. Afterwards follow instructions of the manufacturer.
- Please always ensure that you have original MAHLE-replacement elements in stock: disposable elements (Sm-x) cannot be cleaned.

10.4 Element replacement

- Stop system and relieve filter from pressure.
- Unscrew the filter bowl by turning counter-clockwise. Clean the bowl using a suitable cleaning solvent.
- Remove filter element with a side-to-side motion.
- Check O-ring and back-up ring on the filter bowl for damage. Replace, if necessary.
- Make sure that the part number on the spare element corresponds with the part number of the filter label. Open the plastic bag and push element over the spigot in the filter head. Now remove plastic bag.
- Complete installation by screwing on the bowl, turning clockwise until it comes to a full stop. Back off the bowl $\frac{1}{4}$ to $\frac{1}{2}$ turn.

Subject to technical alteration without prior notice.



KNECHT Filterwerke GmbH

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Float Switches

23

Type-form

Float switch

Sliding tube P= PVC
M= yellow brass
E= high-grade steel

Number of contacts

A - upper contact
B - lower contact
C - auxiliary contact

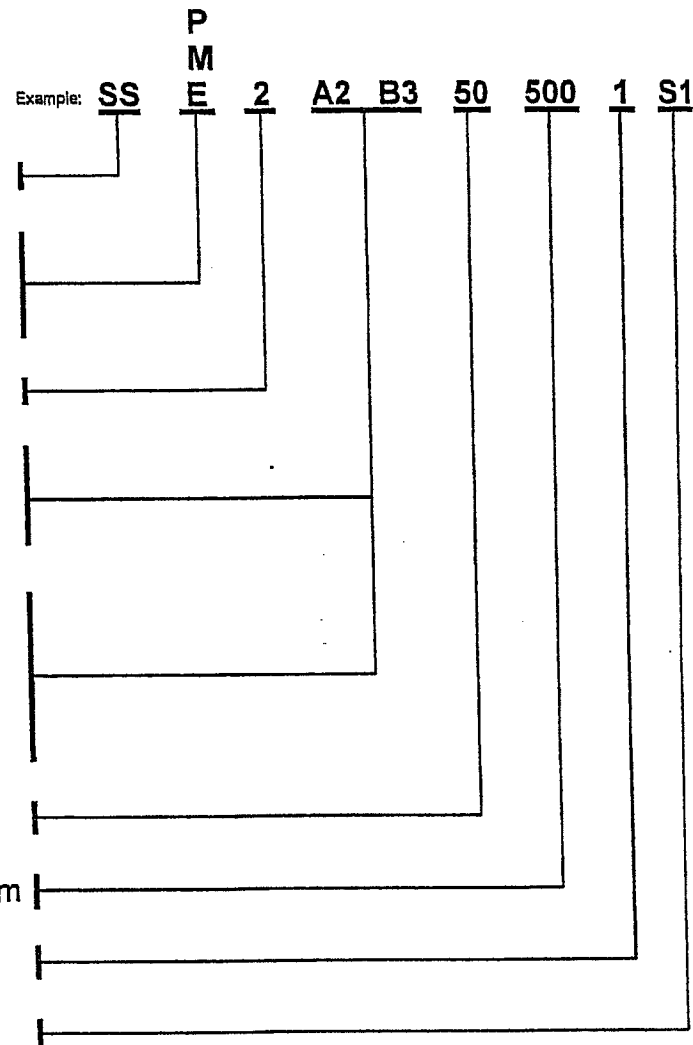
1 - closes if level rises
2 - opens if level rises
3 - closes if level decreases
4 - opens if level decreases
5 - change-over contact

50 -breaking-point-length L1 e.g. 50mm

500 -breaking-point-length L2 e.g.500mm

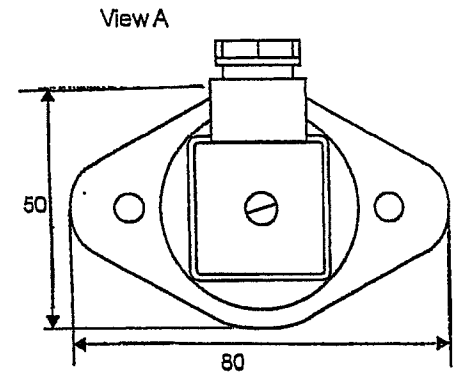
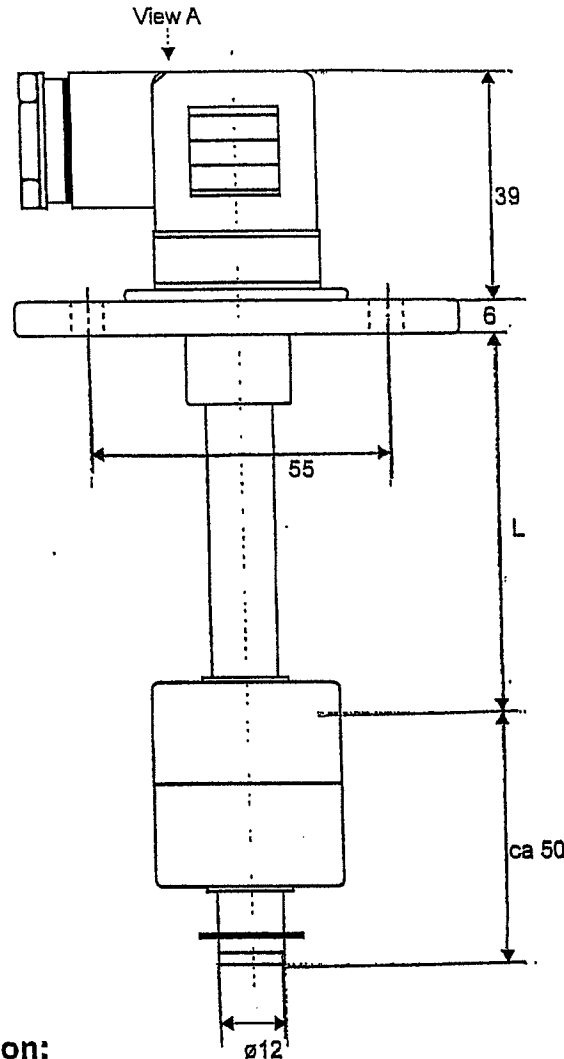
1 - design number

S1 - float e.g. 35x40mm



ENGLER Float Switch, design 5

Pg9



Description:

Connection:

Attachment:

Sliding tube:

Float:

Reed contacts:

Max. Switching voltage:

Max. Switching current:

Max. Switching power:

Max. Temperature:

Protection:

right angle plug, DIN 43650

oval flange, made of makrolon

yellow brass, d= 12mm; high-grade steel

1.4571 d= 12mm; PVC, d= 12mm

S1= float PP, 35x40mm = standard

On request:

S2= float PP 40x40mm

S9= float PP 40x27mm

S4= float high-grade steel 1.4571, 44x53mm

max. 2 normally opened/closed contacts

1 change-over contact.

normally opened/closed contact = 230VAC

change-over contact = 230VAC

normally opened/closed contact = 3A;

change-over contact = 1A

normally opened/closed contact = 60VA;

change-over contact = 60VA

80°C; PVC 60°C. Higher temperatures possible on request

IP65